

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\142-D1-D2-Rev 01.pvdb

Date of Analysis : Aug 12, 2026 5:10pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec :	3650.000 x 2900.000 mm ID
Vessel Design Length, Tangent to Tangent	7225.00 mm
Distance of Bottom Tangent above Grade	5000.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	4700.00 mm
Internal Design Temperature	100 °C
Internal Design Pressure	0.485 MPa
External Design Temperature	0 °C
Maximum Allowable Working Pressure	5 MPa
Shop Test Pressure	0.631 MPa
Field Test Pressure	6.027 MPa
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	42.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-516 70	...	...	K02700	+IT
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-182 F316L	...	<= 130	S31603	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F316L	...	<= 130	S31603	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-3	...	...	...	1.6000	N/A	No
Skirt-2	...	...	...	1.6000	N/A	No
Skirt-1	...	...	...	1.6000	N/A	No
Bottom Head D1	4.972	0.00	5.000	1.0000	No	No
Shell-1	4.963	0.00	5.000	1.0000	N/A	No
Shell-D1'	4.951	0.00	5.000	1.0000	N/A	No
Top Head D1	4.951	0.00	5.000	1.0000	No	No
Int.Skirt-1	...	...	...	...	N/A	No
Int.Skirt-2	...	...	...	...	N/A	No
Int.Skirt-3	...	...	...	...	N/A	No
Bottom Head-D2	4.566	0.00	5.000	3.0000	No	No
Shell-D2	4.558	0.00	5.000	3.0000	N/A	No
Top Head-D2	4.551	0.00	5.000	3.0000	No	No

Liquid Level: 3737.50 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.025

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-3000.0	1700.0	18.0	18.0	...	...	0.70	0.70
Skirt	-1000.0	2000.0	18.0	18.0	...	...	0.70	0.70
Skirt	-0.0	1000.0	18.0	18.0	...	...	0.70	0.50
Ellipse	50.0	50.0	78.0	68.0	67.0	...	1.00	1.00
Cylinder	3050.0	3000.0	69.0	69.0	68.1	...	1.00	1.00
Cylinder	4050.0	1000.0	69.0	69.0	67.9	...	1.00	1.00
Ellipse	4100.0	50.0	78.0	68.0	66.7	...	1.00	1.00
Skirt	4600.0	500.0	18.0	18.0	...	...	0.70	0.70
Skirt	7400.0	2800.0	18.0	18.0	...	...	0.70	0.70
Skirt	7900.0	500.0	18.0	18.0	...	...	0.70	0.70
Ellipse	7950.0	50.0	60.0	52.0	51.1	...	1.00	1.00
Cylinder	10975.0	3025.0	53.0	53.0	52.0	...	1.00	1.00
Ellipse	11025.0	50.0	60.0	52.0	50.9	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_10"	WRC-107/537	0.884	n/a	Passed
N1_36"	WRC-297	0.978	Shell	Passed
N5_4"	WRC-297	0.978	Shell	Passed
N2_30"	WRC-107/537	0.880	n/a	Passed
D2N3_3"	WRC-107/537	0.882	n/a	Passed
D2N1_24"	WRC-297	0.971	Shell	Passed
N2_24"	WRC-107/537	0.879	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	850	44893.3	314468	398494592	3.410E+09
20	30	2700	41816.9	313619	324761120	2.876E+09
30	40	4200	37144.3	311388	245767968	2.250E+09
40	50	4725	35334.6	310121	209513840	1939359872
50	60	6250	31872	290671	207832992	1924333952
60	70	8250	26147.2	222240	120768880	1154654720
70	80	8775	20136.1	207738	97617856	939578560
80	90	9050	20040.7	173388	96613024	930046656
90	100	10700	19137.3	172025	86814552	843658304
100	110	12350	14600.6	162015	39562304	375812384
110	120	12625	13883.6	160541	32438380	295140832
120	130	14162.5	13807.1	139916	31745824	287626272
130	140	15700	5712.36	39953.9	2210679	15462157

Abs Max of the all of the Stress Ratio's : 0.4907

Basing Data : Continuous Top Ring W/Gussets

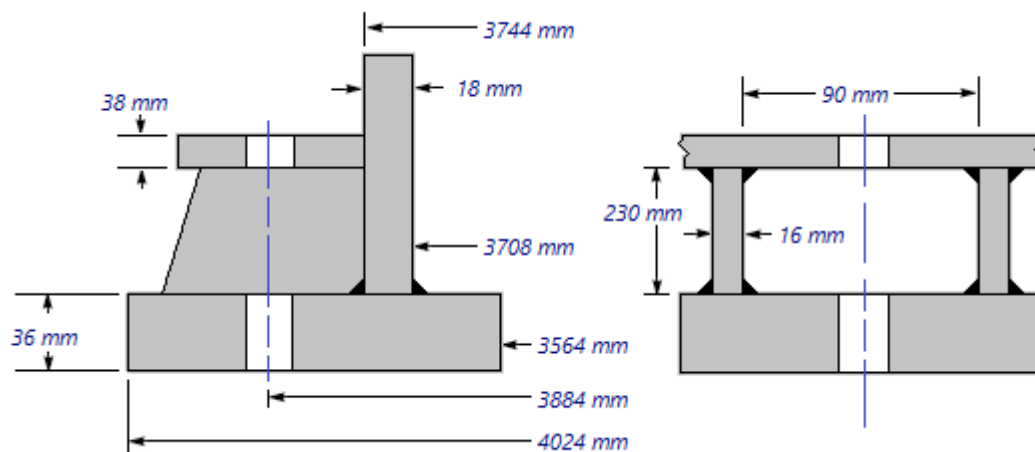
Thickness of Basing	36.0000	mm
Inside Diameter of Basing	3563.9998	mm
Outside Diameter of Basing	4024.0000	mm
Nominal Diameter of Bolts	42.0000	mm
Diameter of Bolt Circle	3884.0000	mm
Number of Bolts	20	

Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	130.0000	mm
Height of Gussets	230.0000	mm
Distance between Gussets	90.0000	mm



Thickness of Top Plate or Ring	38.0000	mm
Radial Width of the Top Plate	140.0000	mm
Number of Gussets	40	

Basering Sketch for Cylindrical Skirts:



Basering Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	167350.	N
Total Earthquake Shear on Support	436924.	N
Wind Moment on Support	2965729792.	N-mm
Earthquake Moment on Support	5976825856.	N-mm
Maximum Initial Bolt Load	204325.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	129382.8	kgm
Shop Test - Fabricated + water (full)	212454.2	kgm
Shipping - Fab. + rem. intls.+ shipping app.	147414.5	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	147864.5	kgm
Empty - Fab. + intls. + details + wghts.	149764.6	kgm
Operating - Empty + operating liquid (no ca)	204521.1	kgm
Field Test - Empty weight + test medium (full)	232149.6	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3_10"	5	...	...	OK	9.10	OK	Passed
N1_36"	5	...	...	OK	17.17	OK	Passed
N5_4"	5	...	...	OK	6.02	OK	Passed
K3_2"	5	...	...	OK	5.80	OK	No Calc[*]
K4_2"	5	...	...	OK	5.80	OK	No Calc[*]
K5_2"	5	...	...	OK	5.80	OK	No Calc[*]
K6_2"	5	...	...	OK	5.80	OK	No Calc[*]
K7_2"	5	...	...	OK	5.80	OK	No Calc[*]
K8_2"	5	...	...	OK	5.80	OK	No Calc[*]
K9_2"	5	...	...	OK	5.80	OK	No Calc[*]
K10_2"	5	...	...	OK	5.80	OK	No Calc[*]
M1_24"	5	...	...	...		OK	Passed
K1_2"	5	...	...	OK	4.80	OK	No Calc[*]
K2_2"	5	...	...	OK	4.80	OK	No Calc[*]
N2_30"	5	...	...	OK	13.47	OK	Passed
D2N3_3"	5	...	...	OK	7.80	OK	Passed
D2N1_24"	5	...	...	OK	12.92	OK	Passed
D2K1_2"	5	...	...	OK	7.80	OK	No Calc[*]
D2K2_2"	5	...	...	OK	7.80	OK	No Calc[*]
D2K3_2"	5	...	...	OK	7.80	OK	No Calc[*]
D2K4_2"	5	...	...	OK	7.80	OK	No Calc[*]
D2K5_2"	5	...	...	OK	7.80	OK	No Calc[*]
D2K6_2"	5	...	...	OK	7.80	OK	No Calc[*]
D2K7_2"	5	...	...	OK	7.80	OK	No Calc[*]
D2K8_2"	5	...	...	OK	7.80	OK	No Calc[*]
D2K9_2"	5	...	...	OK	7.80	OK	No Calc[*]
D2K10_2"	5	...	...	OK	7.80	OK	No Calc[*]
D2M1_24"	5	...	...	...		OK	Passed
N2_24"	5	...	...	OK	12.92	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 5 Nozzle : D2N33"
 Minimum MAWP Shells/Flanges : 5 Element : Bottom Head-D2
 Minimum MAPnc Shells/Flanges : 5 Element : Top Head-D2

Computed Vessel M.A.W.P. : 5 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
40	N3_10"	0.000	0.000	472.000
50	N1_36"	2300.001	0.000	1760.800
50	N5_4"	1350.000	180.000	237.600
50	K3_2"	1200.001	0.000	220.400
50	K4_2"	500.001	0.000	220.400
50	K5_2"	1000.001	200.000	220.400
50	K6_2"	300.001	200.000	220.400
50	K7_2"	1000.001	220.000	220.400
50	K8_2"	300.001	220.000	220.400
50	K9_2"	1000.001	240.000	220.400
50	K10_2"	300.001	240.000	220.400
50	M1_24"	1200.001	270.000	1175.200



60	K1_2"	3800.001	0.000	220.400
60	K2_2"	3400.001	0.000	220.400
70	N2_30"	0.000	0.000	1468.000
110	D2N3_3"	0.000	0.000	174.200
120	D2N1_24"	9525.001	0.000	1175.200
120	D2K1_2"	9500.001	250.000	184.328
120	D2K2_2"	8250.002	250.000	184.328
120	D2K3_2"	9500.001	260.000	184.328
120	D2K4_2"	8250.002	260.000	184.328
120	D2K5_2"	9500.001	270.000	184.328
120	D2K6_2"	8250.002	270.000	184.328
120	D2K7_2"	9500.001	280.000	184.328
120	D2K8_2"	8250.002	280.000	184.328
120	D2K9_2"	10500.002	0.000	184.328
120	D2K10_2"	10300.001	0.000	184.328
120	D2M1_24"	8750.002	180.000	1175.200
130	N2_24"	0.000	0.000	1175.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K3_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	369.49	600
K4_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	369.49	600
K5_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	369.49	600
K6_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	369.49	600
K7_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	369.49	600
K8_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	369.49	600
K9_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	369.49	600
K10_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	369.49	600
K1_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	369.49	600
K2_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	369.49	600
D2K1_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
D2K2_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
D2K3_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
D2K4_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
D2K5_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
D2K6_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
D2K7_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
D2K8_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
D2K9_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
D2K10_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	353.61	300
D2N3_3"	3.000 in	Actual	WNF	3.000	10.000	...	...	202.27	600
N5_4"	4.000 in	Actual	WNF	4.000	12.000	...	...	369.71	600
N3_10"	10.000 in	Actual	WNF	10.000	10.000	...	...	220.42	600
M1_24"	24.000 in	Actual	WNF	24.000	12.000	...	...	394.63	600
D2N1_24"	24.000 in	Actual	WNF	24.000	14.000	...	...	3053.00	300
D2M1_24"	24.000 in	Actual	WNF	24.000	14.000	...	...	385.40	300
N2_24"	24.000 in	Actual	WNF	24.000	14.000	...	...	369.58	300
N2_30"	30.000 in	Actual	WNF	30.000	14.000	...	...	389.81	600
N1_36"	36.000 in	Actual	WNF	36.000	18.000	...	...	3669.00	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K3_2"	SA-182 F316L	69.000	10.000	...	...	...
K4_2"	SA-182 F316L	69.000	10.000	...	...	...
K5_2"	SA-182 F316L	69.000	10.000	...	...	...
K6_2"	SA-182 F316L	69.000	10.000	...	...	...
K7_2"	SA-182 F316L	69.000	10.000	...	...	...
K8_2"	SA-182 F316L	69.000	10.000	...	...	...
K9_2"	SA-182 F316L	69.000	10.000	...	...	...
K10_2"	SA-182 F316L	69.000	10.000	...	...	...
K1_2"	SA-182 F316L	70.000	10.000	...	...	...
K2_2"	SA-182 F316L	70.000	10.000	...	...	...
D2K1_2"	SA-105	53.000	10.000	...	...	...
D2K2_2"	SA-105	53.000	10.000	...	...	...
D2K3_2"	SA-105	53.000	10.000	...	...	...
D2K4_2"	SA-105	53.000	10.000	...	...	...
D2K5_2"	SA-105	53.000	10.000	...	...	...
D2K6_2"	SA-105	53.000	10.000	...	...	...
D2K7_2"	SA-105	53.000	10.000	...	...	...

D2K8_2"	SA-105	53.000	10.000	...	...	...
D2K9_2"	SA-105	53.000	10.000	...	...	...
D2K10_2"	SA-105	53.000	10.000	...	...	...
D2N3_3"	SA-105	52.000	10.000	...	...	...
N5_4"	SA-105	69.000	10.000	...	...	...
N3_10"	SA-105	68.000	10.000	...	...	...
M1_24"	SA-105	69.000	10.000	...	...	...
D2N1_24"	SA-105	53.000	10.000	...	...	...
D2M1_24"	SA-105	53.000	10.000	...	...	...
N2_24"	SA-105	52.000	10.000	...	...	...
N2_30"	SA-105	68.000	10.000	...	...	...
N1_36"	SA-105	69.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K3_2"	1200.000	0.0	300.00	0.00	Shell-1
K4_2"	500.000	0.0	300.00	0.00	Shell-1
K5_2"	1000.001	200.0	300.00	0.00	Shell-1
K6_2"	300.000	200.0	300.00	0.00	Shell-1
K7_2"	1000.001	220.0	300.00	0.00	Shell-1
K8_2"	300.000	220.0	300.00	0.00	Shell-1
K9_2"	1000.001	240.0	300.00	0.00	Shell-1
K10_2"	300.000	240.0	300.00	0.00	Shell-1
K1_2"	3800.000	0.0	300.00	0.00	Shell-D1'
K2_2"	3400.000	0.0	300.00	0.00	Shell-D1'
D2K1_2"	9500.001	250.0	300.00	0.00	Shell-D2
D2K2_2"	8250.002	250.0	300.00	0.00	Shell-D2
D2K3_2"	9500.001	260.0	300.00	0.00	Shell-D2
D2K4_2"	8250.002	260.0	300.00	0.00	Shell-D2
D2K5_2"	9500.001	270.0	300.00	0.00	Shell-D2
D2K6_2"	8250.002	270.0	300.00	0.00	Shell-D2
D2K7_2"	9500.001	280.0	300.00	0.00	Shell-D2
D2K8_2"	8250.002	280.0	300.00	0.00	Shell-D2
D2K9_2"	10500.002	0.0	300.00	0.00	Shell-D2
D2K10_2"	10300.001	0.0	300.00	0.00	Shell-D2
D2N3_3"	...	0.0	150.00	0.00	Bottom Head-D2
N5_4"	1350.000	180.0	300.00	0.00	Shell-1
N3_10"	...	0.0	150.00	0.00	Bottom Head D1
M1_24"	1200.000	270.0	300.00	0.00	Shell-1
D2N1_24"	9525.001	0.0	300.00	2700.00	Shell-D2
D2M1_24"	8750.002	180.0	300.00	0.00	Shell-D2
N2_24"	...	0.0	300.00	0.00	Top Head-D2
N2_30"	...	0.0	300.00	0.00	Top Head D1
N1_36"	2300.000	0.0	300.00	3300.00	Shell-1



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 78.0mm THK X 3650.0mm ID X 50.0mm	SA-516 70
2	ELLIPTICAL HEAD: 2.0 X 1, 60.0mm THK X 2900.0mm ID X 50.0mm	SA-516 70
4	PLATFORM: 1000mm X 1500mm WIDE	
2	LADDER: 2500.0mm	...
2	INSULATION: 962mm X 80mm THK	
2	LINING: 962.500mm X 3.000mm THK	...
1	INSULATION: 3000mm X 80mm THK	
1	LINING: 3000.000mm X 3.000mm THK	...
1	LADDER: 4000.0mm	...
1	INSULATION: 1000mm X 80mm THK	
1	LINING: 1000.000mm X 3.000mm THK	...
2	INSULATION: 775mm X 80mm THK	
1	LADDER: 3500.0mm	...
1	INSULATION: 3025mm X 80mm THK	
1	PLATFORM: 1000mm X 4000mm WIDE	
1	LADDER: 3200.0mm	...
1	CLASS 600 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 4024.0mm OD X 3564.0 ID X 36.0mm	SA-516 70
20	BASERING BOLTS: 42.0mm DIA	SA-193 B7
20	BASERING BOLT NUTS: 42.0mm DIA	...
1	BASERING TOP RING: 4024.0mm OD X 3744.0 ID X 38.0mm	...
40	BASERING GUSSET PLATES: 130.0mm X 130.0mm X 230.0mm X 16.0mm	...
3	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3091.3mm X 1700.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2374.9mm X 1700.0mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3091.3mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2374.9mm X 2000.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3106.4mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3074.0mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2362.1mm X 1000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 69.0mm THK X 3042.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 69.0mm THK X 2337.8mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 69.0mm THK X 3042.9mm X 1000.0mm	SA-516 70



1	CYLINDRICAL SEGMENT 69.0mm THK X 2337.8mm X 1000.0mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3018.6mm X 500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2574.2mm X 500.0mm	SA-516 70
2	CON SKIRT SEGMENT 18.0mm THK X 2455.1mm X 2800.0mm X 3101.0mm	SA-516 70
1	CON SKIRT SEGMENT 18.0mm THK X 2429.6mm X 2800.0mm X 3068.7mm	SA-516 70
1	CON SKIRT SEGMENT 18.0mm THK X 1866.9mm X 2800.0mm X 2358.1mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3011.0mm X 500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 1505.5mm X 500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 1680.4mm X 500.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 53.0mm THK X 3048.0mm X 3025.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 53.0mm THK X 3014.6mm X 3025.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Bottom Head D[10]	B	24	23		0.985	68.000	1.00	Yes
Bottom Head D1[7]	B	26	19		0.873	78.000	1.00	Yes
Shell-1 [8]	B	24	23		0.987	69.000	1.00	Yes
Shell-D1' [8]	B	24	23		0.985	69.000	1.00	Yes
Top Head D1 [10]	B	24	22		0.981	68.000	1.00	Yes
Top Head D1 [7]	B	26	19		0.869	78.000	1.00	Yes
Bottom Head-D[10]	!	-12	-12		1.000	52.000	1.00	Yes
Bottom Head-D2[7]	!	-12	-19		0.877	60.000	1.00	Yes
Shell-D2 [8]	!	-12	-12		0.998	53.000	1.00	Yes
Top Head-D2 [10]	!	-12	-12		0.997	52.000	1.00	Yes
Top Head-D2 [7]	!	-12	-19		0.874	60.000	1.00	Yes
N310" [1]	A	42	36		0.887	68.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
N136" [1]	A	43	42		0.985	69.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
N54" [1]	A	-3	-104	-29	0.163	12.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
M124" [1]	A	43	42		0.985	69.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
Nozzle Flg [5]	A	-18	-48					
Nozzle Flg [5]	A	-18	-48					
N230" [1]	A	42	36		0.883	68.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
D2N33" [1]	A	-8	-104	-29	0.178	10.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
D2N1 24" [1]	A	38	36		0.978	53.000	1.00	Yes
Nozzle Flg [5]	A	-18	-24					
D2K12" [1]	!	-12	-104		0.069	16.764	1.00	No
Nozzle Flg [5]	A	-18	-24					
D2K22" [1]	!	-12	-104		0.069	16.764	1.00	No
Nozzle Flg [5]	A	-18	-24					
D2K32" [1]	!	-12	-104		0.069	16.764	1.00	No
Nozzle Flg [5]	A	-18	-24					
D2K42" [1]	!	-12	-104		0.069	16.764	1.00	No
Nozzle Flg [5]	A	-18	-24					
D2K52" [1]	!	-12	-104		0.069	16.764	1.00	No
Nozzle Flg [5]	A	-18	-24					
D2K62" [1]	!	-12	-104		0.069	16.764	1.00	No
Nozzle Flg [5]	A	-18	-24					
D2K72" [1]	!	-12	-104		0.069	16.764	1.00	No
Nozzle Flg [5]	A	-18	-24					
D2K82" [1]	!	-12	-104		0.069	16.764	1.00	No
Nozzle Flg [5]	A	-18	-24					
D2K92" [1]	!	-12	-104		0.069	16.764	1.00	No
Nozzle Flg [5]	A	-18	-24					
D2K102" [1]	!	-12	-104		0.069	16.764	1.00	No
Nozzle Flg [5]	A	-18	-24					
D2M124" [1]	A	38	36		0.978	53.000	1.00	Yes
Nozzle Flg [5]	A	-18	-24					
N224" [1]	A	37	31		0.881	52.000	1.00	Yes
Nozzle Flg [5]	A	-18	-24					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: 43 42

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature 42.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

[!] - This was an impact tested material.

[1] - Governing Nozzle Weld.



- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	30	50	There is a high jump in the Joint Eff.
Note	70	90	There is a high jump in the Joint Eff.
Note	100	120	There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.4854	MPa
Design Internal Temperature	100.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 97.2	km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration	[Ss] 0.147
Max. Eff. Ground Acceleration	[S] 0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/13
Element From Node	10



Element To Node	20
Element Type	Skirt Support
Description	Skirt-3
Distance "FROM" to "TO"	1700 mm
Skirt Outside Diameter	3744 mm
Diameter of Skirt at Base	3744 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	100 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	179.4 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	239 MPa
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No
Element From Node	10
Detail Type	Weight
Detail ID	TAILING+BRACING
Dist. from "FROM" Node / Offset dist	300 mm
Miscellaneous Weight	12748 N
Offset from Element Centerline	1900 mm

Element#	2/13
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-2
Distance "FROM" to "TO"	2000 mm
Skirt Outside Diameter	3744 mm
Diameter of Skirt at Base	3744 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	100 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element From Node	20
Detail Type	Platform
Detail ID	PLT-1
Dist. from "FROM" Node / Offset dist	500 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	90.0
Platform Wind Area	1499999 mm²
Platform Weight	11830 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	135 mm



Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	2500	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm

Element#	3/13	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Skirt-1	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	3744	mm
Diameter of Skirt at Base	3744	mm
Skirt Thickness	18	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	100	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element#	4/13	
Element From Node	40	
Element To Node	50	
Element Type	Elliptical	
Description	Bottom Head D1	
Distance "FROM" to "TO"	50	mm
Inside Diameter	3650	mm
Element Thickness	68	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	78	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.9507	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Platform	
Detail ID	BTL PLTF	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	4015999	mm ²
Platform Weight	44593	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	2500	mm



Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	-912.5	mm
Height/Length of Liquid	962.5	mm
Liquid Density	0.0009674	kg/cm ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	-912.5	mm
Height/Length of Insulation	962.5	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	40	
Detail Type	Lining	
Detail ID	cladding	
Dist. from "FROM" Node / Offset dist	-912.5	mm
Height/Length of Lining	962.5	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N310"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	10	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Weight	
Detail ID	Anti-Swirl Baff	
Dist. from "FROM" Node / Offset dist	-500	mm
Miscellaneous Weight	147.09	N
Offset from Element Centerline	0	mm

Element#	5/13	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	3650	mm
Element Thickness	69	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	69	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.9507	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	50	



Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1300 mm
Liquid Density	0.0009674 kg/cm ³
Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³
Element From Node	50
Detail Type	Lining
Detail ID	cladding
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³
Element From Node	50
Detail Type	Nozzle
Detail ID	N136"
Dist. from "FROM" Node / Offset dist	2250 mm
Nozzle Diameter	36 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	50
Detail Type	Nozzle
Detail ID	N54"
Dist. from "FROM" Node / Offset dist	1300 mm
Nozzle Diameter	4 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	50
Detail Type	Nozzle
Detail ID	K32"
Dist. from "FROM" Node / Offset dist	1150 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L
Element From Node	50
Detail Type	Nozzle
Detail ID	K42"
Dist. from "FROM" Node / Offset dist	450 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N



Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K52"	
Dist. from "FROM" Node / Offset dist	950	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	200.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K62"	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	200.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K72"	
Dist. from "FROM" Node / Offset dist	950	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	220.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K82"	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	220.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K92"	
Dist. from "FROM" Node / Offset dist	950	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	240.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	
Element From Node	50	



Detail Type	Nozzle
Detail ID	K102"
Dist. from "FROM" Node / Offset dist	250 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	240.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L

Element From Node	50
Detail Type	Nozzle
Detail ID	M124"
Dist. from "FROM" Node / Offset dist	1150 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	50
Detail Type	Weight
Detail ID	Manway Davit
Dist. from "FROM" Node / Offset dist	1150 mm
Miscellaneous Weight	980.6 N
Offset from Element Centerline	2200 mm

Element From Node	50
Detail Type	Weight
Detail ID	Pipe Support
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	0 mm

Element From Node	50
Detail Type	Weight
Detail ID	Nozzle cladding
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	686.42 N
Offset from Element Centerline	0 mm

Element From Node	50
Detail Type	Weight
Detail ID	Cleat weight-1
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	9806 N
Offset from Element Centerline	0 mm

Element From Node	50
Detail Type	Weight
Detail ID	N5 PIPE WEIGHT
Dist. from "FROM" Node / Offset dist	2250 mm
Miscellaneous Weight	66455 N
Offset from Element Centerline	3555 mm

Element#	6/13
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-D1'
Distance "FROM" to "TO"	1000 mm
Inside Diameter	3650 mm
Element Thickness	69 mm



Internal Corrosion Allowance	1	mm
Nominal Thickness	69	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4.9507	MPa
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Platform	
Detail ID	PLT-2	
Dist. from "FROM" Node / Offset dist	950	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	4625600	mm ²
Platform Weight	23322	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	0	mm

Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	60	
Detail Type	Lining	
Detail ID	cladding	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K12"	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-182 F316L	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K22"	
Dist. from "FROM" Node / Offset dist	350	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	



Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-182 F316L

Element From Node 60
 Detail Type Weight
 Detail ID Demister-D1
 Dist. from "FROM" Node / Offset dist 800 mm
 Miscellaneous Weight 2451.5 N
 Offset from Element Centerline 0 mm

Element From Node 60
 Detail Type Weight
 Detail ID Bottom Spt Demi
 Dist. from "FROM" Node / Offset dist 600 mm
 Miscellaneous Weight 490.3 N
 Offset from Element Centerline 0 mm

Element# 7/13
 Element From Node 70
 Element To Node 80
 Element Type Elliptical
 Description Top Head D1
 Distance "FROM" to "TO" 50 mm
 Inside Diameter 3650 mm
 Element Thickness 68 mm
 Internal Corrosion Allowance 1 mm
 Nominal Thickness 78 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 4.9507 MPa
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 70
 Detail Type Insulation
 Detail ID Ins: 70
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 962.5 mm
 Thickness of Insulation 80 mm
 Density 0.000352 kg/cm³

Element From Node 70
 Detail Type Lining
 Detail ID cladding
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Lining 962.5 mm
 Thickness of Lining 3 mm
 Density 0.008027 kg/cm³

Element From Node 70
 Detail Type Nozzle
 Detail ID N230"
 Dist. from "FROM" Node / Offset dist 0 mm
 Nozzle Diameter 30 in.
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1



Nozzle Matl	SA-105
Element From Node	70
Detail Type	Weight
Detail ID	N2 Elbow
Dist. from "FROM" Node / Offset dist	964 mm
Miscellaneous Weight	9806 N
Offset from Element Centerline	0 mm
Element From Node	70
Detail Type	Weight
Detail ID	N2D1 PIPE WEIGH
Dist. from "FROM" Node / Offset dist	950 mm
Miscellaneous Weight	78830 N
Offset from Element Centerline	3480 mm
Element From Node	70
Detail Type	Weight
Detail ID	Top Spt Demiste
Dist. from "FROM" Node / Offset dist	0.0002896 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	0 mm
Element From Node	70
Detail Type	For./Mom.
Detail ID	F&M 24"
Dist. from "FROM" Node / Offset dist	800 mm
Force in X Direction	53604 N
Force in Y Direction	-74221 N
Force in Z Direction	53604 N
Moment about X Axis	123751896 N-mm
Moment about Y Axis	0 N-mm
Moment about Z Axis	123751896 N-mm
Force/Moment Combination Method	SRSS

Element#	8/13
Element From Node	80
Element To Node	90
Element Type	Skirt Support
Description	Int.Skirt-1
Distance "FROM" to "TO"	500 mm
Skirt Outside Diameter	3738 mm
Diameter of Skirt at Base	3738 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	100 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	9/13
Element From Node	90
Element To Node	100
Element Type	Skirt Support
Description	Int.Skirt-2
Distance "FROM" to "TO"	2800 mm
Skirt Outside Diameter	2967 mm
Diameter of Skirt at Base	3738 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	18 mm



External Corrosion Allowance 0 mm
Design Temperature Internal Pressure 100 °C
Design Temperature External Pressure 0 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70 +IT
Efficiency, Longitudinal Seam 0.7
Efficiency, Head-to-Skirt or Circ. Seam 0.7
Weld is pre-Heated No

Element From Node 90
Detail Type Weight
Detail ID Cleat weight-2
Dist. from "FROM" Node / Offset dist 1150 mm
Miscellaneous Weight 9806 N
Offset from Element Centerline 0 mm

Element# 10/13
Element From Node 100
Element To Node 110
Element Type Skirt Support
Description Int.Skirt-3
Distance "FROM" to "TO" 500 mm
Skirt Outside Diameter 2967 mm
Diameter of Skirt at Base 2967 mm
Skirt Thickness 18 mm
Internal Corrosion Allowance 0 mm
Nominal Thickness 18 mm
External Corrosion Allowance 0 mm
Design Temperature Internal Pressure 100 °C
Design Temperature External Pressure 0 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70 +IT
Efficiency, Longitudinal Seam 0.7
Efficiency, Head-to-Skirt or Circ. Seam 0.7
Weld is pre-Heated No

Element# 11/13
Element From Node 110
Element To Node 120
Element Type Elliptical
Description Bottom Head-D2
Distance "FROM" to "TO" 50 mm
Inside Diameter 2900 mm
Element Thickness 52 mm
Internal Corrosion Allowance 3 mm
Nominal Thickness 60 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 4.5507 MPa
Design Temperature Internal Pressure 110 °C
Design External Pressure 0 MPa
Design Temperature External Pressure 0 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70 +IT
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Elliptical Head Factor 2.0
Weld is pre-Heated No

Element From Node 110
Detail Type Liquid
Detail ID Liquid: 110
Dist. from "FROM" Node / Offset dist -725 mm
Height/Length of Liquid 775 mm
Liquid Density 0.001023 kg/cm³

Element From Node 110



Detail Type	Insulation
Detail ID	ins
Dist. from "FROM" Node / Offset dist	-725 mm
Height/Length of Insulation	775 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element From Node	110
Detail Type	Nozzle
Detail ID	D2N33"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	110
Detail Type	Weight
Detail ID	Baffle-D2
Dist. from "FROM" Node / Offset dist	-500 mm
Miscellaneous Weight	147.09 N
Offset from Element Centerline	0 mm

Element#	12/13
Element From Node	120
Element To Node	130
Element Type	Cylinder
Description	Shell-D2
Distance "FROM" to "TO"	3025 mm
Inside Diameter	2900 mm
Element Thickness	53 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	53 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.5507 MPa
Design Temperature Internal Pressure	110 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	120
Detail Type	Platform
Detail ID	PTL-3
Dist. from "FROM" Node / Offset dist	500 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	4422399 mm ²
Platform Weight	20064 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	135 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	3500 mm
Unit Weight of Ladder	0.37 N/mm
Platform Length (top head platform)	0 mm



Element From Node	120
Detail Type	Liquid
Detail ID	Liquid: 120
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	700 mm
Liquid Density	0.001023 kg/cm³

Element From Node	120
Detail Type	Insulation
Detail ID	ins
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3025 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³

Element From Node	120
Detail Type	Nozzle
Detail ID	D2N1 24"
Dist. from "FROM" Node / Offset dist	1575 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	120
Detail Type	Nozzle
Detail ID	D2K12"
Dist. from "FROM" Node / Offset dist	1550 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	250.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	120
Detail Type	Nozzle
Detail ID	D2K22"
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	250.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	120
Detail Type	Nozzle
Detail ID	D2K32"
Dist. from "FROM" Node / Offset dist	1550 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	260.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	120
Detail Type	Nozzle
Detail ID	D2K42"



Dist. from "FROM" Node / Offset dist	300	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	260.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	120	
Detail Type	Nozzle	
Detail ID	D2K52"	
Dist. from "FROM" Node / Offset dist	1550	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	120	
Detail Type	Nozzle	
Detail ID	D2K62"	
Dist. from "FROM" Node / Offset dist	300	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	120	
Detail Type	Nozzle	
Detail ID	D2K72"	
Dist. from "FROM" Node / Offset dist	1550	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	280.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	120	
Detail Type	Nozzle	
Detail ID	D2K82"	
Dist. from "FROM" Node / Offset dist	300	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	280.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	120	
Detail Type	Nozzle	
Detail ID	D2K92"	
Dist. from "FROM" Node / Offset dist	2550	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	



Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	120	
Detail Type	Nozzle	
Detail ID	D2K102"	
Dist. from "FROM" Node / Offset dist	2350	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	120	
Detail Type	Nozzle	
Detail ID	D2M124"	
Dist. from "FROM" Node / Offset dist	800	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	120	
Detail Type	Weight	
Detail ID	Manway Davit	
Dist. from "FROM" Node / Offset dist	800	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	1900	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Demister-D2	
Dist. from "FROM" Node / Offset dist	2825	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Demister-spt1	
Dist. from "FROM" Node / Offset dist	2800	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	demister-spt2	
Dist. from "FROM" Node / Offset dist	2850	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Pipe Support	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Cleat weight-3	
Dist. from "FROM" Node / Offset dist	1515	mm



Miscellaneous Weight	9806	N
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	LIFTING LUG-1	
Dist. from "FROM" Node / Offset dist	2070	mm
Miscellaneous Weight	12748	N
Offset from Element Centerline	1500	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	LIFTING LUG-2	
Dist. from "FROM" Node / Offset dist	2070	mm
Miscellaneous Weight	12748	N
Offset from Element Centerline	1500	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	N1D2 PIPE WEIGH	
Dist. from "FROM" Node / Offset dist	1575	mm
Miscellaneous Weight	65122	N
Offset from Element Centerline	3009	mm

Element#	13/13
Element From Node	130
Element To Node	140
Element Type	Elliptical
Description	Top Head-D2
Distance "FROM" to "TO"	50 mm
Inside Diameter	2900 mm
Element Thickness	52 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	60 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.5507 MPa
Design Temperature Internal Pressure	110 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	130
Detail Type	Platform
Detail ID	TOP PLATFORM
Dist. from "FROM" Node / Offset dist	900 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	5300480 mm ²
Platform Weight	27870 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	4000 mm
Platform Height	1000 mm
Platform Clearance or End Offset	2000 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	3200 mm
Unit Weight of Ladder	0.37 N/mm
Platform Length (top head platform)	4000 mm

Element From Node	130
Detail Type	Insulation



Detail ID	ins	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	775	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	130	
Detail Type	Nozzle	
Detail ID	N224"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	130	
Detail Type	Weight	
Detail ID	N2D2 PIPE WEIGH	
Dist. from "FROM" Node / Offset dist	775	mm
Miscellaneous Weight	50844	N
Offset from Element Centerline	3009	mm
Element From Node	130	
Detail Type	For./Mom.	
Detail ID	F&M24"	
Dist. from "FROM" Node / Offset dist	500	mm
Force in X Direction	32986	N
Force in Y Direction	-49479	N
Force in Z Direction	32986	N
Moment about X Axis	55440792	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	55440792	N-mm
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-3	...	1700	...	1700	...
Skirt-2	...	3700	...	2000	...
Skirt-1	...	4700	...	1000	...
Bottom Head D1	...	4750	...	50	...
Shell-1	...	7750	...	3000	...
Shell-D1'	...	8750	...	1000	...
Top Head D1	...	8800	...	50	...
Int.Skirt-1	...	9300	...	500	...
Int.Skirt-2	...	12100	...	2800	...
Int.Skirt-3	...	12600	...	500	...
Bottom Head-D2	...	12650	...	50	...
Shell-D2	...	15675	...	3025	...
Top Head-D2	...	15725	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-3	...		18	1.6	3744	...	1
Skirt-2	...		18	1.6	3744	...	2
Skirt-1	...		18	1.6	3744	...	3
Bottom Head D1	4.9722		78	1	3650	138	4
Shell-1	4.963		69	1	3650	138	5
Shell-D1'	4.9507		69	1	3650	138	6
Top Head D1	4.9507		78	1	3650	138	7
Int.Skirt-1	...		18	...	3738	...	8
Int.Skirt-2	...		18	...	2967	...	9
Int.Skirt-3	...		18	...	2967	...	10
Bottom Head-D2	4.5655		60	3	2900	138	11
Shell-D2	4.5577		53	3	2900	138	12
Top Head-D2	4.5507		60	3	2900	138	13

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-3	...		No Calc	No Calc	18	No Calc	1
Skirt-2	...		No Calc	No Calc	18	No Calc	2
Skirt-1	...		No Calc	No Calc	18	No Calc	3
Bottom Head D1	4.9507	5.02711	5.12274	68	66.9824		4
Shell-1	4.9507	5.01435	5.10171	69	68.12		5
Shell-D1'	4.9507	5.02669	5.10171	69	67.9494		6
Top Head D1	4.9507	5.0486	5.12274	68	66.6962		7
Int.Skirt-1	...	No Calc	No Calc	18	No Calc		8
Int.Skirt-2	...	No Calc	No Calc	18	No Calc		9
Int.Skirt-3	...	No Calc	No Calc	18	No Calc		10
Bottom Head-D2	4.5507	4.63608	4.93128	52	51.0974		11
Shell-D2	4.5507	4.6457	4.93589	53	51.9585		12
Top Head-D2	4.5507	4.6509	4.93128	52	50.9409		13

Minimum

5

5

Warning: The flange MAWP was less than the design pressure!

MAWP: 5 MPa, limited by: Bottom Head-D2.

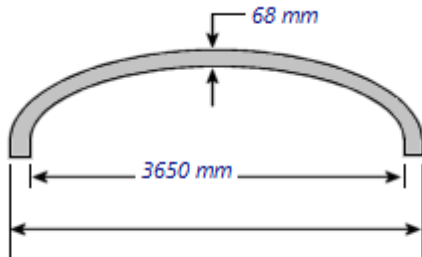
Warning: The computed MAWP is less than the design pressure.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom Head D1

Elliptical Head Design Information:					
Design Pressure	P	4.972 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3650.000 mm	Actual Stress		135.9 MPa
Spec. Min. Thk t,e, or sa		68.0000 mm	Spec. Nominal Thk.		78.000 mm
Surface Area		1630064.0 mm²	SG of Contents		0.969
Element Volume		6889.7 ltr	Weight of Contents		65112.0 N
Empty Weight		102078.7 N	Operating Weight		167190.7 N
			Comp. All. Ope. Stress		141.5 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 45.625 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 22.812 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4.972 \cdot 3652.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.972)$$

$$= 65.9824 + 1.0000 = 66.9824 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.021 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 67.0) / (0.999 \cdot 3652.0 + 0.2 \cdot 67.0)$$

$$= 5 - 0 = 5 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 68.0) / (1.0 \cdot 3650.0 + 0.2 \cdot 68.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4.972 \cdot (0.999 \cdot 3652.0 + 0.2 \cdot 67.0)) / (2 \cdot 1.0 \cdot 67.0)$$

$$= 135.909 \text{ MPa}$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (4.972 \cdot 1826.0) / (138.0 \cdot 1.0 - 0.6 \cdot 4.972) + 1.0$$

$$= 68.246 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.013 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 77.0) / (1826.0 + 0.6 \cdot 77.0)$$

$$= 6 - 0 = 6 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (3652.0 / (2 \cdot 913.5))^2) / 6$$

$$= 0.999270$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 8.870 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 68.0, tr = 65.982, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.985$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	23 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 78.0, tr = 67.246, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.873$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	19 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.972 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	78.000 mm	Basic MDMT	26 °C
Stress/Thk. Ratio	0.873	Temperature Difference	7 °C
UCS-66(c) red.	No	Computed Min. Temp.	19 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

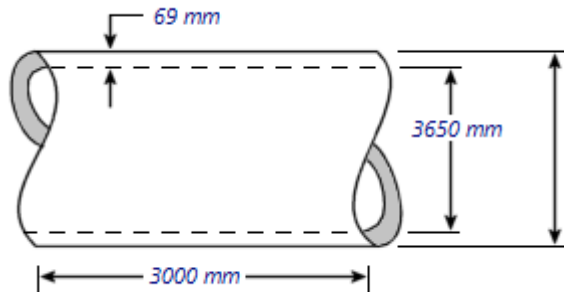
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 45.625 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 22.812 mm

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	4.963 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3650.000 mm	Actual Stress		136.3 MPa
Spec. Min. Thk t,e, or sa		69.0000 mm	Spec. Nominal Thk.		69.000 mm
Surface Area		35701060.0 mm²	SG of Contents		0.969
Element Volume		31396.0 ltr	Weight of Contents		128764.4 N
Empty Weight		183809.7 N	Operating Weight		312574.1 N
			Comp. All. Ope. Stress		141.7 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4.963 \cdot 1826.0) / (138.0 \cdot 1.0 - 0.6 \cdot 4.963)$$

$$= 67.1200 + 1.0000 = 68.1200 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.012 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 68.0) / (1826.0 + 0.6 \cdot 68.0)$$

$$= 5 - 0 = 5 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 69.0) / (1825.0 + 0.6 \cdot 69.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (4.963 \cdot (1826.0 + 0.6 \cdot 68.0)) / (1.0 \cdot 68.0)$$

$$= 136.250 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 1.855 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 69.0, tr = 67.12, c = 1.0 mm, E* = 1.0



Thickness Ratio = $tr * E^* / (tg - c) = 0.987$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 23 °C

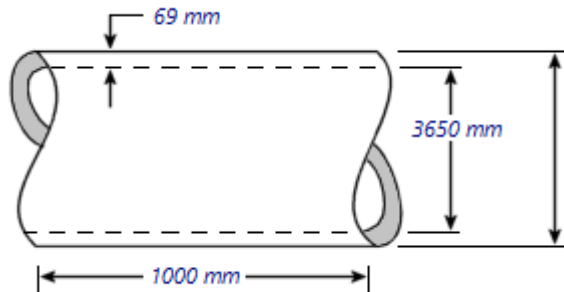
Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.963 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	69.000 mm	Basic MDMT	24 °C
Stress/Thk. Ratio	0.987	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	23 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-D1'

Cylindrical Shell Design Information:					
Design Pressure	P	4.951 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3650.000 mm	Actual Stress		135.9 MPa
Spec. Min. Thk t _e , or s _a		69.0000 mm	Spec. Nominal Thk.		69.000 mm
Surface Area		11900354.0 mm ²	SG of Contents		0.000
Element Volume		10465.3 ltr	Weight of Contents		0.0 N
Empty Weight		61269.9 N	Operating Weight		61269.9 N
			Comp. All. Ope. Stress		141.7 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4.951 \cdot 1826.0) / (138.0 \cdot 1.0 - 0.6 \cdot 4.951)$$

$$= 66.9494 + 1.0000 = 67.9494 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 68.0) / (1826.0 + 0.6 \cdot 68.0)$$

$$= 5 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 69.0) / (1825.0 + 0.6 \cdot 69.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (4.951 \cdot (1826.0 + 0.6 \cdot 68.0)) / (1.0 \cdot 68.0)$$

$$= 135.911 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.855 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 69.0, t_r = 66.949, c = 1.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.985, Temp. Reduction = 1 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 24 °C
Min Metal Temp. at Required thickness (UCS 66.1) 23 °C

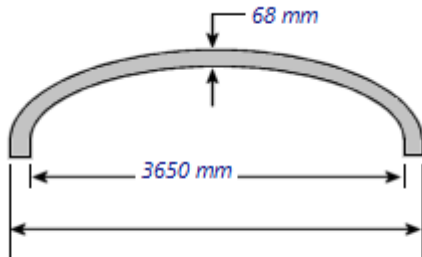
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.951 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	69.000 mm	Basic MDMT	24 °C
Stress/Thk. Ratio	0.985	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	23 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Top Head D1

Elliptical Head Design Information:					
Design Pressure	P	4.951 MPa	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3650.000 mm	Actual Stress		135.3 MPa
Spec. Min. Thk	t,e, or sa	68.0000 mm	Spec. Nominal Thk.		78.000 mm
Surface Area		1630064.0 mm²	SG of Contents		0.000
Element Volume		6889.7 ltr	Weight of Contents		0.0 N
Empty Weight		102078.7 N	Operating Weight		102078.7 N
			Comp. All. Ope. Stress		141.5 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 45.625 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 22.812 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4.951 \cdot 3652.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.951)$$

$$= 65.6962 + 1.0000 = 66.6962 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 67.0) / (0.999 \cdot 3652.0 + 0.2 \cdot 67.0)$$

$$= 5 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 68.0) / (1.0 \cdot 3650.0 + 0.2 \cdot 68.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4.951 \cdot (0.999 \cdot 3652.0 + 0.2 \cdot 67.0)) / (2 \cdot 1.0 \cdot 67.0)$$

$$= 135.322 \text{ MPa}$$

Straight Flange Required Thickness:



$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (4.951 \cdot 1826.0) / (138.0 \cdot 1.0 - 0.6 \cdot 4.951) + 1.0$$

$$= 67.949 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 77.0) / (1826.0 + 0.6 \cdot 77.0)$$

$$= 6 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (3652.0 / (2 \cdot 913.5)))^2 / 6$$

$$= 0.999270$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 8.870 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 68.0, tr = 65.696, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.981$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 22 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 78.0, tr = 66.949, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.869$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 19 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.951 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	78.000 mm	Basic MDMT	26 °C
Stress/Thk. Ratio	0.869	Temperature Difference	7 °C
UCS-66(c) red.	No	Computed Min. Temp.	19 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

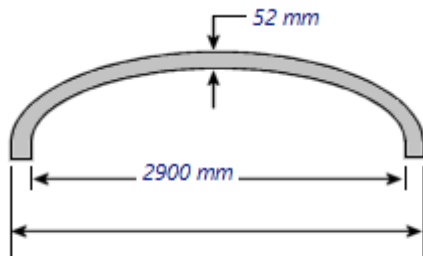
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 45.625 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 22.812 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom Head-D2

Elliptical Head Design Information:							
Design Pressure		P	4.566 MPa	Int. Design Temperature		110 °C	
Material			SA-516 70	Ext. Design Temperature		0 °C	
Length			50.000 mm	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.		c	3.0000 mm	Allowable Stress (ope)		S	138.0 MPa
Ext. Corr. All.		ce	0.0000 mm	Allowable Stress (amb)		Sa	138.0 MPa
Inside Diameter		Di	2900.000 mm	Actual Stress		135.5 MPa	
Spec. Min. Thk t,e, or sa			52.0000 mm	Spec. Nominal Thk.		60.000 mm	
Surface Area			10360760.0 mm²	SG of Contents		1.025	
Element Volume			3523.4 ltr	Weight of Contents		35380.2 N	
Empty Weight			49936.3 N	Operating Weight		85316.5 N	
				Comp. All. Ope. Stress		140.1 MPa	
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.125 mm

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4.566 \cdot 2906.0 \cdot 0.997) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.566)$$

$$= 48.0974 + 3.0000 = 51.0974 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.015 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 49.0) / (0.997 \cdot 2906.0 + 0.2 \cdot 49.0)$$

$$= 5 - 0 = 5 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 52.0) / (1.0 \cdot 2900.0 + 0.2 \cdot 52.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4.566 \cdot (0.997 \cdot 2906.0 + 0.2 \cdot 49.0)) / (2 \cdot 1.0 \cdot 49.0)$$



= 135.466 MPa

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (4.566 \cdot 1453.0) / (138.0 \cdot 1.0 - 0.6 \cdot 4.566) + 3.0 \\ &= 52.044 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 57.0) / (1453.0 + 0.6 \cdot 57.0) \\ &= 5 - 0.01 = 5 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (2906.0 / (2 \cdot 728.0))^2) / 6 \\ &= 0.997256 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 8.604 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

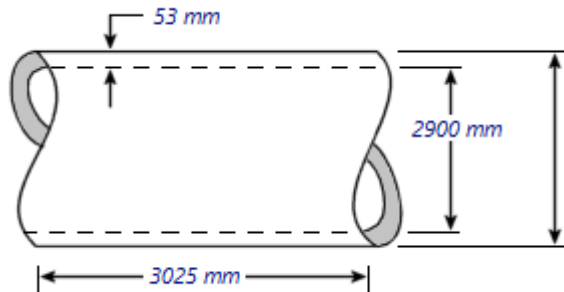
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.250 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.125 mm

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-D2

Cylindrical Shell Design Information:					
Design Pressure	P	4.558 MPa	Int. Design Temperature		110 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3025.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2900.000 mm	Actual Stress		135.2 MPa
Spec. Min. Thk t _e , or s _a		53.0000 mm	Spec. Nominal Thk.		53.000 mm
Surface Area		28566976.0 mm ²	SG of Contents		1.025
Element Volume		19984.3 ltr	Weight of Contents		46434.7 N
Empty Weight		113035.1 N	Operating Weight		159469.8 N
			Comp. All. Ope. Stress		140.5 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4.558 \cdot 1453.0) / (138.0 \cdot 1.0 - 0.6 \cdot 4.558)$$

$$= 48.9585 + 3.0000 = 51.9585 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 50.0) / (1453.0 + 0.6 \cdot 50.0)$$

$$= 5 - 0.01 = 5 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 53.0) / (1450.0 + 0.6 \cdot 53.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (4.558 \cdot (1453.0 + 0.6 \cdot 50.0)) / (1.0 \cdot 50.0)$$

$$= 135.182 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.795 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Note:

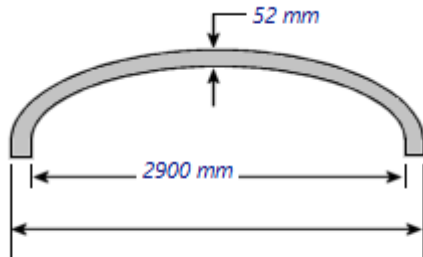
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top Head-D2

Elliptical Head Design Information:					
Design Pressure	P	4.551 MPa	Int. Design Temperature		110 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2900.000 mm	Actual Stress		135.0 MPa
Spec. Min. Thk t _e or s _a		52.0000 mm	Spec. Nominal Thk.		60.000 mm
Surface Area		10360760.0 mm ²	SG of Contents		0.000
Element Volume		3523.4 ltr	Weight of Contents		0.0 N
Empty Weight		49936.3 N	Operating Weight		49936.3 N
			Comp. All. Ope. Stress		140.1 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.125 mm

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4.551 \cdot 2906.0 \cdot 0.997) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.551)$$

$$= 47.9409 + 3.0000 = 50.9409 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 49.0) / (0.997 \cdot 2906.0 + 0.2 \cdot 49.0)$$

$$= 5 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 52.0) / (1.0 \cdot 2900.0 + 0.2 \cdot 52.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4.551 \cdot (0.997 \cdot 2906.0 + 0.2 \cdot 49.0)) / (2 \cdot 1.0 \cdot 49.0)$$

$$= 135.027 \text{ MPa}$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (4.551 \cdot 1453.0) / (138.0 \cdot 1.0 - 0.6 \cdot 4.551) + 3.0 \\ &= 51.881 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 57.0) / (1453.0 + 0.6 \cdot 57.0) \\ &= 5 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (2906.0 / (2 \cdot 728.0))^2) / 6 \\ &= 0.997256 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 8.604 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.250 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.125 mm

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6.027 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.631 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.375 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	5.100 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.694 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.694 MPa
App 27-4	= M.A.W.P.	4.636 MPa

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 0.485 • 1.0
 = 0.631 MPa

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom Head D1	18.0	179.4	0.100	0.67
Shell-1	18.0	179.4	0.101	0.67
Shell-D1'	18.0	179.4	0.101	0.67
Top Head D1	18.0	179.4	0.100	0.67
Bottom Head-D2	18.5	179.4	0.103	0.66
Shell-D2	18.4	179.4	0.103	0.66
Top Head-D2	18.5	179.4	0.103	0.66

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_10"	Nozzle	138.00	138.00	1.000
N1_36"	Nozzle	138.00	138.00	1.000
N5_4"	Nozzle	138.00	138.00	1.000
K3_2"	Nozzle	115.00	115.00	1.000
K4_2"	Nozzle	115.00	115.00	1.000
K5_2"	Nozzle	115.00	115.00	1.000
K6_2"	Nozzle	115.00	115.00	1.000
K7_2"	Nozzle	115.00	115.00	1.000
K8_2"	Nozzle	115.00	115.00	1.000
K9_2"	Nozzle	115.00	115.00	1.000
K10_2"	Nozzle	115.00	115.00	1.000
M1_24"	Nozzle	138.00	138.00	1.000
K1_2"	Nozzle	115.00	115.00	1.000
K2_2"	Nozzle	115.00	115.00	1.000
N2_30"	Nozzle	138.00	138.00	1.000
D2N3_3"	Nozzle	138.00	138.00	1.000
D2N1_24"	Nozzle	138.00	138.00	1.000
D2K1_2"	Nozzle	138.00	138.00	1.000
D2K2_2"	Nozzle	138.00	138.00	1.000
D2K3_2"	Nozzle	138.00	138.00	1.000
D2K4_2"	Nozzle	138.00	138.00	1.000
D2K5_2"	Nozzle	138.00	138.00	1.000
D2K6_2"	Nozzle	138.00	138.00	1.000
D2K7_2"	Nozzle	138.00	138.00	1.000
D2K8_2"	Nozzle	138.00	138.00	1.000
D2K9_2"	Nozzle	138.00	138.00	1.000
D2K10_2"	Nozzle	138.00	138.00	1.000
D2M1_24"	Nozzle	138.00	138.00	1.000
N2_24"	Nozzle	138.00	138.00	1.000

Minimum

1.000



Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom Head D1	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-D1'	138.00	138.00	1.000
Top Head D1	138.00	138.00	1.000
Bottom Head-D2	138.00	138.00	1.000
Shell-D2	138.00	138.00	1.000
Top Head-D2	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_10"	8.20	179.40	0.046
N1_36"	16.67	179.40	0.093
N5_4"	2.56	179.40	0.014
K3_2"	1.41	149.50	0.009
K4_2"	1.41	149.50	0.009
K5_2"	1.41	149.50	0.009
K6_2"	1.41	149.50	0.009
K7_2"	1.41	149.50	0.009
K8_2"	1.41	149.50	0.009
K9_2"	1.41	149.50	0.009
K10_2"	1.41	149.50	0.009
M1_24"	16.67	179.40	0.093
K1_2"	1.41	149.50	0.009
K2_2"	1.41	149.50	0.009
N2_30"	17.88	179.40	0.100
D2N3_3"	2.25	179.40	0.013
D2N1_24"	14.09	179.40	0.079
D2K1_2"	1.39	225.00	0.006
D2K2_2"	1.39	225.00	0.006
D2K3_2"	1.39	225.00	0.006
D2K4_2"	1.39	225.00	0.006
D2K5_2"	1.39	225.00	0.006
D2K6_2"	1.39	225.00	0.006
D2K7_2"	1.39	225.00	0.006
D2K8_2"	1.39	225.00	0.006
D2K9_2"	1.39	225.00	0.006
D2K10_2"	1.39	225.00	0.006
D2M1_24"	14.09	179.40	0.079
N2_24"	14.09	179.40	0.079

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6.027 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.631 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.403 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	5.100 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.694 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.694 MPa
App 27-4	= M.A.W.P.	4.636 MPa

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 4.636 • 1.0

= 6.027 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom Head D1	166.3	179.4	0.927	6.08
Shell-1	166.8	179.4	0.930	6.08
Shell-D1'	166.0	179.4	0.925	6.05
Top Head D1	165.0	179.4	0.920	6.04
Bottom Head-D2	180.2	179.4	1.004	6.07
Shell-D2	179.9	179.4	1.003	6.06
Top Head-D2	179.1	179.4	0.998	6.03

Theoretical Maximum Allowable test pressure is : 6.001 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_10"	Nozzle	138.00	138.00	1.000
N1_36"	Nozzle	138.00	138.00	1.000
N5_4"	Nozzle	138.00	138.00	1.000
K3_2"	Nozzle	115.00	115.00	1.000
K4_2"	Nozzle	115.00	115.00	1.000
K5_2"	Nozzle	115.00	115.00	1.000
K6_2"	Nozzle	115.00	115.00	1.000
K7_2"	Nozzle	115.00	115.00	1.000
K8_2"	Nozzle	115.00	115.00	1.000
K9_2"	Nozzle	115.00	115.00	1.000
K10_2"	Nozzle	115.00	115.00	1.000
M1_24"	Nozzle	138.00	138.00	1.000
K1_2"	Nozzle	115.00	115.00	1.000
K2_2"	Nozzle	115.00	115.00	1.000
N2_30"	Nozzle	138.00	138.00	1.000
D2N3_3"	Nozzle	138.00	138.00	1.000
D2N1_24"	Nozzle	138.00	138.00	1.000
D2K1_2"	Nozzle	138.00	138.00	1.000
D2K2_2"	Nozzle	138.00	138.00	1.000
D2K3_2"	Nozzle	138.00	138.00	1.000
D2K4_2"	Nozzle	138.00	138.00	1.000
D2K5_2"	Nozzle	138.00	138.00	1.000
D2K6_2"	Nozzle	138.00	138.00	1.000
D2K7_2"	Nozzle	138.00	138.00	1.000
D2K8_2"	Nozzle	138.00	138.00	1.000
D2K9_2"	Nozzle	138.00	138.00	1.000
D2K10_2"	Nozzle	138.00	138.00	1.000
D2M1_24"	Nozzle	138.00	138.00	1.000
N2_24"	Nozzle	138.00	138.00	1.000



Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom Head D1	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-D1'	138.00	138.00	1.000
Top Head D1	138.00	138.00	1.000
Bottom Head-D2	138.00	138.00	1.000
Shell-D2	138.00	138.00	1.000
Top Head-D2	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_10"	74.85	179.40	0.417
N1_36"	151.89	179.40	0.847
N5_4"	23.29	179.40	0.130
K3_2"	12.83	149.50	0.086
K4_2"	12.83	149.50	0.086
K5_2"	12.83	149.50	0.086
K6_2"	12.83	149.50	0.086
K7_2"	12.83	149.50	0.086
K8_2"	12.83	149.50	0.086
K9_2"	12.83	149.50	0.086
K10_2"	12.83	149.50	0.086
M1_24"	151.89	179.40	0.847
K1_2"	12.77	149.50	0.085
K2_2"	12.77	149.50	0.085
N2_30"	161.86	179.40	0.902
D2N3_3"	20.70	179.40	0.115
D2N1_24"	129.60	179.40	0.722
D2K1_2"	12.83	225.00	0.057
D2K2_2"	12.83	225.00	0.057
D2K3_2"	12.83	225.00	0.057
D2K4_2"	12.83	225.00	0.057
D2K5_2"	12.83	225.00	0.057
D2K6_2"	12.83	225.00	0.057
D2K7_2"	12.83	225.00	0.057
D2K8_2"	12.83	225.00	0.057
D2K9_2"	12.83	225.00	0.057
D2K10_2"	12.83	225.00	0.057
D2M1_24"	129.60	179.40	0.722
N2_24"	128.97	179.40	0.719

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	8500	...	...	No Calc	No Calc
20	30	8500	...	...	No Calc	No Calc
30	40	8500	...	...	No Calc	No Calc
40	50	No Calc	3786	67	0.0024579	108.144
50	60	4708.33	3788	68	0.0025961	109.223
60	70	4708.33	3788	68	0.0025961	109.223
70	80	No Calc	3786	67	0.0024579	108.144
80	90	8500	...	...	No Calc	No Calc
90	100	8500	...	...	No Calc	No Calc
100	110	8500	...	...	No Calc	No Calc
110	120	No Calc	3004	49	0.0022655	106.47
120	130	3608.33	3006	50	0.0023845	107.531
130	140	No Calc	3004	49	0.0022655	106.47

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	68	No Calc	...	2.12644
50	60	69	No Calc	...	2.61428
60	70	69	No Calc	...	2.61428
70	80	68	No Calc	...	2.12644
80	90	...	No Calc	...	No Calc
90	100	...	No Calc	...	No Calc
100	110	...	No Calc	...	No Calc
110	120	52	No Calc	...	1.92965
120	130	53	No Calc	...	2.3848
130	140	52	No Calc	...	1.92965

Minimum

2

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	8500	No Calc	No Calc	No Calc
20	30	8500	No Calc	No Calc	No Calc
30	40	8500	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	4708.33	No Calc	No Calc	No Calc
60	70	4708.33	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	8500	No Calc	No Calc	No Calc
90	100	8500	No Calc	No Calc	No Calc
100	110	8500	No Calc	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc
120	130	3608.33	No Calc	No Calc	No Calc
130	140	No Calc	No Calc	No Calc	No Calc





Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	4192.28	...	3945.55	...	293.46
20	30	3265.89	...	2975.63	...	228.612
30	40	1632.94	...	1487.81	...	114.306
40	50	10409.8	6889.68	10276.4	6900.73	728.688
50	60	18744.6	31396	18477.9	31430.4	1312.12
60	70	6248.21	10465.3	6159.31	10476.8	437.374
70	80	10409.8	6889.68	10276.4	6900.73	728.688
80	90	815.163	...	815.163	...	57.0614
90	100	4130.25	...	4130.25	...	289.117
100	110	646.211	...	646.211	...	45.2348
110	120	5092.42	3523.4	4837.8	3544.63	356.47
120	130	11527.1	19984.3	10885.7	20067.1	806.9
130	140	5092.42	3523.4	4837.8	3544.63	356.47
Total		82207	82671.85	79751	82865.07	5754

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	1300	-1900	300	...	TAILING+BRACING
20	Plat	1290.89	1787.28	500	-1787.28	PLT-1
40	Plat	4865.89	0.24609E-04	...	...	BTL PLTF
40	Liqd	6640.01	...	-456.25	...	Liquid: 40
40	Insl	579.026	...	-431.25	...	Ins: 40
40	Lini	397.603	...	-431.25	...	cladding
40	Nozl	209.561	...	-1003.51	...	N3_10"
40	Wght	15	...	-500	...	Anti-Swirl Baff
50	Liqd	13131.2	...	650	...	Liquid: 50
50	Insl	1027.78	...	1500	...	Ins: 50
50	Lini	828.681	...	1500	...	cladding
50	Nozl	18543.7	1975	2250	...	N1_36"
50	Nozl	47.6017	-1975	1300	...	N5_4"
50	Nozl	16.8528	1975	1150	...	K3_2"
50	Nozl	16.8528	1975	450	...	K4_2"
50	Nozl	16.8528	-1855.89	950	675.49	K5_2"
50	Nozl	16.8528	-1855.89	250	675.49	K6_2"
50	Nozl	16.8528	-1512.94	950	1269.51	K7_2"
50	Nozl	16.8528	-1512.94	250	1269.51	K8_2"
50	Nozl	16.8528	-987.5	950	1710.4	K9_2"
50	Nozl	16.8528	-987.5	250	1710.4	K10_2"
50	Nozl	1656.56	...	1150	1975	M1_24"
50	Wght	100	...	1150	2200	Manway Davit
50	Wght	50	...	1500	...	Pipe Support
50	Wght	70	...	1500	...	Nozzle cladding
50	Wght	1000	...	1500	...	Cleat weight-1
50	Wght	6777	3555	2250	...	N5 PIPE WEIGHT
60	Plat	2544.83	...	950	-1769.29	PLT-2
60	Insl	342.592	...	500	...	Ins: 60
60	Lini	276.227	...	500	...	cladding
60	Nozl	16.8528	1975	750.001	...	K1_2"
60	Nozl	16.8528	1975	350.001	...	K2_2"
60	Wght	250	...	800.001	...	Demister-D1
60	Wght	50	...	600.001	...	Bottom Spt Demi
70	Insl	579.026	...	481.25	...	Ins: 70
70	Lini	397.603	...	481.25	...	cladding
70	Nozl	1471.67	...	1078.5	...	N2_30"
70	Wght	1000	...	964	...	N2 Elbow
70	Wght	8039	3480	950	...	N2D1 PIPE WEIGH
70	Wght	50	...	0.00028956	...	Top Spt Demiste
70	Forc	...	...	800	...	F&M 24"



90	Wght	1000	...	1150	...
110	Liqd	3608.02	...	-362.5	...
110	Insl	372.18	...	-337.5	...
110	Nozl	19.0593	...	-824.006	...
110	Wght	15	...	-500	...
120	Plat	2189.31	...	500	-1537.63
120	Liqd	4735.34	...	350	...
120	Insl	826.823	...	1512.5	...
120	Nozl	8736.15	1600	1575	...
120	Nozl	14.9943	-547.232	1550	1503.51
120	Nozl	14.9943	-547.232	300.002	1503.51
120	Nozl	14.9943	-277.837	1550	1575.69
120	Nozl	14.9943	-277.837	300.002	1575.69
120	Nozl	14.9943	...	1550	1600
120	Nozl	14.9943	...	300.002	1600
120	Nozl	14.9943	277.837	1550	1575.69
120	Nozl	14.9943	277.837	300.002	1575.69
120	Nozl	14.9943	1600	2550	...
120	Nozl	14.9943	1600	2350	...
120	Nozl	1273.94	-1600	800.002	...
120	Wght	50	-1900	800.002	...
120	Wght	200	...	2825	...
120	Wght	50	...	2800	...
120	Wght	50	...	2850	...
120	Wght	50	...	1500	...
120	Wght	1000	...	1515	...
120	Wght	1300	...	2070	1500
120	Wght	1300	...	2070	-1500
120	Wght	6641	3009	1575	...
130	Plat	3041.09	...	900	...
130	Insl	372.18	...	387.5	...
130	Nozl	694.408	...	899	...
130	Wght	5185	3009	775	...
130	Forc	...	...	500	...

Cleat weight-2
 Liquid: 110
 ins
 D2N3_3"
 Baffle-D2
 PTL-3
 Liquid: 120
 ins
 D2N1_24"
 D2K1_2"
 D2K2_2"
 D2K3_2"
 D2K4_2"
 D2K5_2"
 D2K6_2"
 D2K7_2"
 D2K8_2"
 D2K9_2"
 D2K10_2"
 D2M1_24"
 Manway Davit
 Demister-D2
 Demister-spt1
 demister-spt2
 Pipe Support
 Cleat weight-3
 LIFTING LUG-1
 LIFTING LUG-2
 N1D2 PIPE WEIGH
 TOP PLATFORM
 ins
 N2_24"
 N2D2 PIPE WEIGH
 F&M_24"

Total Weight of Each Detail Type:

Platforms	13932.0
Liquid	28114.6
Insulation	4099.6
Lining	1900.1
Nozzles	32971.1
Weights	35542.0

Sum of the Detail Weights 116559.4 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	87961.7	87961.7	87961.7	87961.7	87961.7	87961.7
Nozzles	32971.1	32971.1	32971.1	32971.1	32971.1	32971.1
Wld Weights	8450.0	8450.0	8450.0	8450.0	8450.0	8450.0
Platforms	...	...	13932.0	13932.0	13932.0	13932.0
Insulation	...	...	4099.6	4099.6	4099.6	4099.6
Lining	...	...	...	...	1900.1	1900.1
Empty Weights	...	...	...	450.0	450.0	...
Test Weights	...	450.0	...	...	...	...
Ope Weights	...	...	...	...	...	27092.0
Ope. Liquid	...	...	...	...	...	28114.6
Test Liquid	...	82621.4	...	...	...	...
Totals	129382.8	212454.2	147414.5	147864.5	149764.6	204521.1

Field Installation Options:

* Lining installed after lifting.

Miscellaneous Weight Percent: 7.0 %



Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	129382.8 kgm
Shop Test Wt.	- Fabricated weight + water (full)	212454.2 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	147414.5 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	147864.5 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	149764.6 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	204521.1 kgm
Field Test Wt.	- Empty weight + test medium (full)	232149.6 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	67103.1 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	72627.2 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	48165.0 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	19995610	19803346
20	30	23524246	23298052
30	40	11762123	11649026
40	50	16300064	15014736
50	60	35701060	34400440
60	70	11900354	11466814
70	80	16300064	15014736
80	90	5871637	5815087
90	100	26096108	25779436
100	110	4660553	4604004
110	120	10360760	9571848
120	130	28566976	27559624
130	140	10360760	9571848

Total	221400336.0	213548992.0 mm ²
Total	221.4	213.5 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	5785.74	5785.74	5785.74	24230632	5785.74
20	30	4785.39	4785.39	4785.39	22633358	4785.39
30	40	1747.25	1747.25	1747.25	...	1747.25
40	50	16808	23845.6	24091.1	1.1747	17205.6
50	60	42687.2	63424.1	74892.8	593913472	50292.9
60	70	9906.71	10182.9	20641.9	653036	10182.9
70	80	14239.2	22675.8	21522.3	274441056	22675.8
80	90	872.224	872.224	872.224	...	872.224
90	100	5419.36	5419.36	5419.36	...	5419.36
100	110	691.446	691.446	691.446	...	691.446
110	120	5855.13	9463.15	9376.39	...	5855.13
120	130	29510.2	40886.5	49482.3	312534784	36151.2
130	140	9556.57	14741.6	13077.8	153051904	14741.6

Empty Support Force + the Sum of the Y forces	1573658.38	N
Operating Support Force + the Sum of the Y forces	2129233.50	N
Hydro Support Force + the Sum of the Y forces	2402475.75	N

Cumulative Vessel Weight



From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	176407	204521	232386
20	30	170621	198735	226600
30	40	165835	193950	221815
40	50	164088	192203	220068
50	60	146883	168357	195976
60	70	96589.7	104933	121084
70	80	86406.8	94750.1	100442
80	90	63730.9	72074.3	78919.6
90	100	62858.7	71202.1	78047.3
100	110	57439.4	65782.7	72628
110	120	56747.9	65091.3	71936.5
120	130	50892.8	55628.1	62560.1
130	140	14741.6	14741.6	13077.8

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	1381458176	1381458176	1381458176
20	30	1357227520	1357227520	1357227520
30	40	1334594304	1334594304	1334594304
40	50	1334594304	1334594304	1334594304
50	60	1334594304	1334594304	1334594304
60	70	740680768	740680768	740680768
70	80	740027712	740027712	740027712
80	90	465586688	465586688	465586688
90	100	465586688	465586688	465586688
100	110	465586688	465586688	465586688
110	120	465586688	465586688	465586688
120	130	465586688	465586688	465586688
130	140	153051904	153051904	153051904

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N3_10"	9.14	10.20	130	600	GR 1.1	...	...	...	...
N1_36"	9.14	10.20	130	600	GR 1.1	...	...	...	...
N5_4"	9.14	10.20	130	600	GR 1.1	...	...	...	...
K3_2"	6.57	8.23	130	600	GR 2.3	...	...	...	...
K4_2"	6.57	8.23	130	600	GR 2.3	...	...	...	...
K5_2"	6.57	8.23	130	600	GR 2.3	...	...	...	...
K6_2"	6.57	8.23	130	600	GR 2.3	...	...	...	...
K7_2"	6.57	8.23	130	600	GR 2.3	...	...	...	...
K8_2"	6.57	8.23	130	600	GR 2.3	...	...	...	...
K9_2"	6.57	8.23	130	600	GR 2.3	...	...	...	...
K10_2"	6.57	8.23	130	600	GR 2.3	...	...	...	...
M1_24"	9.14	10.20	130	600	GR 1.1	...	...	...	...
K1_2"	9.14	10.20	130	600	GR 1.1	...	...	...	...
K2_2"	9.14	10.20	130	600	GR 1.1	...	...	...	...
N2_30"	9.14	10.20	130	600	GR 1.1	...	...	...	...
D2N3_3"	9.27	10.20	110	600	GR 1.1	5.320	8.592	9.275	8.592
D2N1_24"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2K1_2"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2K2_2"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2K3_2"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2K4_2"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2K5_2"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2K6_2"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2K7_2"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2K8_2"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2K9_2"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2K10_2"	4.64	5.10	110	300	GR 1.1	...	...	...	...
D2M1_24"	4.64	5.10	110	300	GR 1.1	...	...	...	...
N2_24"	4.64	5.10	110	300	GR 1.1	...	...	...	...
Min Rating	4.638	5.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
D2N3_3"	111.125	0.50	1267078	5998	4.636	8.592	Pass

Evaluating Flange on Nozzle [D2N33"]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR] \\
 &= 16(1267078) + 4 \cdot \text{abs}(5998)111.125 \leq \pi \cdot 111.125^3 \cdot [(9.275 - 4.636) + .500 \cdot 9.275] \\
 &= 0.22940E+08 \leq 0.40002E+08 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle D2N33" [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(9.275, 0.500 \cdot 9.275 + 9.275 - 22940234.0 / (\pi \cdot 111.125^3)) \\
 &= \min(9.275, 8.592) \\
 &= 8.592 \text{ MPa}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition



Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 4.93464 Hz.

The Natural Frequency for the Vessel (Ope...) is 4.66421 Hz.

The Natural Frequency for the Vessel (Filled) is 4.46111 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	75807.6	235682272
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	46649.2	101739192

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	75807.6	235682272
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	46649.2	101739192

User Input Forces and Moments:

From Node	Distance From	- - - Fx	Forces - - - Fy	- - - Fz	- - - Mx	Moments - - - My	- - - Mz
70	800.00	53604.	-74221.	53604.	0.124E+09	0.124E+09	
130	500.00	32986.	-49479.	32986.	0.554E+08	0.554E+08	



Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.202 \cdot 0.925) / (1 + 1.7 \cdot 3.4 \cdot 0.202)))$
 $= \min(0.85, 0.888)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	4.664 Hz
Natural Frequency of Vessel (Empty)	4.935 Hz
Natural Frequency of Vessel (Test)	4.461 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.562
Structure Height to Diameter ratio	4.741
Height to top of Structure	16582.000 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Z_g = 274320.0$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 850.0 + 300.0 = 1150.0$ mm
 $= 3.773$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (3.773 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$
 $= 2.01(15 / 274320.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [0.766] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.562 \cdot 82.212$$

$$= 628.8 \text{ lbs. } [2796.7] \text{ N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-3	1150.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-2	3000.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-1	4500.0	0.000	0.000	0.000	0.849	1.000	0.766
Bottom Head D1	5025.0	0.000	0.000	0.000	0.866	1.000	0.766
Shell-1	6550.0	0.000	0.000	0.000	0.916	1.000	0.766
Shell-D1'	8550.0	0.000	0.000	0.000	0.968	1.000	0.766
Top Head D1	9075.0	0.000	0.000	0.000	0.981	1.000	0.766
Int.Skirt-1	9350.0	0.000	0.000	0.000	0.987	1.000	0.766
Int.Skirt-2	10945.8	0.000	0.000	0.000	1.020	1.000	0.766
Int.Skirt-3	12650.0	0.000	0.000	0.000	1.052	1.000	0.766
Bottom Head-D2	12925.0	0.000	0.000	0.000	1.056	1.000	0.766
Shell-D2	14462.5	0.000	0.000	0.000	1.082	1.000	0.766
Top Head-D2	16361.8	0.000	0.000	0.000	1.110	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0.0/15725.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[11.77/51.59^2] \cdot 10^4 = 44.222$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.000004$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 1979771 / (15725.0 \cdot 3184.38^2)$$



= 0.12416E-04

Since Vp is > 0.000004 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLT-1	1499999.88	2500.00	0.77	1149.07	1.00
BTL PLTF	4015999.75	5000.00	0.77	3076.44	1.00
PLT-2	4625600.00	9000.00	0.77	3543.42	1.00
PTL-3	4422399.50	13450.00	0.77	3387.76	1.00
TOP PLATFORM	5300480.00	16875.00	0.77	4060.41	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
TAILING+BRACI	0.00	600.00	0.77	0.00
Anti-Swirl Ba	0.00	4500.00	0.77	0.00
Manway Davit	0.00	6200.00	0.77	0.00
Pipe Support	0.00	6550.00	0.77	0.00
Nozzle claddi	0.00	6550.00	0.77	0.00
Cleat weight-	0.00	6550.00	0.77	0.00
N5 PIPE WEIGH	0.00	7300.00	0.77	0.00
Demister-D1	0.00	8850.00	0.77	0.00
Bottom Spt De	0.00	8650.00	0.77	0.00
N2 Elbow	0.00	10014.00	0.77	0.00
N2D1 PIPE WEI	0.00	10000.00	0.77	0.00
Top Spt Demis	0.00	9050.00	0.77	0.00
Cleat weight-	0.00	10750.00	0.77	0.00
Baffle-D2	0.00	12400.00	0.77	0.00
Manway Davit	0.00	13750.00	0.77	0.00
Demister-D2	0.00	15775.00	0.77	0.00
Demister-spt1	0.00	15750.00	0.77	0.00
demister-spt2	0.00	15800.00	0.77	0.00
Pipe Support	0.00	14450.00	0.77	0.00
Cleat weight-	0.00	14465.00	0.77	0.00
LIFTING LUG-1	0.00	15020.00	0.77	0.00
LIFTING LUG-2	0.00	15020.00	0.77	0.00
N1D2 PIPE WEI	0.00	14525.00	0.77	0.00
N2D2 PIPE WEI	0.00	16750.00	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 4.66421 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1150	4492.8	7637759	0.76606	3076.41
20	30	3000	4492.8	8985600	0.76606	4672.62
30	40	4500	4492.8	4492800	0.76606	1809.65
40	50	5025	4735.2	236760	0.76606	3462.61
50	60	6550	4737.6	14212801	0.76606	5724.76
60	70	8550	4737.6	4737600	0.76606	6011.16
70	80	9075	4735.2	236760	0.76606	95.3644
80	90	9350	4485.6	2242800	0.76606	903.376
90	100	10945.8	4022.59	11263260	0.76606	4536.72
100	110	12650	3560.4	1780200	0.76606	717.045
110	120	12925	3796.8	189840	0.76606	76.4655
120	130	14462.5	3799.2	11492581	0.76606	8094.74
130	140	16361.8	3796.8	2745414	0.76606	5712.36



The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= 0.020(53.96980.75)
= 0.398 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/4.664)
= 0.214

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of (1.7 * 0.398, 0.214) per 12.8.2
= 0.214

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= 0.157/(2.0/2.0)
= 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= 0.104/(0.214(2.0/2.0))
= 0.485

Minimum value of C_s per equation 15.4-1 [Cs]:
= max((0.044 * SDS * I_e), 0.030)
= max((0.044 * 0.1568 * 2.0), 0.030)
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= 0.1568 * 2005534
= 314467.656 N

Vertical load per 12.4-4, [Ev]:
= 0.2 * Sds * W
= 0.2 * 0.157 * 2005534



= 62893.5 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 4.66421 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	850	56734.9	848.973	863.804	-60806.3
20	30	2700	46925.5	2230.47	2269.44	-62585.5
30	40	4200	17133.5	1266.84	1288.97	-64057.1
40	50	4725	233830	19450.3	13949.4	-64594.4
50	60	6250	621937	68430.6	46861.4	-71927.3
60	70	8250	99853.9	14502.5	14355.6	-91431.2
70	80	8775	222359	34349.9	21946.8	-94562.7
80	90	9050	8553.03	1362.68	1386.48	-27315
90	100	10700	53142.3	10010.3	10185.2	-27583.2
100	110	12350	6780.32	1474.15	1499.9	-29249.8
110	120	12625	92795.6	20624.5	12983.9	-29462.4
120	130	14162.5	400933	99962.4	73409.1	-32372.5
130	140	15700	144556	39953.9	26353.6	-44945.7



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	850	122457	122457	1732530048	1732530048
20	30	2700	122457	122457	1524269056	1524269056
30	40	4200	122457	122457	1279256064	1279256064
40	50	4725	122457	122457	1156749568	1156749568
50	60	6250	122457	122457	1150624128	1150624128
60	70	8250	122457	122457	783104832	783104832
70	80	8775	122457	122457	660598400	660598400
80	90	9050	46649.2	46649.2	422582752	422582752
90	100	10700	46649.2	46649.2	399248672	399248672
100	110	12350	46649.2	46649.2	268577856	268577856
110	120	12625	46649.2	46649.2	245243776	245243776
120	130	14162.5	46649.2	46649.2	242910368	242910368
130	140	15700	46649.2	46649.2	101739192	101739192

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	850	44893.3	314468	398494592	3.410E+09
20	30	2700	41816.9	313619	324761120	2.876E+09
30	40	4200	37144.3	311388	245767968	2.250E+09
40	50	4725	35334.6	310121	209513840	1939359872
50	60	6250	31872	290671	207832992	1924333952
60	70	8250	26147.2	222240	120768880	1154654720
70	80	8775	20136.1	207738	97617856	939578560
80	90	9050	20040.7	173388	96613024	930046656
90	100	10700	19137.3	172025	86814552	843658304
100	110	12350	14600.6	162015	39562304	375812384
110	120	12625	13883.6	160541	32438380	295140832
120	130	14162.5	13807.1	139916	31745824	287626272
130	140	15700	5712.36	39953.9	2210679	15462157

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	122457	0.0086976	0.034079	0.39256E-04
20	30	122457	0.083854	0.15351	0.79013E-04
30	40	122457	0.19524	0.2413	0.96285E-04
40	50	122457	0.24371	0.24612	0.96496E-04
50	60	122457	0.39528	0.55228	0.00010695
60	70	122457	0.60609	0.66056	0.00010955
70	80	122457	0.6633	0.66604	0.00010963
80	90	46649.2	0.69363	0.72156	0.00011242
90	100	46649.2	0.88595	1.06271	0.0001301
100	110	46649.2	1.09547	1.12866	0.00013362
110	120	46649.2	1.132	1.13534	0.00013374
120	130	46649.2	1.34007	1.54865	0.00013883
130	140	46649.2	1.55212	1.55559	0.00013883

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 78.625 Actual deflection : 1.556 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	44893.3	0.0020126	0.007792	0.88654E-05
20	30	41816.9	0.018903	0.034118	0.17080E-04
30	40	37144.3	0.04309	0.052884	0.20365E-04
40	50	35334.6	0.053393	0.053903	0.20404E-04
50	60	31872	0.08531	0.11807	0.22213E-04
60	70	26147.2	0.12923	0.14049	0.22616E-04
70	80	20136.1	0.14106	0.14162	0.22634E-04
80	90	20040.7	0.14732	0.1531	0.23270E-04
90	100	19137.3	0.18716	0.22354	0.26619E-04
100	110	14600.6	0.23023	0.23698	0.27120E-04
110	120	13883.6	0.23766	0.23834	0.27136E-04
120	130	13807.1	0.27967	0.32133	0.27587E-04
130	140	5712.36	0.32202	0.32271	0.27587E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	376.18	2351.13
20	30	376.18	2351.13
30	40	376.18	2351.13
40	50	396.476	2477.98
50	60	396.677	2479.23
60	70	396.677	2479.23
70	80	396.476	2477.98
80	90	375.577	2347.36
90	100	336.81	2105.06
100	110	298.111	1863.19
110	120	317.904	1986.9
120	130	318.105	1988.16



130	140	317.904	1986.9
-----	-----	---------	--------

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 78.625 Actual deflection : 0.323 mm

Total Deflection in the Operating Condition + Applied Forces:
Allowable deflection : 78.625 Actual deflection : 1.878 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	210701	191973	195328912	178042912
20	30	210701	191973	195328912	178042912
30	40	210701	191973	195328912	178042912
40	50	794269	782800	725255680	715160384
50	60	806168	794698	736134976	726044992
60	70	806168	794698	736134976	726044992
70	80	794269	782800	725255680	715160384
80	90	210362	210362	194698960	194698960
90	100	168326	168326	123335632	123335632
100	110	166762	166762	122203856	122203856
110	120	482247	454887	349846368	330657216
120	130	491687	464327	356703136	337526752
130	140	482247	454887	349846368	330657216



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	115.92	150.696	-105.668	-105.668
20	30	115.92	150.696	-105.668	-105.668
30	40	82.8	107.64	-105.668	-105.668
40	50	165.597	215.276	-141.458	-141.458
50	60	165.597	215.276	-141.68	-141.68
60	70	165.597	215.276	-141.68	-141.68
70	80	165.597	215.276	-141.458	-141.458
80	90	115.92	150.696	-108.959	-108.959
90	100	115.92	150.696	-116.494	-116.494
100	110	115.92	150.696	-116.49	-116.49
110	120	165.6	215.28	-140.132	-140.132
120	130	165.6	215.28	-140.458	-140.458
130	140	165.6	215.28	-140.132	-140.132

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	66.4724	...	80.9225
50	60	65.4803	...	79.7147
60	70	65.4803	...	79.7147
70	80	66.4724	...	80.9225
80	90	...	...	...
90	100	...	...	...
100	110	...	...	...
110	120	66.561	...	88.1528
120	130	65.2115	...	86.3656
130	140	66.561	...	88.1528

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-9.01163	-10.4479	-11.8713	7.75664	7.75664
20	30	-8.71607	-10.1523	-11.5758	7.62059	7.62059
30	40	-8.47161	-9.90783	-11.3313	7.49351	7.49351
40	50	-2.05568	-2.1602	-2.05568	1.86555	1.86555
50	60	-1.81258	-1.91554	-1.81258	1.83758	1.83758
60	70	-1.19195	-1.29491	-1.19195	1.01983	1.01983
70	80	-1.08249	-1.18702	-1.08249	1.03444	1.03444
80	90	-2.97107	-3.36002	-3.67914	2.39055	2.39055
90	100	-3.34647	-3.83257	-4.23138	3.77375	3.77375
100	110	-3.37786	-3.86851	-4.27107	3.8087	3.8087
110	120	-1.22342	-1.22342	-1.22342	1.40761	1.40761
120	130	-1.07488	-1.07488	-1.07488	1.37896	1.37896
130	140	-0.31781	-0.31781	-0.31781	0.46272	0.46272

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	7.75664	2.23748	19.1443	1.11874	...
20	30	7.62059	1.82348	16.1454	0.91174	...
30	40	7.49351	1.37994	12.6347	0.68997	...
40	50	1.86555	0.29287	2.71091	0.14643	...
50	60	1.83758	0.28616	2.64958	0.14308	...
60	70	1.01983	0.16628	1.58983	0.083142	...
70	80	1.03444	0.13645	1.31338	0.068227	...
80	90	2.39055	0.49606	4.77531	0.24803	...
90	100	3.77375	0.32067	3.04609	0.16033	...
100	110	3.8087	0.32364	3.07431	0.16182	...
110	120	1.40761	0.098071	0.8923	0.049036	...
120	130	1.37896	0.094024	0.85189	0.047012	...
130	140	0.46272	0.0066836	0.046747	0.0033418	...



Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	13.7563
20	30	...	...	...	11.5894
30	40	...	...	...	9.05773
40	50	...	...	...	1.94232
50	60	...	...	...	1.89833
60	70	...	...	...	1.13157
70	80	...	...	...	0.93212
80	90	...	...	...	3.38869
90	100	...	...	...	2.13263
100	110	...	...	...	2.15238
110	120	...	...	...	0.62384
120	130	...	...	...	0.5954
130	140	...	...	...	0.030834

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	-0.64442	-0.31677
20	30	-0.64442	-0.32604
30	40	-0.64442	-0.33371
40	50	-0.15804	-0.082524
50	60	-0.15567	-0.090517
60	70	-0.15567	-0.11506
70	80	-0.15804	-0.12081
80	90	-0.23523	-0.12986
90	100	-0.29397	-0.16388
100	110	-0.29673	-0.17541
110	120	-0.10878	-0.064774
120	130	-0.10657	-0.069725
130	140	-0.10878	-0.098815

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	9.72785	9.72785	8.86697	8.86697
20	30	8.5585	8.5585	7.8011	7.8011
30	40	7.1828	7.1828	6.54714	6.54714
40	50	1.61695	1.61695	1.59444	1.59444
50	60	1.58428	1.58428	1.56256	1.56256
60	70	1.07824	1.07824	1.06347	1.06347
70	80	0.92341	0.92341	0.91056	0.91056
80	90	2.16975	2.16975	2.16975	2.16975
90	100	2.17692	2.17692	2.17692	2.17692
100	110	2.19708	2.19708	2.19708	2.19708
110	120	0.74145	0.74145	0.70078	0.70078
120	130	0.71945	0.71945	0.68077	0.68077
130	140	0.30759	0.30759	0.29072	0.29072

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
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10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	10.07	115.92	-29.38	105.67	0.0868	0.2780
20	8.64	115.92	-27.36	105.67	0.0746	0.2590
30	6.94	82.80	-25.17	105.67	0.0838	0.2382
40	1.56	165.60	-5.99	141.46	0.0094	0.0423
50	1.74	165.60	-5.68	141.68	0.0105	0.0401
60	0.92	165.60	-3.61	141.68	0.0055	0.0255
70	0.85	165.60	-3.33	141.46	0.0052	0.0236
80	1.85	115.92	-8.26	108.96	0.0160	0.0758
90	2.63	115.92	-9.91	116.49	0.0227	0.0851
100	2.65	115.92	-10.00	116.49	0.0229	0.0859
110	0.91	165.60	-3.58	140.13	0.0055	0.0255
120	1.01	165.60	-3.37	140.46	0.0061	0.0240
130	0.35	165.60	-1.20	140.13	0.0021	0.0086

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	21.91	115.92	-40.57	105.67	0.1890	0.3839
20	18.73	115.92	-36.81	105.67	0.1615	0.3484
30	14.93	82.80	-32.54	105.67	0.1803	0.3079
40	3.29	165.60	-7.56	141.46	0.0198	0.0535
50	3.42	165.60	-7.22	141.68	0.0206	0.0510
60	1.92	165.60	-4.54	141.68	0.0116	0.0320



70	1.69	165.60	-4.09	141.46	0.0102	0.0289
80	4.85	115.92	-11.05	108.96	0.0418	0.1014
90	4.57	115.92	-11.59	116.49	0.0394	0.0995
100	4.60	115.92	-11.71	116.49	0.0397	0.1005
110	1.48	165.60	-4.06	140.13	0.0090	0.0290
120	1.55	165.60	-3.84	140.46	0.0094	0.0273
130	0.38	165.60	-1.22	140.13	0.0023	0.0087

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.63	115.92	-30.81	105.67	0.0744	0.2916
20	7.21	115.92	-28.80	105.67	0.0622	0.2725
30	5.50	82.80	-26.61	105.67	0.0665	0.2518
40	1.46	165.60	-6.09	141.46	0.0088	0.0431
50	1.64	165.60	-5.78	141.68	0.0099	0.0408
60	0.81	165.60	-3.71	141.68	0.0049	0.0262
70	0.75	165.60	-3.44	141.46	0.0045	0.0243
80	1.46	115.92	-8.65	108.96	0.0126	0.0794
90	2.14	115.92	-10.40	116.49	0.0185	0.0893
100	2.16	115.92	-10.49	116.49	0.0187	0.0901
110	0.91	165.60	-3.58	140.13	0.0055	0.0255
120	1.01	165.60	-3.37	140.46	0.0061	0.0240
130	0.35	165.60	-1.20	140.13	0.0021	0.0086

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	25.86	115.92	-47.39	105.67	0.2231	0.4485
20	21.85	115.92	-42.80	105.67	0.1885	0.4051
30	17.07	82.80	-37.55	105.67	0.2062	0.3554
40	3.95	165.60	-8.44	141.46	0.0239	0.0596
50	4.07	165.60	-8.08	141.68	0.0245	0.0570
60	2.28	165.60	-5.10	141.68	0.0138	0.0360
70	1.96	165.60	-4.58	141.46	0.0119	0.0324
80	5.85	115.92	-12.83	108.96	0.0504	0.1177
90	5.00	115.92	-12.99	116.49	0.0431	0.1115
100	5.04	115.92	-13.12	116.49	0.0434	0.1127
110	1.75	165.60	-4.33	140.13	0.0106	0.0309
120	1.81	165.60	-4.09	140.46	0.0109	0.0292
130	0.40	165.60	-1.23	140.13	0.0024	0.0088

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		150.70	-20.75	105.67		0.1963
20		150.70	-20.11	105.67		0.1903
30		107.64	-19.51	105.67		0.1847
40		215.28	-4.07	141.46		0.0288
50	0.17	215.28	-3.79	141.68	0.0008	0.0268
60		215.28	-2.29	141.68		0.0162
70	0.02	215.28	-2.19	141.46	0.0001	0.0154
80		150.70	-6.32	108.96		0.0580
90		150.70	-8.17	116.49		0.0701
100		150.70	-8.24	116.49		0.0707
110	0.23	215.28	-2.68	140.13	0.0011	0.0191
120	0.35	215.28	-2.50	140.46	0.0016	0.0178
130	0.15	215.28	-0.78	140.13	0.0007	0.0056

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		150.70	-19.63	105.67		0.1858
20		150.70	-19.20	105.67		0.1817
30		107.64	-18.82	105.67		0.1782
40		215.28	-3.92	141.46		0.0277



50	0.03	215.28	-3.65	141.68	0.0001	0.0258
60		215.28	-2.21	141.68		0.0156
70		215.28	-2.12	141.46		0.0150
80		150.70	-6.07	108.96		0.0557
90		150.70	-8.01	116.49		0.0687
100		150.70	-8.08	116.49		0.0694
110	0.18	215.28	-2.63	140.13	0.0009	0.0188
120	0.30	215.28	-2.45	140.46	0.0014	0.0175
130	0.14	215.28	-0.78	140.13	0.0007	0.0056

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.63	115.92	-30.81	105.67	0.0744	0.2916
20	7.21	115.92	-28.80	105.67	0.0622	0.2725
30	5.50	82.80	-26.61	105.67	0.0665	0.2518
40	67.93	165.60		141.46	0.4102	
50	67.12	165.60		141.68	0.4053	
60	66.29	165.60		141.68	0.4003	
70	67.22	165.60		141.46	0.4059	
80	1.46	115.92	-8.65	108.96	0.0126	0.0794
90	2.14	115.92	-10.40	116.49	0.0185	0.0893
100	2.16	115.92	-10.49	116.49	0.0187	0.0901
110	67.48	165.60		140.13	0.4075	
120	66.22	165.60		140.46	0.3999	
130	66.91	165.60		140.13	0.4041	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	25.86	115.92	-47.39	105.67	0.2231	0.4485
20	21.85	115.92	-42.80	105.67	0.1885	0.4051
30	17.07	82.80	-37.55	105.67	0.2062	0.3554
40	70.42	165.60		141.46	0.4253	
50	69.55	165.60		141.68	0.4200	
60	67.76	165.60		141.68	0.4092	
70	68.44	165.60		141.46	0.4133	
80	5.85	115.92	-12.83	108.96	0.0504	0.1177
90	5.00	115.92	-12.99	116.49	0.0431	0.1115
100	5.04	115.92	-13.12	116.49	0.0434	0.1127
110	68.31	165.60		140.13	0.4125	
120	67.02	165.60		140.46	0.4047	
130	66.96	165.60		140.13	0.4044	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.63	115.92	-30.81	105.67	0.0744	0.2916
20	7.21	115.92	-28.80	105.67	0.0622	0.2725
30	5.50	82.80	-26.61	105.67	0.0665	0.2518
40	1.46	165.60	-6.09	141.46	0.0088	0.0431
50	1.64	165.60	-5.78	141.68	0.0099	0.0408
60	0.81	165.60	-3.71	141.68	0.0049	0.0262
70	0.75	165.60	-3.44	141.46	0.0045	0.0243
80	1.46	115.92	-8.65	108.96	0.0126	0.0794
90	2.14	115.92	-10.40	116.49	0.0185	0.0893
100	2.16	115.92	-10.49	116.49	0.0187	0.0901
110	0.91	165.60	-3.58	140.13	0.0055	0.0255
120	1.01	165.60	-3.37	140.46	0.0061	0.0240
130	0.35	165.60	-1.20	140.13	0.0021	0.0086

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	25.86	115.92	-47.39	105.67	0.2231	0.4485
20	21.85	115.92	-42.80	105.67	0.1885	0.4051



30	17.07	82.80	-37.55	105.67	0.2062	0.3554
40	3.95	165.60	-8.44	141.46	0.0239	0.0596
50	4.07	165.60	-8.08	141.68	0.0245	0.0570
60	2.28	165.60	-5.10	141.68	0.0138	0.0360
70	1.96	165.60	-4.58	141.46	0.0119	0.0324
80	5.85	115.92	-12.83	108.96	0.0504	0.1177
90	5.00	115.92	-12.99	116.49	0.0431	0.1115
100	5.04	115.92	-13.12	116.49	0.0434	0.1127
110	1.75	165.60	-4.33	140.13	0.0106	0.0309
120	1.81	165.60	-4.09	140.46	0.0109	0.0292
130	0.40	165.60	-1.23	140.13	0.0024	0.0088

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		150.70	-20.75	105.67		0.1963
20		150.70	-20.11	105.67		0.1903
30		107.64	-19.51	105.67		0.1847
40	80.88	215.28		141.46	0.3757	
50	79.88	215.28		141.68	0.3711	
60	79.63	215.28		141.68	0.3699	
70	80.94	215.28		141.46	0.3760	
80		150.70	-6.32	108.96		0.0580
90		150.70	-8.17	116.49		0.0701
100		150.70	-8.24	116.49		0.0707
110	88.39	215.28		140.13	0.4106	
120	86.72	215.28		140.46	0.4028	
130	88.30	215.28		140.13	0.4102	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		150.70	-19.63	105.67		0.1858
20		150.70	-19.20	105.67		0.1817
30		107.64	-18.82	105.67		0.1782
40	80.73	215.28		141.46	0.3750	
50	79.74	215.28		141.68	0.3704	
60	79.54	215.28		141.68	0.3695	
70	80.87	215.28		141.46	0.3757	
80		150.70	-6.07	108.96		0.0557
90		150.70	-8.01	116.49		0.0687
100		150.70	-8.08	116.49		0.0694
110	88.34	215.28		140.13	0.4103	
120	86.67	215.28		140.46	0.4026	
130	88.30	215.28		140.13	0.4102	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-16.77	105.67		0.1587
20		115.92	-16.34	105.67		0.1546
30		82.80	-15.97	105.67		0.1511
40	66.28	165.60		141.46	0.4003	
50	65.51	165.60		141.68	0.3956	
60	65.31	165.60		141.68	0.3944	
70	66.42	165.60		141.46	0.4011	
80		115.92	-5.36	108.96		0.0492
90	0.43	115.92	-7.12	116.49	0.0037	0.0611
100	0.43	115.92	-7.19	116.49	0.0037	0.0617
110	66.75	165.60		140.13	0.4031	
120	65.52	165.60		140.46	0.3956	
130	66.71	165.60		140.13	0.4028	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		115.92	-9.52	105.67		0.0901
20		115.92	-9.25	105.67		0.0875
30		82.80	-9.03	105.67		0.0854
40	64.34	165.60		141.46	0.3886	
50	63.59	165.60		141.68	0.3840	
60	64.20	165.60		141.68	0.3877	
70	65.30	165.60		141.46	0.3943	
80		115.92	-3.36	108.96		0.0308
90		115.92	-3.83	116.49		0.0329
100		115.92	-3.87	116.49		0.0332
110	65.41	165.60		140.13	0.3950	
120	64.20	165.60		140.46	0.3877	
130	66.26	165.60		140.13	0.4001	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-18.20	105.67		0.1723
20		115.92	-17.77	105.67		0.1682
30		82.80	-17.40	105.67		0.1647
40	66.18	165.60		141.46	0.3996	
50	65.40	165.60		141.68	0.3949	
60	65.21	165.60		141.68	0.3938	
70	66.32	165.60		141.46	0.4005	
80		115.92	-5.75	108.96		0.0528
90		115.92	-7.61	116.49		0.0653
100		115.92	-7.68	116.49		0.0659
110	66.75	165.60		140.13	0.4031	
120	65.52	165.60		140.46	0.3956	
130	66.71	165.60		140.13	0.4028	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-16.77	105.67		0.1587
20		115.92	-16.34	105.67		0.1546
30		82.80	-15.97	105.67		0.1511
40	66.28	165.60		141.46	0.4003	
50	65.51	165.60		141.68	0.3956	
60	65.31	165.60		141.68	0.3944	
70	66.42	165.60		141.46	0.4011	
80		115.92	-5.36	108.96		0.0492
90	0.43	115.92	-7.12	116.49	0.0037	0.0611
100	0.43	115.92	-7.19	116.49	0.0037	0.0617
110	66.75	165.60		140.13	0.4031	
120	65.52	165.60		140.46	0.3956	
130	66.71	165.60		140.13	0.4028	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-18.20	105.67		0.1723
20		115.92	-17.77	105.67		0.1682
30		82.80	-17.40	105.67		0.1647
40		165.60	-4.03	141.46		0.0285
50		165.60	-3.75	141.68		0.0265
60		165.60	-2.31	141.68		0.0163
70		165.60	-2.22	141.46		0.0157
80		115.92	-5.75	108.96		0.0528
90		115.92	-7.61	116.49		0.0653
100		115.92	-7.68	116.49		0.0659
110	0.18	165.60	-2.63	140.13	0.0011	0.0188
120	0.30	165.60	-2.45	140.46	0.0018	0.0175
130	0.14	165.60	-0.78	140.13	0.0009	0.0056

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10	6.72	96.60	-28.25	88.06	0.0696	0.3208
20	5.70	96.60	-26.66	88.06	0.0590	0.3027
30	4.43	69.00	-24.92	88.06	0.0643	0.2830
40	67.71	138.00		117.88	0.4907	
50	66.90	138.00		118.07	0.4848	
60	66.17	138.00		118.07	0.4795	
70	67.12	138.00		117.88	0.4864	
80	1.07	96.60	-8.05	90.80	0.0111	0.0887
90	1.95	96.60	-9.95	97.08	0.0202	0.1025
100	1.96	96.60	-10.05	97.08	0.0203	0.1035
110	67.42	138.00		116.78	0.4886	
120	66.17	138.00		117.05	0.4795	
130	66.91	138.00		116.78	0.4849	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.72	96.60	-28.25	88.06	0.0696	0.3208
20	5.70	96.60	-26.66	88.06	0.0590	0.3027
30	4.43	69.00	-24.92	88.06	0.0643	0.2830
40	1.24	138.00	-5.73	117.88	0.0090	0.0486
50	1.42	138.00	-5.43	118.07	0.0103	0.0460
60	0.69	138.00	-3.51	118.07	0.0050	0.0297
70	0.65	138.00	-3.27	117.88	0.0047	0.0277
80	1.07	96.60	-8.05	90.80	0.0111	0.0887
90	1.95	96.60	-9.95	97.08	0.0202	0.1025
100	1.96	96.60	-10.05	97.08	0.0203	0.1035
110	0.86	138.00	-3.44	116.78	0.0062	0.0294
120	0.95	138.00	-3.24	117.05	0.0069	0.0277
130	0.35	138.00	-1.19	116.78	0.0026	0.0102

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-18.20	105.67		0.1723
20		115.92	-17.77	105.67		0.1682
30		82.80	-17.40	105.67		0.1647
40	66.18	165.60		141.46	0.3996	
50	65.40	165.60		141.68	0.3949	
60	65.21	165.60		141.68	0.3938	
70	66.32	165.60		141.46	0.4005	
80		115.92	-5.75	108.96		0.0528
90		115.92	-7.61	116.49		0.0653
100		115.92	-7.68	116.49		0.0659
110	66.75	165.60		140.13	0.4031	
120	65.52	165.60		140.46	0.3956	
130	66.71	165.60		140.13	0.4028	

Absolute Maximum of the all of the Stress Ratio's 0.4907

Governing Element: Bottom Head D1
 Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-3	...	-10.8	7.8	9.7	6.7	+ Tens
Skirt-3	...	-10.8	-7.8	-9.7	-28.2	- Comp
Skirt-2	...	-10.5	7.6	8.6	5.7	+ Tens
Skirt-2	...	-10.5	-7.6	-8.6	-26.7	- Comp
Skirt-1	...	-10.2	7.5	7.2	4.4	+ Tens
Skirt-1	...	-10.2	-7.5	-7.2	-24.9	- Comp



Bottom Head D1	66.5	-2.2	1.9	1.6	67.7	+ Tens
Bottom Head D1	66.5	-2.2	-1.9	-1.6	...	- Comp
Shell-1	65.5	-2.0	1.8	1.6	66.9	+ Tens
Shell-1	65.5	-2.0	-1.8	-1.6	...	- Comp
Shell-D1'	65.5	-1.4	1.0	1.1	66.2	+ Tens
Shell-D1'	65.5	-1.4	-1.0	-1.1	...	- Comp
Top Head D1	66.5	-1.3	1.0	0.9	67.1	+ Tens
Top Head D1	66.5	-1.3	-1.0	-0.9	...	- Comp
Int.Skirt-1	...	-3.5	2.4	2.2	1.1	+ Tens
Int.Skirt-1	...	-3.5	-2.4	-2.2	-8.1	- Comp
Int.Skirt-2	...	-4.0	3.8	2.2	2.0	+ Tens
Int.Skirt-2	...	-4.0	-3.8	-2.2	-9.9	- Comp
Int.Skirt-3	...	-4.0	3.8	2.2	2.0	+ Tens
Int.Skirt-3	...	-4.0	-3.8	-2.2	-10.0	- Comp
Bottom Head-D2	66.6	-1.3	1.4	0.7	67.4	+ Tens
Bottom Head-D2	66.6	-1.3	-1.4	-0.7	...	- Comp
Shell-D2	65.2	-1.1	1.4	0.7	66.2	+ Tens
Shell-D2	65.2	-1.1	-1.4	-0.7	...	- Comp
Top Head-D2	66.6	-0.4	0.5	0.3	66.9	+ Tens
Top Head-D2	66.6	-0.4	-0.5	-0.3	...	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

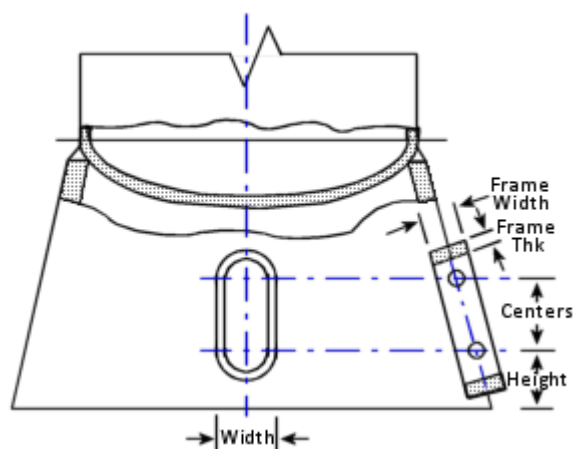
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-3

Total vessel (empty) weight	Fg	1585863.375 N
Weight of vessel contents (liquid)	Ff	275691.281 N
Vessel Gross Weight	Fg + Ff	1861554.750 N
Skirt Mean Diameter Top	DzTop	3726.000 mm
Skirt Mean Diameter Bottom	DzBot	3726.000 mm
Skirt Analysis Thickness (Corroded)	ez	16.400 mm
Skirt Height	H	1700.000 mm
Additional Axial Force at section 1-1	F1	-123699.836 N
Moment at section 1-1	M1	4399770112.000 N-mm
Applied Moment - Bottom of Skirt	Mb	5142120448.000 N-mm
Weight at Bottom of Skirt	W	2005533.500 N



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	600.0000	90.0	1000.00	0.00	80.000	12.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	80.7333	99.2667

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
--------------------	------------------



-260.733 80.733

Height of Weakest Section	H	999.479	mm
Skirt mean diameter at height H	Dz	3726.000	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	0.000	mm
Centroid Position in Y direction	h	-80.440	mm
Principle Moment of Inertia	Ixx	304441294848.000	mm ⁴
Principle Moment of Inertia	Iyy	333012369408.000	mm ⁴
Distance to Extreme Fiber	ybar	1951.640	mm
Distance to Extreme Fiber	xbar	1871.200	mm
Minimum Section Modulus	W4	155992688.000	mm ³
Metal Area at the Cutting Plane	A4	184008.406	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:
 = sin⁻¹(Largest Opening Width / Dz)
 = sin⁻¹(600.0/3726.0)
 = 0.1617 rad (9.2667 deg)

Skirt Mean Radius at Section 4-4 [r]:
 = Dz / 2
 = 3726.0/2
 = 1863.0 mm

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / ez)}$$

$$= 0.162 \cdot \sqrt{(1863.0/16.4)}$$

$$= 1.724$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:
 = ex(new) - ci - cext
 = 18.0 - 0.8 - 0.8
 = 16.4 mm

Find the moment acting at the Critical Height [M4]:
 = (Mbottom - Mtop) * (H - CutHeight) / H + Mtop
 = (.51421E+10 - 0.43998E+10) * (1700.0 - 999.479) / 1700.0 + 0.43998E+10
 = 0.47057E+10 N-mm

Compute the section modulus [Zxx and Zyy]:
 $Zxx = Ixx / ybar = 0.30444E+12 / 1951.64 = 0.15599E+09 \text{ mm}^3$
 $Zyy = Iyy / xbar = 0.33301E+12 / 1871.2 = 0.17797E+09 \text{ mm}^3$
 $Z = \min(Zxx, Zyy) = \min(.15599E+09, .17797E+09) = 0.15599E+09 \text{ mm}^3$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(Ixx, Iyy) / \text{Area}(A4))}$$

$$= \sqrt{(\min(0.30444E+12, 0.33301E+12) / 184008.406)}$$

$$= 1286.273 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:
 = rg / L
 = 1286.273 / 1700.0
 = 0.757

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{ex^2 + ey^2}$$

$$= \sqrt{(0.0^2 + -80.44^2)}$$

$$= 0.000 \text{ mm}$$

Where ex = 0.0 : ey = -80.44

Moment Increment at Centroid [ΔM_4]:

$$= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e$$

$$= \text{abs}(-123699.84 - 1585863 - 275691.28) \cdot 80.44$$

$$= 159758416.000 \text{ N-mm}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:

$$= \text{abs}((l_t - ea_3)/2)$$

$$= \text{abs}(80.0 - 16.4)/2$$

$$= 31.800 \text{ mm}$$

Mean Diameter of Opening, Figure 16.12-5 [d]:

$$= \text{opening width} - e_t$$

$$= 600.0 - 12.0$$

$$= 588.000 \text{ mm}$$

Angle, 16.12-72 [δ]:

$$= \sin^{-1}(d/D_3)$$

$$= \sin^{-1}(588.0/3726.0)$$

$$= 0.158 \text{ radians}$$

Area in the Frame [At]:

$$= 2 \cdot l_t \cdot e_t$$

$$= 2 \cdot 80.0 \cdot 12.0$$

$$= 1920.000 \text{ mm}^2$$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:

$$= (\pi - \delta) \cdot D_3 \cdot ea_3$$

$$= (\pi - 0.158) \cdot 3726.0 \cdot 16.4$$

$$= 182287.734 \text{ mm}^2$$

Combined Area at Opening, 16.12-73 [A4]:

$$= AS + AT$$

$$= 182287.734 + 1920.0$$

$$= 184207.734 \text{ mm}^2$$

Computed Offset, 16.12-74(b) [yt]:

$$= 0.5 \cdot D_3 \cdot \cos(\delta) + h_t - 0.5 \cdot l_t$$

$$= 0.5 \cdot 3726.0 \cdot \cos(0.158) + 31.8 - 0.5 \cdot 80.0$$

$$= 1831.456 \text{ mm}$$

Distance to Neutral Axis, 16.12-74 [yG]:

$$= (0.5 \cdot D_3 \cdot ea_3 \cdot d - 2 \cdot l_t \cdot e_a t \cdot y_t)/A_4$$

$$= (0.5 \cdot 3726.0 \cdot 16.4 \cdot 588.0 - 2 \cdot 80.0 \cdot 12.0 \cdot 1831.456)/184207.734$$

$$= 78.438 \text{ mm}$$

Computed Offset [ymax]:

$$= \max[\text{abs}(0.5 \cdot d_3 \cdot \cos(\delta) + h_t + y_G); \text{abs}(0.5 \cdot D_3 - y_G)]$$

$$= \max[\text{abs}(0.5 \cdot 3726.0 \cdot \cos(0.158) + 31.8 + 78.438); \text{abs}(0.5 \cdot 3726.0 - 78.438)]$$

$$= 1949.894 \text{ mm}$$

Inertia Value, 16.12-75(b) [IS]:

$$= (\pi - \delta - \sin(\delta) \cdot \cos(\delta)) \cdot ea_3 \cdot (D_3/2)^3$$

$$= (\pi - 0.158 - \sin(0.158) \cdot \cos(0.158)) \cdot 16.4 \cdot (3726.0/2)^3$$

$$= 299814092800.0 \text{ mm}^4$$

Section Modulus [W4]:

$$= (I_s + A_t(y_t^2 + l_t^2/12) - A_4 \cdot y_G^2)/y_{\max}$$

$$= (0.29981E+12 + 1920.0(1831.456^2 + 80.0^2/12) - 184207.734 \cdot 78.438^2)/1949.894$$

$$= 156481440.000 \text{ mm}^3$$

Weakening Factor, 16.12-70(b) [psi1]:

$$= \min(1; A_4/(\pi \cdot D_3 \cdot ea_3))$$

$$= \min(1; 184207.734/(\pi \cdot 3726.0 \cdot 16.4))$$

$$= 0.960$$

Weakening Factor, 16.12-70(c) [psi2]:

$$= \min(1; 4 \cdot W_4/(\pi \cdot D^2/3/(\pi \cdot D_3 \cdot ea_3)))$$



$$= \min(1; 4.0 \cdot 0.15648E+09 / (\pi \cdot 3726.0^2 \cdot 16.4)) \\ = 0.875$$

Maximum Bending Moment [Mmax]:

$$= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ = \pi/4 \cdot 3726.0^2 \cdot 16.4 \cdot 105.7 \\ = 18901665792.0 \text{ N-mm}$$

Maximum Compressive Force [Fc,max]:

$$= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ = \pi \cdot 3726.0 \cdot 16.4 \cdot 105.7 \\ = 20283420.0 \text{ N}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$= F4 / (\psi_1 \cdot F_{c,max}) + (\text{abs}(M4) + F4 \cdot \text{abs}(y_g)) / (\psi_2 \cdot M_{max}) \\ = 1985254 / (0.96 \cdot 20283420) + (\text{abs}(0.47057E+10) + 1985254 \cdot \text{abs}(78.438)) / \\ (0.875 \cdot 0.18902E+11) \\ = 0.396 \text{ must be } \leq \text{ to } 1, \text{ Passed}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

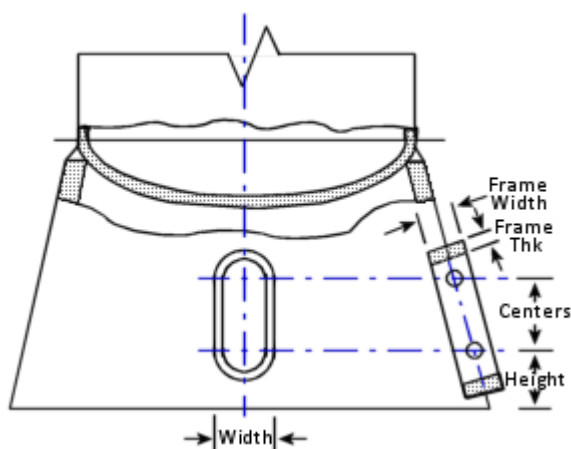
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-2

Total vessel (empty) weight	Fg	1585863.375 N
Weight of vessel contents (liquid)	Ff	275691.281 N
Vessel Gross Weight	Fg + Ff	1861554.750 N
Skirt Mean Diameter Top	DzTop	3726.000 mm
Skirt Mean Diameter Bottom	DzBot	3726.000 mm
Skirt Analysis Thickness (Corroded)	ez	16.400 mm
Skirt Height	H	2000.000 mm
Additional Axial Force at section 1-1	F1	-123699.836 N
Moment at section 1-1	M1	3529497088.000 N-mm
Applied Moment - Bottom of Skirt	Mb	4399770112.000 N-mm
Weight at Bottom of Skirt	W	1948798.625 N



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	500.0000	0.0	550.00	0.00	80.000	12.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-7.7119	7.7119

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
--------------------	------------------



-352.288 -7.712

Height of Weakest Section	H	549.603	mm
Skirt mean diameter at height H	Dz	3726.000	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	-63.188	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	333086425088.000	mm ⁴
Principle Moment of Inertia	Iyy	310572908544.000	mm ⁴
Distance to Extreme Fiber	ybar	1871.200	mm
Distance to Extreme Fiber	xbar	1934.388	mm
Minimum Section Modulus	W4	160553664.000	mm ³
Metal Area at the Cutting Plane	A4	185666.609	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:
 $= \sin^{-1}(\text{Largest Opening Width} / Dz)$
 $= \sin^{-1}(500.0/3726.0)$
 $= 0.1346 \text{ rad (7.7119 deg)}$

Skirt Mean Radius at Section 4-4 [r]:
 $= Dz / 2$
 $= 3726.0/2$
 $= 1863.0 \text{ mm}$

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:
 $= \delta \cdot \sqrt{(r / ez)}$
 $= 0.135 \cdot \sqrt{(1863.0/16.4)}$
 $= 1.435$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:
 $= ex(\text{new}) - ci - cext$
 $= 18.0 - 0.8 - 0.8$
 $= 16.4 \text{ mm}$

Find the moment acting at the Critical Height [M4]:
 $= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}}$
 $= (.43998\text{E}+10 - 0.35295\text{E}+10) \cdot (2000.0 - 549.603) / 2000.0 + 0.35295\text{E}+10$
 $= 0.41606\text{E}+10 \text{ N-mm}$

Compute the section modulus [Zxx and Zyy]:
 $Zxx = Ixx / ybar = 0.33309\text{E}+12 / 1871.2 = 0.17801\text{E}+09 \text{ mm}^3$
 $Zyy = Iyy / xbar = 0.31057\text{E}+12 / 1934.388 = 0.16055\text{E}+09 \text{ mm}^3$
 $Z = \min(Zxx, Zyy) = \min(.17801\text{E}+09, .16055\text{E}+09) = 0.16055\text{E}+09 \text{ mm}^3$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:
 $= \sqrt{(\min(Ixx, Iyy) / \text{Area}(A4))}$
 $= \sqrt{(\min(0.33309\text{E}+12, 0.31057\text{E}+12) / 185666.609)}$
 $= 1293.347 \text{ mm}$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:
 $= rg / L$
 $= 1293.347 / 2000.0$
 $= 0.647$

Location of the Skirt Centroid at critical height [e]:
 $= \sqrt{ex^2 + ey^2}$
 $= \sqrt{(-63.188^2 + 0.0^2)}$
 $= 80.440 \text{ mm}$

Where $ex = -63.188$: $ey = 0.0$

Moment Increment at Centroid [ΔM_4]:

$$= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e$$

$$= \text{abs}(-123699.84 - 1585863 - 275691.28) \cdot 63.188$$

$$= 125495896.000 \text{ N-mm}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:

$$= \text{abs}((l_t - ea_3)/2)$$

$$= \text{abs}(80.0 - 16.4)/2$$

$$= 31.800 \text{ mm}$$

Mean Diameter of Opening, Figure 16.12-5 [d]:

$$= \text{opening width} - e_t$$

$$= 500.0 - 12.0$$

$$= 488.000 \text{ mm}$$

Angle, 16.12-72 [δ]:

$$= \sin^{-1}(d/D_3)$$

$$= \sin^{-1}(488.0/3726.0)$$

$$= 0.131 \text{ radians}$$

Area in the Frame [At]:

$$= 2 \cdot l_t \cdot e_t$$

$$= 2 \cdot 80.0 \cdot 12.0$$

$$= 1920.000 \text{ mm}^2$$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:

$$= (\pi - \delta) \cdot D_3 \cdot ea_3$$

$$= (\pi - 0.131) \cdot 3726.0 \cdot 16.4$$

$$= 183945.172 \text{ mm}^2$$

Combined Area at Opening, 16.12-73 [A4]:

$$= AS + AT$$

$$= 183945.172 + 1920.0$$

$$= 185865.172 \text{ mm}^2$$

Computed Offset, 16.12-74(b) [yt]:

$$= 0.5 \cdot D_3 \cdot \cos(\delta) + h_t - 0.5 \cdot l_t$$

$$= 0.5 \cdot 3726.0 \cdot \cos(0.131) + 31.8 - 0.5 \cdot 80.0$$

$$= 1838.752 \text{ mm}$$

Distance to Neutral Axis, 16.12-74 [yG]:

$$= (0.5 \cdot D_3 \cdot ea_3 \cdot d - 2 \cdot l_t \cdot e_a t \cdot y_t)/A_4$$

$$= (0.5 \cdot 3726.0 \cdot 16.4 \cdot 488.0 - 2 \cdot 80.0 \cdot 12.0 \cdot 1838.752)/185865.172$$

$$= 61.225 \text{ mm}$$

Computed Offset [ymax]:

$$= \max[\text{abs}(0.5 \cdot d_3 \cdot \cos(\delta) + h_t + y_G); \text{abs}(0.5 \cdot D_3 - y_G)]$$

$$= \max[\text{abs}(0.5 \cdot 3726.0 \cdot \cos(0.131) + 31.8 + 61.225); \text{abs}(0.5 \cdot 3726.0 - 61.225)]$$

$$= 1939.977 \text{ mm}$$

Inertia Value, 16.12-75(b) [IS]:

$$= (\pi - \delta - \sin(\delta) \cdot \cos(\delta)) \cdot ea_3 \cdot (D_3/2)^3$$

$$= (\pi - 0.131 - \sin(0.131) \cdot \cos(0.131)) \cdot 16.4 \cdot (3726.0/2)^3$$

$$= 305446322176.0 \text{ mm}^4$$

Section Modulus [W4]:

$$= (I_s + A_t(y_t^2 + l_t^2/12) - A_4 \cdot y_G^2)/y_{\max}$$

$$= (0.30545E+12 + 1920.0(1838.752^2 + 80.0^2/12) - 185865.172 \cdot 61.225^2)/1939.977$$

$$= 160436144.000 \text{ mm}^3$$

Weakening Factor, 16.12-70(b) [psi1]:

$$= \min(1; A_4/(\pi \cdot D_3 \cdot ea_3))$$

$$= \min(1; 185865.172/(\pi \cdot 3726.0 \cdot 16.4))$$

$$= 0.968$$

Weakening Factor, 16.12-70(c) [psi2]:

$$= \min(1; 4 \cdot W_4/(\pi \cdot D^3/(\pi \cdot D_3 \cdot ea_3)))$$



$$= \min(1; 4.0 \cdot 0.16044\text{E}+09 / (\pi \cdot 3726.0^2 \cdot 16.4)) \\ = 0.897$$

Maximum Bending Moment [Mmax]:

$$= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{\text{call}} \\ = \pi/4 \cdot 3726.0^2 \cdot 16.4 \cdot 105.7 \\ = 18901665792.0 \text{ N-mm}$$

Maximum Compressive Force [Fc,max]:

$$= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{\text{call}} \\ = \pi \cdot 3726.0 \cdot 16.4 \cdot 105.7 \\ = 20283420.0 \text{ N}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$= F_4 / (\psi_1 \cdot F_{c,\text{max}}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_g)) / (\psi_2 \cdot M_{\text{max}}) \\ = 1985254 / (0.968 \cdot 20283420) + (\text{abs}(0.41606\text{E}+10) + 1985254 \cdot \text{abs}(61.225)) / \\ (0.897 \cdot 0.18902\text{E}+11) \\ = 0.354 \text{ must be } \leq \text{ to } 1, \text{ Passed}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

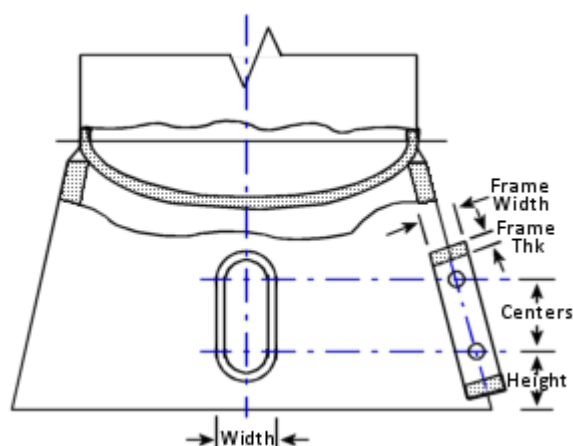
The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt-1

Total vessel (empty) weight	Fg	1585863.375 N
Head Weight below section 2-2	DeltaFg	168718.000 N
Weight of vessel contents (liquid)	Ff	275691.281 N
Vessel Gross Weight	Fg + Ff	1861554.750 N

Skirt Mean Diameter Top	End	DzTop	3726.000 mm
Skirt Mean Diameter Bottom	End	DzBot	3726.000 mm
Skirt Analysis Thickness (Corroded)	ez	16.400 mm	
Skirt Height	H	1000.000 mm	

Additional Axial Force at section 1-1	F1	-123699.836 N
Moment at section 1-1	M1	3096109568.000 N-mm
Applied Moment - Bottom of Skirt	Mb	3529497088.000 N-mm
Weight at Bottom of Skirt	W	1901873.000 N



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	80.0000	0.0	500.00	0.00	80.000	12.000
2	80.0000	45.0	500.00	0.00	80.000	12.000
3	80.0000	90.0	500.00	0.00	80.000	12.000
4	80.0000	135.0	500.00	0.00	80.000	12.000
5	80.0000	180.0	500.00	0.00	80.000	12.000
6	80.0000	225.0	500.00	0.00	80.000	12.000
7	80.0000	270.0	500.00	0.00	80.000	12.000
8	80.0000	315.0	500.00	0.00	80.000	12.000

Interference check:

Access	Access	Pitch mm	Ligament mm
1	2	1463.1722	1383.1722
2	3	1463.1722	1383.1722
3	4	1463.1722	1383.1722
4	5	1463.1722	1383.1722
5	6	1463.1722	1383.1722



6	7	1463.1722	1383.1722
7	8	1463.1722	1383.1722
1	8	1463.1722	1383.1722

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 1463.1722 mm
 Minimum Ligament between Openings is: 1383.1722 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(3726.0 \cdot 16.4)} \text{ mm}$$

$$= 494.394 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 8

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-1.2303	1.2303
2	43.7697	46.2303
3	88.7697	91.2303
4	133.7697	136.2303
5	178.7697	181.2303
6	223.7697	226.2303
7	268.7697	271.2303
8	313.7697	316.2303

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
1.230	43.770
46.230	88.770
91.230	133.770
136.230	178.770
181.230	223.770
226.230	268.770
271.230	313.770
-43.770	-1.230

Height of Weakest Section	H	499.988	mm
Skirt mean diameter at height H	Dz	3726.000	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	-0.000	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	341584216064.000	mm ⁴
Principle Moment of Inertia	Iyy	341584216064.000	mm ⁴
Distance to Extreme Fiber	ybar	1871.200	mm
Distance to Extreme Fiber	xbar	1871.200	mm
Minimum Section Modulus	W4	182548352.000	mm ³
Metal Area at the Cutting Plane	A4	196834.672	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / Dz)$$

$$= \sin^{-1}(80.0/3726.0)$$

$$= 0.0215 \text{ rad (1.2303 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= Dz / 2$$

$$= 3726.0/2$$

$$= 1863.0 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.



Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
1920.0	1115.3	...	...
1920.0	1115.3	...	...
1920.0	1115.3	...	...
1920.0	1115.3	...	...
1920.0	1115.3	...	...
1920.0	1115.3	...	...
1920.0	1115.3	...	...
1920.0	1115.3	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / ez)}$$

$$= 0.021 \cdot \sqrt{(1863.0/16.4)}$$

$$= 0.229$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= ex(new) - ci - cext$$

$$= 18.0 - 0.8 - 0.8$$

$$= 16.4 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{bottom} - M_{top}) \cdot (H - CutHeight) / H + M_{top}$$

$$= (.35295E+10 - 0.30961E+10) \cdot (1000.0 - 499.988) / 1000.0 + 0.30961E+10$$

$$= 0.33128E+10 \text{ N-mm}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{bar} = 0.34158E+12 / 1871.2 = 0.18255E+09 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{bar} = 0.34158E+12 / 1871.2 = 0.18255E+09 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(.18255E+09, .18255E+09) = 0.18255E+09 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / Area(A4))}$$

$$= \sqrt{(\min(0.34158E+12, 0.34158E+12) / 196834.672)}$$

$$= 1317.341 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 1317.341 / 1000.0$$

$$= 1.317$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(ex^2 + ey^2)}$$

$$= \sqrt{(-0.0^2 + 0.^2)}$$

$$= 63.188 \text{ mm}$$

Where $ex = -0.0$: $ey = 0$.

Moment Increment at Centroid [$\Delta M4$]:

$$= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e$$

$$= \text{abs}(-123699.84 - 1585863 - 275691.28) \cdot 0.0$$

$$= 0.000 \text{ N-mm}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:

$$= \min(1; A4 / (\pi \cdot D3 \cdot ea3))$$



$$= \min(1; 196834.672 / (\pi \cdot 3726.0 \cdot 16.4)) \\ = 1.000$$

Weakening Factor, 16.12-70(c) [psi2]:

$$= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / (\pi \cdot D3 \cdot ea3)) \\ = \min(1; 4.0 \cdot 0.18255E+09 / (\pi \cdot 3726.0^2 \cdot 16.4)) \\ = 1.000$$

Maximum Bending Moment [Mmax]:

$$= \pi/4 \cdot d3^2 \cdot ea3 \cdot \sigma_{call} \\ = \pi/4 \cdot 3726.0^2 \cdot 16.4 \cdot 105.7 \\ = 18901665792.0 \text{ N-mm}$$

Maximum Compressive Force [Fc,max]:

$$= \pi \cdot d3 \cdot ea3 \cdot \sigma_{call} \\ = \pi \cdot 3726.0 \cdot 16.4 \cdot 105.7 \\ = 20283420.0 \text{ N}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$= F4 / (\psi1 \cdot Fc,max) + (abs(M4) + F4 \cdot abs(yq)) / (\psi2 \cdot Mmax) \\ = 1985254 / (1.0 \cdot 20283420) + (abs(0.33128E+10) + 1985254 \cdot abs(0.0)) / \\ (1.0 \cdot 0.18902E+11) \\ = 0.273 \text{ must be } \leq \text{ to } 1, \text{ Passed}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Int.Skirt-2

Mean skirt diameter at the critical height is computed later.

This is a Conical type Support Skirt , Wider at the Bottom

Half Apex Angle of the skirt a 7.839 deg

Total vessel (empty) weight Fg 1585863.375 N

Weight of vessel contents (liquid) Ff 275691.281 N

Vessel Gross Weight Fg + Ff 1861554.750 N

Skirt Mean Diameter Top End DzTop 2948.830 mm

Skirt Mean Diameter Bottom End DzBot 3719.830 mm

Skirt Analysis Thickness (Corroded) ez 18.000 mm

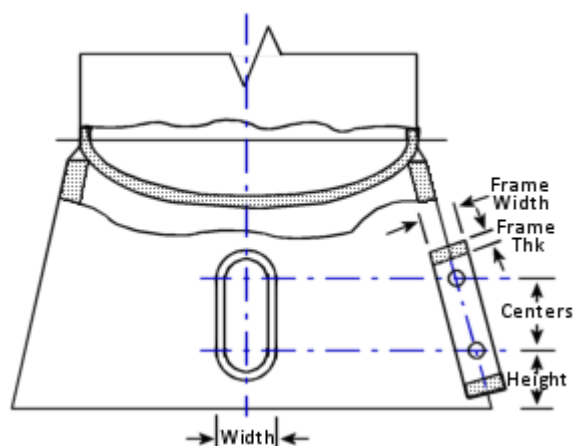
Skirt Height H 2800.000 mm

Additional Axial Force at section 1-1 F1 -123699.836 N

Moment at section 1-1 M1 644390272.000 N-mm

Applied Moment - Bottom of Skirt Mb 1242906880.000 N-mm

Weight at Bottom of Skirt W 698207.188 N



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Oround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	200.0000	0.0	1850.00	0.00	80.000	10.000
2	600.0000	180.0	1150.00	0.00	80.000	10.000

Interference check:

Access	Access	Pitch mm	Ligament mm
1	2	5885.5679	5485.5679
1	2	5885.5679	5485.5679

Interference Check: No Overlap Detected between Openings



Minimum Pitch between openings is: 5885.5679 mm
 Minimum Ligament between Openings is: 5485.5679 mm

Recommended minimum Ligament [IMin]:
 $= 2 \cdot \sqrt{(D \cdot t)}$
 $= 2 \cdot \sqrt{(3719.83 \cdot 18.0)} \text{ mm}$
 $= 517.521 \text{ mm}$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

No Openings Encountered at this Height

Height of Weakest Section	H	2799.902	mm
Skirt mean diameter at height H	Dz	2948.857	mm
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	11.620	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	182966173696.000	mm ⁴
Principle Moment of Inertia	Iyy	182966173696.000	mm ⁴
Distance to Extreme Fiber	ybar	1600.128	mm
Distance to Extreme Fiber	xbar	1611.748	mm
Minimum Section Modulus	W4	124093040.000	mm ³
Metal Area at the Cutting Plane	A4	168326.984	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:
 $= \sin^{-1}(\text{Largest Opening Width} / Dz)$
 $= \sin^{-1}(600.0/2948.857)$
 $= 0.2049 \text{ rad (11.7399 deg)}$

Skirt Mean Radius at Section 4-4 [r]:
 $= Dz / 2$
 $= 2948.857/2$
 $= 1474.4 \text{ mm}$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
1600.0	3422.4	...	...
1600.0	10692.2	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:
 $= \delta \cdot \sqrt{(r / ez)}$
 $= 0.205 \cdot \sqrt{(1474.429/18.0)}$
 $= 1.854$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:
 $= ex(\text{new}) - ci - cext$
 $= 18.0 - 0.0 - 0.0$
 $= 18.0 \text{ mm}$

The half apex angle of the skirt is: 7.839 deg

WARNING: Skirt Angle Exceeds Limits (7 deg).

Find the moment acting at the Critical Height [M4]:
 $= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}}$
 $= (.12429\text{E}+10 - 0.64439\text{E}+09) \cdot (2800.0 - 2800.0) / 2800.0 + 0.64439\text{E}+09$



= 0.64439E+09 N-mm

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx}/y_{bar} = 0.18297E+12/1600.128 = 0.11434E+09 \text{ mm}^3$$

$$Z_{yy} = I_{yy}/x_{bar} = 0.18297E+12/1611.748 = 0.11352E+09 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(0.11434E+09, 0.11352E+09) = 0.11352E+09 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{\min(I_{xx}, I_{yy}) / \text{Area}(A4)}$$

$$= \sqrt{\min(0.18297E+12, 0.18297E+12) / 168326.984}$$

$$= 1042.578 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 1042.578/2800.0$$

$$= 0.372$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{e_x^2 + e_y^2}$$

$$= \sqrt{11.62^2 + 0.2^2}$$

$$= 0.000 \text{ mm}$$

Where $e_x = 11.62$: $e_y = 0$.

Moment Increment at Centroid [$\Delta M4$]:

$$= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e$$

$$= \text{abs}(-123699.84 - 1585863 - 275691.28) \cdot 11.62$$

$$= 23077032.000 \text{ N-mm}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:

$$= \min(1; A4 / (\pi \cdot D3 \cdot ea3))$$

$$= \min(1; 168326.984 / (\pi \cdot 2948.83 \cdot 18.0))$$

$$= 1.000$$

Weakening Factor, 16.12-70(c) [psi2]:

$$= \min(1; 4 \cdot W4 / (\pi \cdot D^3 / (\pi \cdot D3 \cdot ea3)))$$

$$= \min(1; 4.0 \cdot 0.14475E+09 / (\pi \cdot 2948.83^2 \cdot 18.0))$$

$$= 1.000$$

Maximum Bending Moment [Mmax]:

$$= \pi/4 \cdot d3^2 \cdot ea3 \cdot \sigma_{call}$$

$$= \pi/4 \cdot 2948.83^2 \cdot 18.0 \cdot 105.7$$

$$= 12993991680.0 \text{ N-mm}$$

Maximum Compressive Force [Fc,max]:

$$= \pi \cdot d3 \cdot ea3 \cdot \sigma_{call}$$

$$= \pi \cdot 2948.83 \cdot 18.0 \cdot 105.7$$

$$= 17618818.0 \text{ N}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$= F4 / (\psi1 \cdot F_{c,max}) + (\text{abs}(M4) + F4 \cdot \text{abs}(y_g)) / (\psi2 \cdot M_{max})$$

$$= 1985254 / (1.0 \cdot 17618818) + (\text{abs}(0.64439E+09) + 1985254 \cdot \text{abs}(11.62)) / (1.0 \cdot 0.12994E+11)$$

$$= 0.164 \text{ must be } \leq \text{ to } 1, \text{ Passed}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Int.Skirt-3

Total vessel (empty) weight	Fg	1585863.375 N
Head Weight below section 2-2	DeltaFg	57415.398 N
Weight of vessel contents (liquid)	Ff	275691.281 N
Vessel Gross Weight	Fg + Ff	1861554.750 N

Skirt Mean Diameter Top	End	DzTop	2949.000 mm
Skirt Mean Diameter Bottom	End	DzBot	2949.000 mm
Skirt Analysis Thickness (Corroded)		ez	18.000 mm
Skirt Height		H	500.000 mm

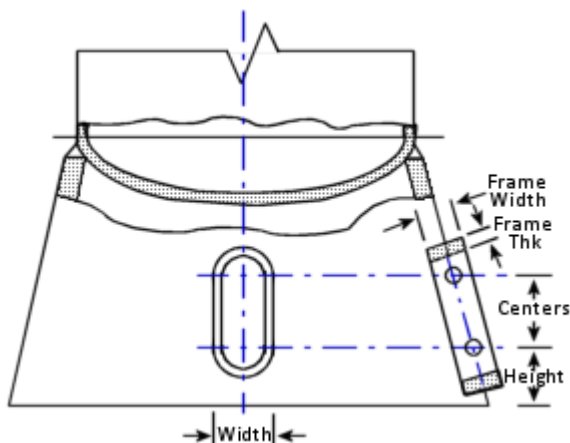
Additional Axial Force at section 1-1	F1	-123699.836 N
Moment at section 1-1	M1	540384640.000 N-mm
Applied Moment - Bottom of Skirt	Mb	644390272.000 N-mm
Weight at Bottom of Skirt	W	645064.938 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	2955.000 mm
Head Corroded Thickness	eb	49.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 728.0 \text{ mm} \\
 K &= D_i / (2 * h_i) = 2906.0 / (2 * 728.0) = 1.9959 \\
 R &= D_i (0.44 * K + 0.02) = 2906.0 * (0.44 * 1.996 + 0.02) = 2610.1309 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 2906.0 * ((0.5 / 1.996) - 0.08) = 495.5201 \text{ mm}
 \end{aligned}$$



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	80.0000	0.0	250.00	0.00	80.000	12.000
2	80.0000	60.0	250.00	0.00	80.000	12.000
3	80.0000	120.0	250.00	0.00	80.000	12.000
4	80.0000	180.0	250.00	0.00	80.000	12.000
5	80.0000	240.0	250.00	0.00	80.000	12.000
6	80.0000	300.0	250.00	0.00	80.000	12.000

Interference check:

Access	Access	Pitch mm	Ligament mm
1	2	1544.0670	1464.0670
2	3	1544.0670	1464.0670
3	4	1544.0670	1464.0670
4	5	1544.0670	1464.0670
5	6	1544.0670	1464.0670
1	6	1544.0670	1464.0670

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 1544.0670 mm
 Minimum Ligament between Openings is: 1464.0670 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(2949.0 \cdot 18.0)} \text{ mm}$$

$$= 460.791 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 6

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-1.5545	1.5545
2	58.4455	61.5545
3	118.4455	121.5545
4	178.4455	181.5545
5	238.4455	241.5545
6	298.4455	301.5545

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
1.554	58.445
61.555	118.446
121.554	178.445
181.555	238.445
241.555	298.445
-58.445	-1.554

Height of Weakest Section	H	249.944	mm
Skirt mean diameter at height H	Dz	2949.000	mm
Rotation of Principle Axes	a	-9.352	deg.
Centroid Position in X direction	g	0.000	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	184412880896.000	mm ⁴
Principle Moment of Inertia	Iyy	184412913664.000	mm ⁴
Distance to Extreme Fiber	ybar	1483.500	mm
Distance to Extreme Fiber	xbar	1483.500	mm
Minimum Section Modulus	W4	124309416.000	mm ³
Metal Area at the Cutting Plane	A4	169641.000	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / Dz)$$

$$= \sin^{-1}(80.0/2949.0)$$

$$= 0.0271 \text{ rad (1.5545 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= Dz / 2$$



$$= 2949.0/2$$

$$= 1474.5 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
1920.0	1224.1	...	...
1920.0	1224.1	...	...
1920.0	1224.1	...	...
1920.0	1224.1	...	...
1920.0	1224.1	...	...
1920.0	1224.1	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / ez)}$$

$$= 0.027 \cdot \sqrt{(1474.5/18.0)}$$

$$= 0.246$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= ex(new) - ci - cext$$

$$= 18.0 - 0.0 - 0.0$$

$$= 18.0 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{bottom} - M_{top}) \cdot (H - CutHeight) / H + M_{top}$$

$$= (.64439E+09 - 0.54038E+09) \cdot (500.0 - 249.944) / 500.0 + 0.54038E+09$$

$$= 0.59240E+09 \text{ N-mm}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{bar} = 0.18441E+12 / 1483.5 = 0.12431E+09 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{bar} = 0.18441E+12 / 1483.5 = 0.12431E+09 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(.12431E+09, .12431E+09) = 0.12431E+09 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / Area(A4))}$$

$$= \sqrt{(\min(0.18441E+12, 0.18441E+12) / 169641.0)}$$

$$= 1042.63 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 1042.63 / 500.0$$

$$= 2.085$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(ex^2 + ey^2)}$$

$$= \sqrt{(0.^2 + 0.^2)}$$

$$= 11.620 \text{ mm}$$

Where $ex = 0.$: $ey = 0.$

Moment Increment at Centroid [$\Delta M4$]:

$$= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e$$

$$= \text{abs}(-123699.84 - 1585863 - 275691.28) \cdot 0.0$$

$$= 0.000 \text{ N-mm}$$

16.12.3.4 Checking at connection areas:



Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= -123699.836 - 1585863 - 275691.281 + 4 \cdot 0.54038E+09/2949.0$$

$$= -1252578.2 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= -123699.836 - 1585863 - 275691.281 - 4 \cdot 0.54038E+09/2949.0$$

$$= -2717930.8 \text{ N}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-0.1E+07 + 57415.398 + 275691.281) / (\pi \cdot 2955.0 \cdot 49.0) + 4.551 \cdot 2955.0 / (4 \cdot 49.0)$$

$$= 66.59 \text{ MPa}$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-0.3E+07 + 57415.398 + 275691.281) / (\pi \cdot 2955.0 \cdot 49.0) + 4.551 \cdot 2955.0 / (4 \cdot 49.0)$$

$$= 63.37 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (275691.281 + 57415.398) / (\pi \cdot 2955.0 \cdot 49.0) + 4.551 \cdot 2955.0 / (4 \cdot 49.0)$$

$$= 69.34 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= -1252578.250 / (\pi \cdot 2949.0 \cdot 18.0)$$

$$= -7.51 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -2717930.750 / (\pi \cdot 2949.0 \cdot 18.0)$$

$$= -16.30 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$

$$= 1 - 0.5(2955.0 + 49.0 - 2949.0 + 18.0) / (495.52 + 49.0)$$

$$= 0.933$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.933) = 21.098 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(e_b^2 + e_z^2 + 2 \cdot e_b \cdot e_z \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(49.0^2 + 18.0^2 + 2 \cdot 49.0 \cdot 18.0 \cdot 0.933)}$$

$$= 33.056 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 33.05 \cdot -1252578.2 = -41.4218 \times 10^6 \text{ N-mm}$$

$$M_q = a \cdot F_{zq} = 33.05 \cdot -2717930.8 = -89.8799 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057(e_b/e_z)^2$$

$$= 0.63 - 0.057(49.0/18.0)^2$$

$$= 0.208$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:



$$\begin{aligned}
 &= C \cdot 6 \cdot Mp / (\pi \cdot Db \cdot eb^2) \\
 &= 0.208 \cdot 6 \cdot -0.4E+08 / (\pi \cdot 2955.0 \cdot 49.0^2) \\
 &= -2.31 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Stress } [\sigma_{1qb(o)} \text{ \& } \sigma_{2qb(o)}]: \\
 &= C \cdot 6 \cdot Mq / (\pi \cdot Db \cdot eb^2) \\
 &= 0.208 \cdot 6 \cdot -0.9E+08 / (\pi \cdot 2955.0 \cdot 49.0^2) \\
 &= -5.02 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Stress } [\sigma_{3pb(o)}]: \\
 &= C \cdot 6 \cdot Mp / (\pi \cdot Dz \cdot ez^2) \\
 &= 0.208 \cdot 6 \cdot -0.4E+08 / (\pi \cdot 2949.0 \cdot 18.0^2) \\
 &= -17.18 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Stress } [\sigma_{3qb(o)}]: \\
 &= C \cdot 6 \cdot Mq / (\pi \cdot Dz \cdot ez^2) \\
 &= 0.208 \cdot 6 \cdot -0.9E+08 / (\pi \cdot 2949.0 \cdot 18.0^2) \\
 &= -37.29 \text{ MPa}
 \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$eb/Db = 1.929/116.339 = 0.0166$$

$$y = 125 \cdot eb/Db = 125 \cdot 49.0/2955.0 = 2.0728$$

As this an elliptical head, it will be treated a a Korbbogen type head

$$\begin{aligned}
 &\text{Geometric value } [\alpha]: \\
 &= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 2.073 = 3.7854
 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 $[\sigma_{1b(p)} \text{ \& } \sigma_{2b(p)}]$:

$$\begin{aligned}
 &= (P + Ph) \cdot Db ((\gamma / \gamma_A) \cdot \alpha - 1) / (4 \cdot eb) \\
 &= (4.551 + 0.0) \cdot 2955.0 ((21.1/40.0) \cdot 3.785 - 1) / (4 \cdot 49.0) \\
 &= 68.38 \text{ MPa}
 \end{aligned}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

$$\begin{aligned}
 &\text{Stress } [\sigma_{1pitot}]: \\
 &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\
 &= 66.587 - -2.314 + 68.375 \\
 &= 137.28 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Stress } [\sigma_{1potot}]: \\
 &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\
 &= 66.587 + -2.314 - 68.375 \\
 &= -4.10 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Stress } [\sigma_{1qitot}]: \\
 &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\
 &= 63.366 - -5.021 + 68.375 \\
 &= 136.76 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Stress } [\sigma_{1qotot}]: \\
 &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\
 &= 63.366 + -5.021 - 68.375 \\
 &= -10.03 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Stress } [\sigma_{2pitot}]: \\
 &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\
 &= 69.341 + -2.314 + 68.375 \\
 &= 135.40 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Stress } [\sigma_{2potot}]: \\
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 69.341 - -2.314 - 68.375 \\
 &= 3.28 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Stress } [\sigma_{2qitot}]: \\
 &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\
 &= 69.341 + -5.021 + 68.375
 \end{aligned}$$



= 132.70 MPa

Stress [σ_{2qotot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 69.341 - -5.021 - 68.375$
 $= 5.99 \text{ MPa}$

Stress [σ_{3pitot}]:
 $= \sigma_{3pm} - \sigma_{3pb}$
 $= -7.512 - -17.183$
 $= 9.67 \text{ MPa}$

Stress [σ_{3potot}]:
 $= \sigma_{3pm} + \sigma_{3pb}$
 $= -7.512 + -17.183$
 $= -24.69 \text{ MPa}$

Stress [σ_{3qitot}]:
 $= \sigma_{3qm} - \sigma_{3qb}$
 $= -16.3 - -37.285$
 $= 20.99 \text{ MPa}$

Stress [σ_{3qotot}]:
 $= \sigma_{3qm} + \sigma_{3qb}$
 $= -16.3 + -37.285$
 $= -53.58 \text{ MPa}$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$
 $= 138.0(3 - (1/1.5)) \cdot (66.587/138.0)^2$
 $= 392.58 \text{ MPa}$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$
 $= 138.0(3 - (1/1.5)) \cdot (63.366/138.0)^2$
 $= 394.60 \text{ MPa}$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$
 $= 138.0(3 - (1/1.5)) \cdot (69.341/138.0)^2$
 $= 390.77 \text{ MPa}$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$
 $= 138.0(3 - (1/1.5)) \cdot (69.341/138.0)^2$
 $= 390.77 \text{ MPa}$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2$
 $= 138.0(3 - (1/1.5)) \cdot (-7.512/138.0)^2$
 $= 413.73 \text{ MPa}$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2$
 $= 138.0(3 - (1/1.5)) \cdot (-16.3/138.0)^2$
 $= 412.72 \text{ MPa}$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:
 $= \min(1; A4 / (\pi \cdot D3 \cdot ea3))$
 $= \min(1; 169641.0 / (\pi \cdot 2949.0 \cdot 18.0))$
 $= 1.000$

Weakening Factor, 16.12-70(c) [psi2]:
 $= \min(1; 4 \cdot W4 / (\pi \cdot D^23 / (\pi \cdot D3 \cdot ea3)))$
 $= \min(1; 4.0 \cdot 0.12431E+09 / (\pi \cdot 2949.0^2 \cdot 18.0))$
 $= 1.000$



Maximum Bending Moment [Mmax]:

$$= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{call}$$

$$= \pi/4 \cdot 2949.0^2 \cdot 18.0 \cdot 105.7$$

$$= 12995488768.0 \text{ N-mm}$$

Maximum Compressive Force [Fc,max]:

$$= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{call}$$

$$= \pi \cdot 2949.0 \cdot 18.0 \cdot 105.7$$

$$= 17619834.0 \text{ N}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$= F4 / (\psi_1 \cdot F_{c,max}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_q)) / (\psi_2 \cdot M_{max})$$

$$= 1985254 / (1.0 \cdot 17619834) + (\text{abs}(0.59240E+09) + 1985254 \cdot \text{abs}(0.0)) /$$

$$(1.0 \cdot 0.12995E+11)$$

$$= 0.158 \text{ must be } \leq \text{ to } 1, \text{ Passed}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	137.28	392.58	Pass	46
Stot1po	-4.10	392.58	Pass	47
Stot1qi	136.76	394.60	Pass	48
Stot1qo	-10.03	394.60	Pass	49
Stot2pi	135.40	390.77	Pass	50
Stot2po	3.28	390.77	Pass	51
Stot2qi	132.70	390.77	Pass	52
Stot2qo	5.99	390.77	Pass	53
Stot3pi	9.67	413.73	Pass	54
Stot3po	-24.69	413.73	Pass	55
Stot3qi	20.99	412.72	Pass	56
Stot3qo	-53.58	412.72	Pass	57

The stresses are satisfactory in the region without openings.
 The stresses are satisfactory in the region with the opening(s).

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**Shop/Field Installation Options :**

Platform(s) installed in the Shop.

Insulation is installed in the Shop.

Lining is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	9118.941 mm
Center of Gravity of the Liquid	7284.462 mm
Center of Gravity of the Insulation	9590.856 mm
Center of Gravity of the Lining	6750.000 mm
Center of Gravity of the Nozzles	9442.672 mm
Center of Gravity of the Added Weights (Operating)	11114.152 mm
Center of Gravity of the Added Weights (Empty)	10094.439 mm
Center of Gravity of Bare Shell New and Cold	8457.865 mm
Center of Gravity of Bare Shell Corroded	8441.128 mm
Vessel CG in the Operating Condition	8968.390 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	8869.666 mm
Vessel CG in the Test Condition	8978.636 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	3744.0000	mm
Skirt Thickness	STHK	18.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	[Impact Tested]

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	36.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	MPa
Basing Yield Stress		262.00	MPa
Inside Diameter of Basing	DI, D4	3563.9998	mm
Outside Diameter of Basing	DOU, D5	4024.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	42.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	138.00	MPa
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	3884.0000	mm
Bolt Allowable Shear Stress		82.800	MPa
Ultimate Comp. Strength of Concrete	FPC	41.0	MPa
Allowable Comp. Strength of Concrete	FC	22.5	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	130.0000	mm
Width of Gussets at Base Plate	BWDT	130.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	262.0	MPa
Height of Chair	h1	304.0000	mm
Distance between Gussets	RG, b6	90.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	38.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	140.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	58.0000	mm
External Corrosion Allowance	CA	1.6000	mm
Dead Weight of Vessel	DW	1592290.8	N
Operating Weight of Vessel	ROW	2129233.5	N
Earthquake Moment on Basing	EQMOM	4244295424.0	N-mm
Wind Moment on Basing	WIMOM	1233199744.0	N-mm
User Moment at the Base [Earthquake Case]		1732530048.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		1732530048.0	N-mm (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 0.59768E+10/3884.0) - 2066340) / 20$$

$$= 204324.8 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.29657E+10/3884.0) - 1592291) / 20$$

$$= 73038.8 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/3884.0) - 2129234) / 20$$

$$= -106461.7 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 855.6120 mm²
 Area Available in all the Bolts Abss * RN : 17112.2402 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.300	0.150	1.511	2.441	0.438	0.781	811685.2	2878025.0	78767.0
4	0.225	0.263	1.297	2.605	0.454	0.777	773081.2	2839421.0	70303.3
6	0.282	0.272	1.459	2.482	0.442	0.780	801833.1	2868173.0	76530.6
8	0.268	0.265	1.420	2.513	0.445	0.780	794532.2	2860872.0	74908.5
10	0.266	0.267	1.417	2.515	0.445	0.780	793927.2	2860267.2	74775.3
12	0.267	0.267	1.418	2.515	0.445	0.780	794078.2	2860418.2	74808.6

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 794078 / (1.402 \cdot 3884.0 \cdot 2.515)$$

$$= 115.963 \text{ MPa, Should be } \leq 138.0 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 855.612 / (\pi \cdot 3884.0)$$

$$= 1.402 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 4.287[(2 \cdot 0.2667 \cdot 3884.0 + 230.0) / (2 \cdot 0.2667 \cdot 3884.0)]$$

$$= 4.763 \text{ MPa, Should be } \leq 22.5 \text{ MPa}$$

Values for table 10.3, l = 140.0, b = 305.049, l/b = 0.458943

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(10638.388, 32285.143) = 32285.143 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 32285.143 / 172.9)} + 1.6$$

$$= 35.072 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 140.0, b = 90.0, b/l = 0.642857

As b/l (0.643) is less than 1, inverting b/l = 1.556.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 118064.42 / (4 \cdot \pi) [1.3(\ln((2 \cdot 140.0 \cdot \sin(\pi \cdot 70.0/140.0)) / (\pi \cdot 35.0))) + 1] - [(0.7 - 0.065) \cdot 118064.42 / (4 \cdot \pi)]$$

$$= 14841.2168 \text{ N}$$



Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(14841.22) / 174.66) + 1.6}$$

$$= 24.1802 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 118064.42 / (124.11 \cdot 130.0) + 1.6$$

$$= 11.125 \text{ (not less than } 9.525 + 1.6 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3884.0 / 20 = 610.097 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (118064 \cdot 70.0 / (610.1 \cdot 266.0 \cdot 207))^{2/3} \cdot 1872.0^{1/3} + \text{Ca}$$

$$= 8.517 + 1.6 = 10.117 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 436924.5 / (855.612 \cdot 20)$$

$$= 25.5 \text{ MPa. Must be less than } 82.8 \text{ MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	35.0716	mm
Actual Basing Thickness as entered by user	36.0000	mm
Required Thickness of Chair Cap	24.1802	mm
Actual Top Ring Thickness as entered by user	38.0000	mm
Required Gusset thickness, + CA	11.1250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	10.1169	mm
Given Thickness of Skirt	18.0000	mm
Required Gusset Height to meet local stress	63.5499	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 47.393 \cdot (18.0 - 1.6)$$

$$= 777.19 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 777 / [(0.4 \cdot 239) \cdot 2 \cdot 0.707]$$

$$= 5.75 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 118064.4 / (180.0 + 2 \cdot (230.0 + 38.0))$$

$$= 164.894 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 118064.4 \cdot 70.0 / (180.0 \cdot (268.0) + 0.6667 \cdot (268.0)^2)$$

$$= 85.979 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{164.89^2 + 85.98^2}$$

$$= 185.964 \text{ N/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$



$$= 185.96 / [(0.4 \cdot 239) \cdot 2 \cdot 0.707]$$
$$= 1.376 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$
$$= 118064.4 / (2 \cdot 130.0)$$
$$= 454.094 \text{ N/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$
$$= 118064.4 \cdot 70.0 / (2 \cdot 268.0 \cdot 130.0)$$
$$= 118.607 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$
$$= \sqrt{454.09^2 + 118.61^2}$$
$$= 469.328 \text{ N/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 469.33 / [(0.4 \cdot 239) \cdot 2 \cdot 0.707]$$
$$= 3.472 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	3.4722	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N310"

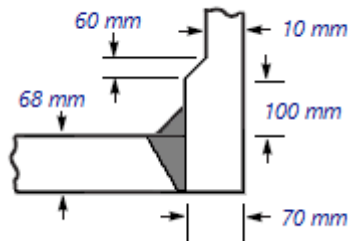
From: 40

Pressure for Reinforcement Calculations	P	4.9722	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	3650.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	68.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		70.0000	mm
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	68.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2500.0	mm
Centerline Distance to Tangent		2500.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N310"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.000 in.
Actual Thickness Used in Calculation	0.394 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 4.97 \cdot 3652.0 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.97)$
 $= 59.3964 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.97 \cdot 127.0 / (138 \cdot 1.0 + 0.4 \cdot 4.97)$
 $= 4.5108 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 60.0 / 0.5773$
 $= 103.9231 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

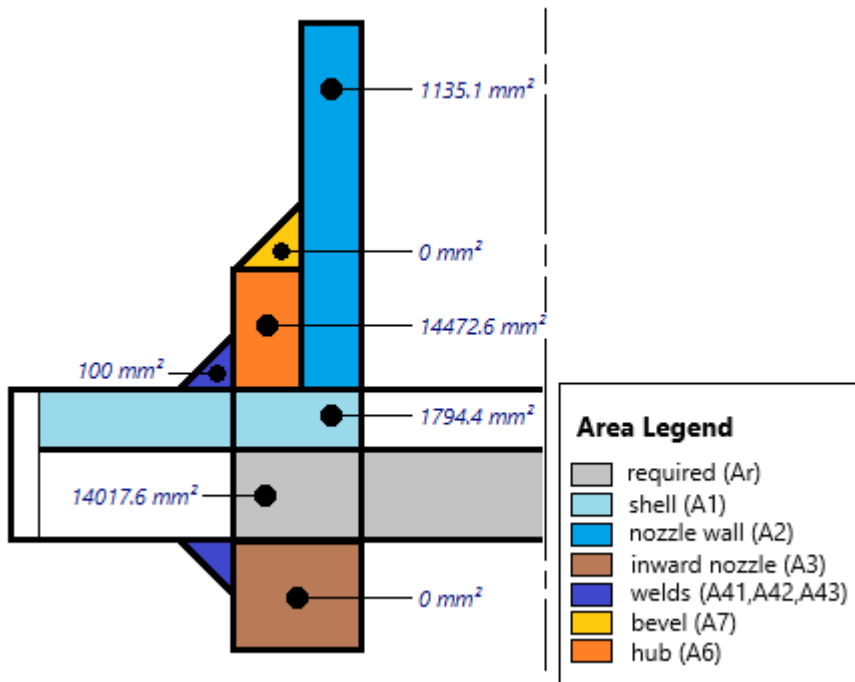
Parallel to Vessel Wall (Diameter Limit)	D1	472.0000	mm
Parallel to Vessel Wall, opening length	d	236.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	126.4231	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$

= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14017.552	NA	NA
Area in Shell	A1	1794.449	NA	NA
Area in Nozzle Wall	A2	1135.065	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	14472.595	NA	NA
TOTAL AREA AVAILABLE	Atot	17502.109	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (236.0 \cdot 59.3964 \cdot 1.0 + 2 \cdot 69.0 \cdot 59.3964 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14017.552 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 236.0(1.0 \cdot 67.0 - 1.0 \cdot 59.396) - 2 \cdot 69.0$
 $(1.0 \cdot 67.0 - 1.0 \cdot 59.3964) \cdot (1 - 1.0)$
 $= 1794.449 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 126.42)(9.0 - 4.51)1.0$
 $= 1135.065 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$



= 100.000 mm²

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(126.4, 150.0, 100.0)) \cdot (70.0 - 10.0) \cdot 1.0$$

$$= 14472.595 \text{ mm}^2 \text{ Includes Bevel Area of } 2472.594 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.5108 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 66.9824 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 66.9824 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 9.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.103, \max(66.9824, 2.5)]$$

$$= 9.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.5108, 9.1026)$$

$$= 9.1026 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 71.3,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 273.7 MPa	Passed
Occasional	: 31.4,	Allowable	: 183.5 MPa	Passed
Shear	: 25.1,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govern. thk, tg = 10.0, tr = 4.511, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.501, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-40 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govern. thk, tg = 68.0, tr = 59.396, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.887, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	36 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 4.97/10.20 = 0.487

Weld Size Calculations, Description: N310"

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (14017.5518 - 1794.4495 + 2 • 69.0 • 1.0 •
(1.0 • 67.0 - 59.3964))138)
= 1831402.12 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (15607.6602 + 100.0 - 0.0 • 1.0) • 138
= 2167433.75 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (15607.6602 + 0.0 + 100.0 + (9246.001)) • 138
= 3443250.25 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (15607.6602 + 0.0 + 100.0 + 0.0 + (9246.001)) • 138
= 3443250.25 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 254.0 • 10.0 • 0.49 • 138
= 269764. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 122.5) • (70.0 - 1.0) • 0.7 • 138
= 2564928. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 254.0 • (68.0 - 1.0) • 0.74 • 138
= 2729620. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (269764 + 2564929) = 2834693 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (269764 + 0 + 2729621 + 0) = 2999385 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (269764 + 2729621 + 0) = 2999385 N

Summary of Failure Path Calculations:

Path 1-1 = 2834692 N	, must exceed W = 1831402 N	or W1 = 2167433 N
Path 2-2 = 2999384 N	, must exceed W = 1831402 N	or W2 = 3443250 N
Path 3-3 = 2999384 N	, must exceed W = 1831402 N	or W3 = 3443250 N

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 5 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.4157 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.4158 mm

Input Echo, WRC107/537 Item 1, Description: N310" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	6570.000	mm
Vessel Thickness	Tv	68.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.0000	mm
Nozzle Diameter	Dn	374.000	mm
Nozzle Thickness	Tn	70.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Design Internal Pressure	Dp	4.972	MPa
Include Pressure Thrust		No	

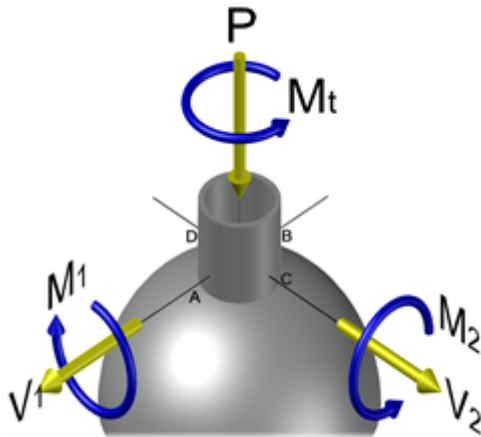
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-22500.0	N
Longitudinal Shear (SUS)	(V1) V1	16250.0	N
Circumferential Shear (SUS)	(Vc) V2	16250.0	N
Circumferential Moment (SUS)	(Mc) M1	12500000.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	12500000.0	N-mm
Torsional Moment (SUS)	Mt	14375019.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 374.0 + 2 \cdot 1.65 \cdot \sqrt{(3319.5 (68.0 - 1.0))}$
 $= 1930.280 \text{ mm}$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-22500.0	N
Circumferential Shear	(VC)	V2	16250.0	N
Longitudinal Shear	(VL)	V1	16250.0	N
Circumferential Moment	(MC)	M1	12500000.0	N-mm
Longitudinal Moment	(ML)	M2	12500000.0	N-mm
Torsional Moment		MT	14375019.0	N-mm

Dimensionless Param: U = 0.40 TAU = 5.00 (2.21) RHO = 0.97

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.08542	(A,B,C,D)
$M(x) / P$	SP 1	0.10891	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.17148	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.31033	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.17148	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.31033	(A,B,C,D)
$N(y) * T / P$	SP 1	0.15398	(A,B,C,D)
$M(y) / P$	SP 1	0.04165	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.06341	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.13439	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.06341	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.13439	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Rad. Bend.	P	3.3	-3.3	3.3	-3.3	3.3	-3.3	3.3	-3.3
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-11.0	11.0	11.0	-11.0
Rad. Memb.	ML	-1.0	-1.0	1.0	1.0	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-11.0	11.0	11.0	-11.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-8.3	7.1	15.7	-12.8	-8.3	7.1	15.7	-12.8

Tang. Memb. P	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Tang. Bend. P	1.3	-1.3	1.3	-1.3	1.3	-1.3	1.3	-1.3
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Tang. Bend. MC	0.0	0.0	0.0	0.0	-4.8	4.8	4.8	-4.8
Tang. Memb. ML	-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Tang. Bend. ML	-4.8	4.8	4.8	-4.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-3.1	3.9	7.2	-4.9	-3.1	3.9	7.2	-4.9
Shear VC	0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tot. Shear	1.4	1.4	0.6	0.6	0.6	0.6	1.4	1.4
Str. Int.	8.6	7.6	15.7	12.9	8.4	7.2	15.9	13.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		122.0	122.0	122.0	122.0	122.0	122.0	122.0	122.0
Rad. Pl (SUS)		-0.6	-0.6	1.4	1.4	-0.6	-0.6	1.4	1.4
Rad. Q (SUS)		-7.7	7.7	14.3	-14.3	-7.7	7.7	14.3	-14.3
Long. Pm (SUS)		122.0	122.0	122.0	122.0	122.0	122.0	122.0	122.0
Long. Pl (SUS)		0.4	0.4	1.1	1.1	0.4	0.4	1.1	1.1
Long. Q (SUS)		-3.5	3.5	6.0	-6.0	-3.5	3.5	6.0	-6.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear Q (SUS)		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pm (SUS)		122.0	122.0	122.0	122.0	122.0	122.0	122.0	122.0
Pm+Pl (SUS)		122.5	122.5	123.7	123.7	122.5	122.5	123.7	123.7
Pm+Pl+Q (Total)		119.2	129.6	137.7	117.1	118.9	129.2	137.9	117.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.96	138.00	Passed
Pm+Pl (SUS)	123.69	207.00	Passed
Pm+Pl+Q (TOTAL)	137.89	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N136"

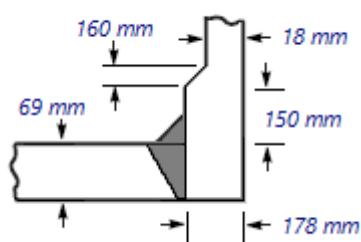
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Pressure for Reinforcement Calculations	P	4.9507	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2300.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	160.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		178.0000	mm
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		69.0000	mm
Inside Projection	h	3300.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N136"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 0.709 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.95 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.95)$
= 66.9494 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4.95 \cdot 457.2 / (138 \cdot 1.0 + 0.4 \cdot 4.95)$
= 16.1699 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 160.0/0.5773
= 277.1282 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

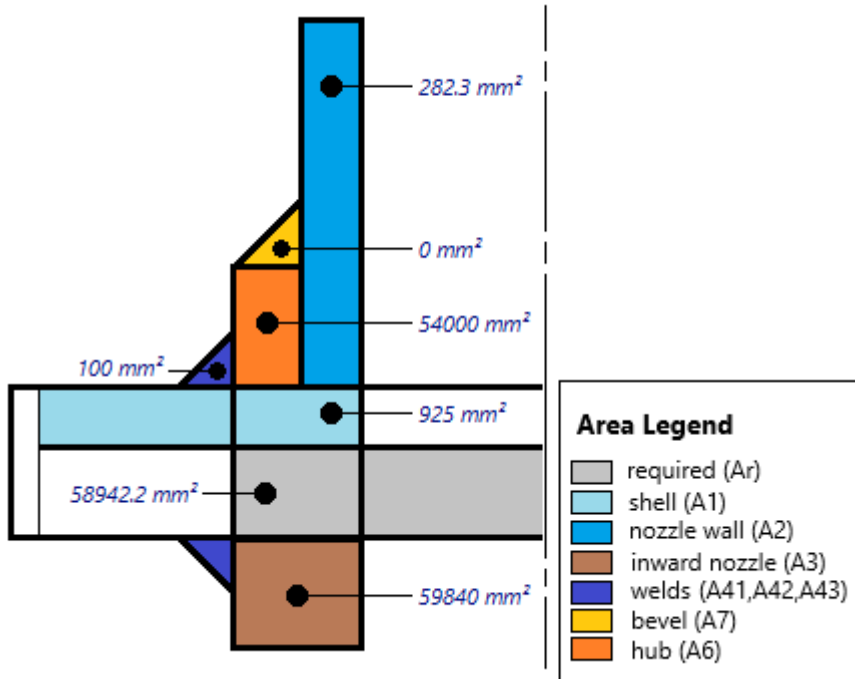
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1760.7998	mm
Parallel to Vessel Wall, opening length	d	880.3999	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	170.0000	mm
Normal to Vessel Wall, Inward		40.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	58942.223	NA	NA
Area in Shell	A1	924.971	NA	NA
Area in Nozzle Wall	A2	282.251	NA	NA
Area in Inward Nozzle	A3	59840.004	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	54000.027	NA	NA
TOTAL AREA AVAILABLE	Atot	115147.250	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (880.3999 \cdot 66.9494 \cdot 1.0 + 2 \cdot 177.0 \cdot 66.9494 \cdot 1.0 \cdot (1 - 1.0))$
 $= 58942.223 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 880.4(1.0 \cdot 68.0 - 1.0 \cdot 66.949) - 2 \cdot 177.0$
 $(1.0 \cdot 68.0 - 1.0 \cdot 66.9494) \cdot (1 - 1.0)$
 $= 924.971 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 170.0)(17.0 - 16.17)1.0$
 $= 282.251 \text{ mm}^2$

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 176.0 \cdot (170.0) \cdot 1.0$
 $= 59840.004 \text{ mm}^2$



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2 \\ &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{hp}, h_o, Hubht)) \cdot (Hubtk - tn) \cdot fr_2 \\ &= (2 \cdot \min(170.0, 300.0, 150.0)) \cdot (178.0 - 18.0) \cdot 1.0 \\ &= 54000.027 \text{ mm}^2 \text{ Includes Bevel Area of } 6000.029 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad ta = 17.1698 \text{ mm} \\ \text{Wall Thickness per UG16(b),} & \quad tr16b = 2.5000 \text{ mm} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad trb1 = 67.9494 \text{ mm} \\ \text{Wall Thickness} & \quad tb1 = \max(trb1, tr16b) = 67.9494 \text{ mm} \\ \text{Wall Thickness} & \quad tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm} \\ \text{Wall Thickness per table UG-45} & \quad tb3 = 9.3312 \text{ mm} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.331, \max(67.9494, 2.5)] \\ &= 9.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(17.1698, 9.3312) \\ &= 17.1698 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	82.9,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.1 MPa	Passed
Occasional	:	62.9,	Allowable	:	183.5 MPa	Passed
Shear	:	12.5,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 18.0, tr = 16.17, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.951, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	7 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 69.0, tr = 66.949, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.985, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 4.95/10.20 = 0.485$$

Weld Size Calculations, Description: N136"



Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (58942.2227 - 924.9713 + 2 \cdot 177.0 \cdot 1.0 \cdot (1.0 \cdot 68.0 - 66.9494)) \cdot 138)$
 $= 8056875.50 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (54282.2773 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 7503980.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (54282.2773 + 59840.0039 + 100.0 + (24072.0)) \cdot 138$
 $= 19082644.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (54282.2773 + 59840.0039 + 100.0 + 0.0 + (24072.0)) \cdot 138$
 $= 19082644.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 914.4 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 971150. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 448.7) \cdot (178.0 - 1.0) \cdot 0.7 \cdot 138$
 $= 24100128. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 914.4 \cdot (69.0 - 1.0) \cdot 0.74 \cdot 138$
 $= 9973300. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (971150 + 24100128) = 25071278 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (971150 + 0 + 9973300 + 0) = 10944451 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (971150 + 9973300 + 0) = 10944451 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 25071278N	, must exceed W = 8056875 N	or W1 = 7503980 N
Path 2-2 = 10944451N	, must exceed W = 8056875 N	or W2 = 19082644 N
Path 3-3 = 10944451N	, must exceed W = 8056875 N	or W3 = 19082644 N



Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 58.1968 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 3669.0000 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 880.40 mm

Input Echo, WRC297 Item 1, Description: N136" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3650.000	mm
Shell Thickness	69.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	1234.400	mm
Nozzle Thickness	178.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc 138.00	MPa
Nozzle Hot S.I. Allowable	SNmh 138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4.95	MPa
Radial Load	P	-80968.28	N
Circumferential Shear	Vc	58476.96	N
Longitudinal Shear	Vl	58476.96	N
Circumferential Moment	Mc	0.162E+09	N-mm
Longitudinal Moment	Ml	0.162E+09	N-mm
Torsional Moment	Mt	0.186E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.935) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 1234.4 + 2 \cdot 1.65 \cdot \sqrt{(1860.0(69.0 - 1.0))} \\
 &= 2408.013 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3720.000 mm
Nozzle Outside Diameter	(d) =	1234.400 mm

Vessel Thickness used (T) = 68.000 mm
 Nozzle Thickness used (t) = 177.000 mm
 T / t = 0.384
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 2.454$$

Nr/P = 0.063
 Mr/P = 0.123
 M0/P = 0.037
 N0/P = 0.083
 MrD/Mc = 0.276
 NrDL/Mc = 0.097
 M0d/Mc = 0.084
 N0DL/Mc = 0.080
 MrD/Ml = 0.100
 NrDL/Ml = 0.064
 M0D/Ml = 0.030
 N0DL/Ml = 0.103

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-5	5	5	-5
Normal Pressure Stress	130	135	130	135
Outplane Stress Summary	127	135	143	130

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-17	17	17	-17
Inplane Pressure Stress	65	65	65	65
Inplane Stress Summary	60	68	98	38

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	127	135	143	130

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-14	14	14	-14
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65	65	65	65
Outplane Stress Summary	54	74	87	50

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-47	47	47	-47
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130	135	130	135
Inplane Stress Summary	94	167	194	79

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	1	1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	94	167	194	79

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	18	12	18
Hoop Stress Summary	10	16	16	22

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	4	5	10	4

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	10	16	16	22

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	18	12	18
Hoop Stress Summary	11	17	16	21

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	0	9	14	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	11	17	16	21

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130	135	130	135	130	135	130	135
Circ. Pl (SUS)	-1	-1	3	3	-1	-1	3	3
Circ. Q (SUS)	-2	2	8	-8	-35	35	59	-59
Long. Pm (SUS)	65	65	65	65	65	65	65	65
Long. Pl (SUS)	0	0	2	2	-1	-1	3	3
Long. Q (SUS)	-5	5	29	-29	-11	11	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	130.0	135.0	130.0	135.0	130.0	135.0	130.0	135.0
Pm+Pl (SUS)	129.0	134.0	133.0	138.0	129.0	134.0	133.0	138.0
Pm+Pl+Q (Total)	127.0	136.0	141.0	130.0	94.0	169.0	192.0	79.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	135.00	138.00	Passed
Pm+Pl (SUS)	138.00	207.00	Passed
Pm+Pl+Q (TOTAL)	192.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	18	12	18	12	18	12	18
Circ. Pl (SUS)	-1	-1	3	3	-1	-1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-4	4	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	12.0	18.0	12.0	18.0	12.0	18.0	12.0	18.0
Pm+Pl (SUS)	11.0	17.0	15.0	21.0	11.0	17.0	15.0	21.0
Pm+Pl+Q (Total)	11.0	17.0	15.0	21.0	11.0	17.0	15.0	21.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	18.00	138.00	Passed
Pm+Pl (SUS)	21.00	207.00	Passed
Pm+Pl+Q (TOTAL)	21.00	414.00	Passed



*Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.*

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Input, Nozzle Desc: N54"

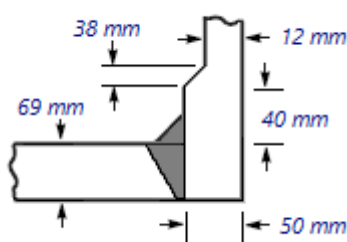
From: 50

Pressure for Reinforcement Calculations	P	4.9507	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1350.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	40.0000	mm
Height of Beveled Transition	L'	38.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		69.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N54"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 4.000 in.
Actual Thickness Used in Calculation 0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.95 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.95)$
= 66.9494 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4.95 \cdot 50.8 / (138 \cdot 1.0 + 0.4 \cdot 4.95)$
= 1.7967 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 38.0/0.5773
= 65.8180 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

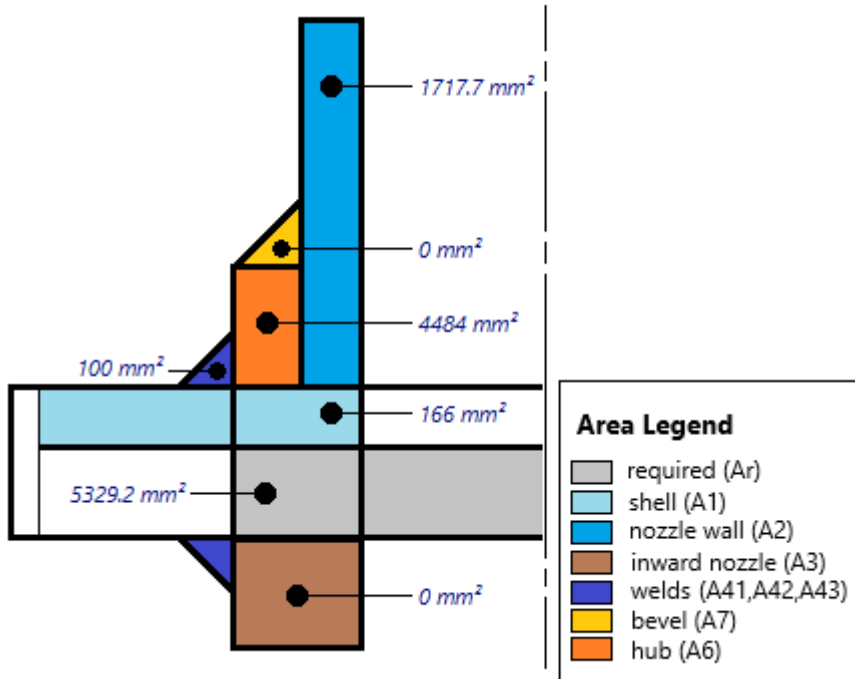
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	237.6000	mm
Parallel to Vessel Wall	Rn+tn+t	118.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	93.3180	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5329.170	NA	NA
Area in Shell	A1	165.999	NA	NA
Area in Nozzle Wall	A2	1717.676	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	4484.000	NA	NA
TOTAL AREA AVAILABLE	Atot	6467.674	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (79.6 \cdot 66.9494 \cdot 1.0 + 2 \cdot 49.0 \cdot 66.9494 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 5329.170 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 158.0(1.0 \cdot 68.0 - 1.0 \cdot 66.949) - 2 \cdot 49.0$$

$$(1.0 \cdot 68.0 - 1.0 \cdot 66.9494) \cdot (1 - 1.0)$$

$$= 165.999 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 93.32)(11.0 - 1.8)1.0$$

$$= 1717.676 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 100.000 \text{ mm}^2$$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(93.3, 300.0, 40.0)) \cdot (50.0 - 12.0) \cdot 1.0 \\ &= 4484.000 \text{ mm}^2 \text{ Includes Bevel Area of } 1444.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad t_a = 2.7967 \text{ mm} \\ \text{Wall Thickness per UG16(b),} & \quad t_{r16b} = 2.5000 \text{ mm} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad t_{rb1} = 67.9494 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b1} = \max(t_{rb1}, t_{r16b}) = 67.9494 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b2} = \max(t_{rb2}, t_{r16b}) = 2.5000 \text{ mm} \\ \text{Wall Thickness per table UG-45} & \quad t_{b3} = 6.0200 \text{ mm} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ &= \min[6.02, \max(67.9494, 2.5)] \\ &= 6.0200 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(2.7967, 6.02) \\ &= 6.0200 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	49.1,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	295.9 MPa	Passed
Occasional	:	7.9,	Allowable	:	183.5 MPa	Passed
Shear	:	27.7,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 12.0$, $t_r = 1.797$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.163$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 50.0$, $t_r = 1.797$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	36 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 4.95/10.20 = 0.485$$

Weld Size Calculations, Description: N54"

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (5329.1704 - 166.0 + 2 \cdot 49.0 \cdot 1.0 \cdot (1.0 \cdot 68.0 - 66.9494)) \cdot 138)$
 $= 726651.50 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (6201.6753 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 869541.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (6201.6753 + 0.0 + 100.0 + (6664.0005)) \cdot 138$
 $= 1789078.75 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (6201.6753 + 0.0 + 100.0 + 0.0 + (6664.0005)) \cdot 138$
 $= 1789078.75 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 101.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 107906. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 45.3) \cdot (50.0 - 1.0) \cdot 0.7 \cdot 138$
 $= 673572. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 101.6 \cdot (69.0 - 1.0) \cdot 0.74 \cdot 138$
 $= 1108144. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (107906 + 673572) = 781478 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (107906 + 0 + 1108144 + 0) = 1216050 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (107906 + 1108144 + 0) = 1216050 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 781478 N	, must exceed W = 726651 N	or W1 = 869541 N
Path 2-2 = 1216050 N	, must exceed W = 726651 N	or W2 = 1789078 N
Path 3-3 = 1216050 N	, must exceed W = 726651 N	or W3 = 1789078 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 MPa



Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.7071 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.7072 mm

Input Echo, WRC297 Item 1, Description: N54" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	3650.000	mm	
Shell Thickness	69.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	1.0000	mm	
Nozzle Diameter	177.600	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	4.95	MPa
Radial Load	P	-8996.48	N
Circumferential Shear	Vc	6497.37	N
Longitudinal Shear	Vl	6497.37	N
Circumferential Moment	Mc	0.200E+07	N-mm
Longitudinal Moment	Ml	0.200E+07	N-mm
Torsional Moment	Mt	0.230E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (2.574) is < 5
 check the limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (3.552) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 177.6 + 2 \cdot 1.65 \cdot \sqrt{(1860.0(69.0 - 1.0))}$
 $= 1351.213 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3720.000	mm
Nozzle Outside Diameter	(d) =	177.600	mm
Vessel Thickness used	(T) =	68.000	mm
Nozzle Thickness used	(t) =	49.000	mm
T / t	=	1.388	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.353$$

Nr/P = 0.214
 Mr/P = 0.290
 M0/P = 0.126
 N0/P = 0.185
 MrD/Mc = 0.269
 NrDL/Mc = 0.164
 M0d/Mc = 0.143
 N0DL/Mc = -0.117
 MrD/M1 = 0.256
 NrDL/M1 = 0.164
 M0D/M1 = 0.134
 N0DL/M1 = -0.052

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-1	1	1	-1
Normal Pressure Stress	130	135	130	135
Outplane Stress Summary	130	136	134	132

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	65	65	65	65
Inplane Stress Summary	64	65	73	58

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	130	136	134	132

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0



Outplane Bending (Mc)	-2	2	2	-2
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65	65	65	65

Outplane Stress Summary	65	66	68	61
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-3	3	3	-3
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130	135	130	135

Inplane Stress Summary	129	136	138	128
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	1	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	129	136	138	128
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	8	2	8

Hoop Stress Summary	3	9	3	9
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-2	-1	14	-3
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	6	10	14	12

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	8	2	8
Hoop Stress Summary	3	9	3	8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-2	0	15	-3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	6	10	15	12

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	130	135	130	135	130	135	130	135
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	2	-2	0	0	6	-6
Long. Pm (SUS)	65	65	65	65	65	65	65	65
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	6	-6	-1	1	3	-3
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	130.0	135.0	130.0	135.0	130.0	135.0	130.0	135.0
Pm+Pl (SUS)	130.0	135.0	130.0	135.0	130.0	135.0	130.0	135.0
Pm+Pl+Q (Total)	130.0	135.0	132.0	133.0	130.0	135.0	136.0	129.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	135.00	138.00	Passed
Pm+Pl (SUS)	135.00	207.00	Passed
Pm+Pl+Q (TOTAL)	136.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	8	2	8	2	8	2	8
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	0	0	8	-8	-1	1	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	2.0	8.0	2.0	8.0	2.0	8.0	2.0	8.0
Pm+Pl (SUS)	4.0	10.0	4.0	8.0	4.0	10.0	4.0	8.0
Pm+Pl+Q (Total)	4.0	10.0	12.0	12.0	5.0	9.0	13.0	13.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	10.00	207.00	Passed
Pm+Pl+Q (TOTAL)	13.00	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K32"

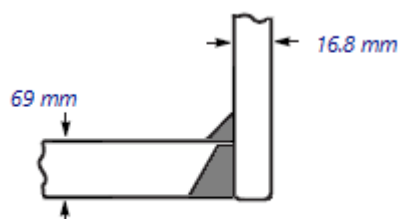
From: 50

Pressure for Reinforcement Calculations	P	4.9521	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1200.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	69.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.95)$
 $= 66.9691 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([4.95 / (115.0 \cdot 1.0)] - 1))$
 $= 1.1617 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	220.4000	mm
Parallel to Vessel Wall	Rn+tn+tt	110.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K32".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.1617	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 67.9691	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 67.9691	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(67.9691, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1617, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K32"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.8000	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4880 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.4880 mm



This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 110.20 mm

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Input, Nozzle Desc: K42"

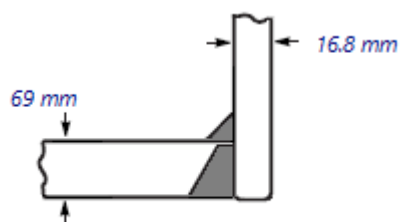
From: 50

Pressure for Reinforcement Calculations	P	4.9588	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		500.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	69.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.96 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.96)$
 $= 67.0610 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([4.96 / (115.0 \cdot 1.0)] - 1))$
 $= 1.1633 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	220.4000	mm
Parallel to Vessel Wall	Rn+tn+tt	110.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K42".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.1633	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 68.0610	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 68.0610	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(68.061, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1633, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K42"

Intermediate Calc. for nozzle/shell Welds Tmin 15.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4880 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.4880 mm



This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 110.20 mm

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Input, Nozzle Desc: K52"

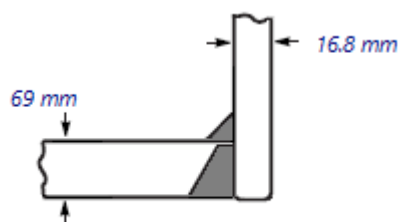
From: 50

Pressure for Reinforcement Calculations	P	4.9540	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1000.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		200.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	69.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.95)$
 $= 66.9953 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([4.95 / (115.0 \cdot 1.0)] - 1))$
 $= 1.1621 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	220.4000	mm
Parallel to Vessel Wall	Rn+tn+tt	110.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K52".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.1621	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 67.9953	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 67.9953	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(67.9953, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1621, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K52"

Intermediate Calc. for nozzle/shell Welds Tmin 15.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4880 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.4880 mm

PV Elite 28 U2 Licensee: SL User
FileName : 142-D1-D2-Rev 01
Nozzle Calcs.: K5_2"

Nozl: 64 5:12pm Aug 12,2026

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Input, Nozzle Desc: K62"

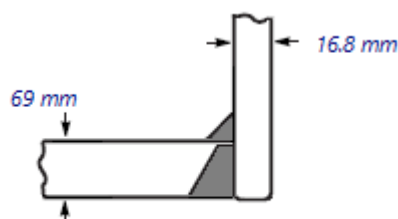
From: 50

Pressure for Reinforcement Calculations	P	4.9607	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		200.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	69.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.96 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.96)$
 $= 67.0872 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([4.96 / (115.0 \cdot 1.0)] - 1))$
 $= 1.1637 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	220.4000	mm
Parallel to Vessel Wall	Rn+tn+tt	110.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K62".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.1637	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 68.0872	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 68.0872	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(68.0872, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1637, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K62"

Intermediate Calc. for nozzle/shell Welds Tmin 15.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4880 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.4880 mm

PV Elite 28 U2 Licensee: SL User
FileName : 142-D1-D2-Rev 01
Nozzle Calcs.: K6_2"

Nozl: 65 5:12pm Aug 12, 2026

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Input, Nozzle Desc: K72"

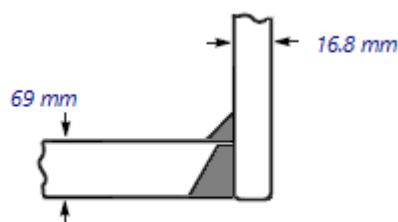
From: 50

Pressure for Reinforcement Calculations	P	4.9540	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1000.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		220.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	69.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K72"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.95)$
 $= 66.9953 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([4.95 / (115.0 \cdot 1.0)] - 1))$
 $= 1.1621 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	220.4000	mm
Parallel to Vessel Wall	Rn+tn+tt	110.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K72".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.1621	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 67.9953	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 67.9953	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(67.9953, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1621, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K72"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.8000	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4880 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.4880 mm

PV Elite 28 U2 Licensee: SL User
FileName : 142-D1-D2-Rev 01
Nozzle Calcs.: K7_2"

Nozl: 66 5:12pm Aug 12, 2026

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Input, Nozzle Desc: K82"

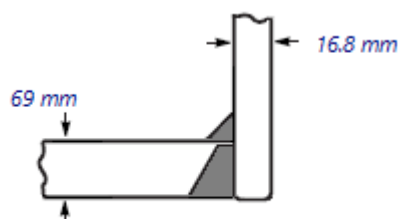
From: 50

Pressure for Reinforcement Calculations	P	4.9607	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		220.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	69.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K82"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.96 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.96)$
 $= 67.0872 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([4.96 / (115.0 \cdot 1.0)] - 1))$
 $= 1.1637 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	220.4000	mm
Parallel to Vessel Wall	Rn+tn+tt	110.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K82".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.1637	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 68.0872	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 68.0872	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(68.0872, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1637, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K82"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.8000	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4880 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.4880 mm

PV Elite 28 U2 Licensee: SL User
FileName : 142-D1-D2-Rev 01
Nozzle Calcs.: K8_2"

Nozl: 67 5:12pm Aug 12, 2026

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Input, Nozzle Desc: K92"

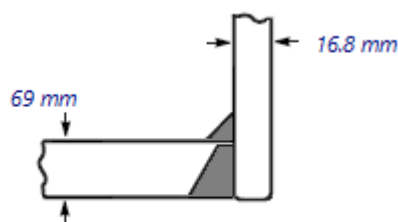
From: 50

Pressure for Reinforcement Calculations	P	4.9540	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1000.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		240.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	69.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K92"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.95)$
 $= 66.9953 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([4.95 / (115.0 \cdot 1.0)] - 1))$
 $= 1.1621 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	220.4000	mm
Parallel to Vessel Wall	Rn+tn+tt	110.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K92".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.1621	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 67.9953	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 67.9953	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(67.9953, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1621, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K92"

Intermediate Calc. for nozzle/shell Welds Tmin 15.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4880 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.4880 mm

PV Elite 28 U2 Licensee: SL User
FileName : 142-D1-D2-Rev 01
Nozzle Calcs.: K9_2"

Nozl: 68 5:12pm Aug 12, 2026

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Input, Nozzle Desc: K102"

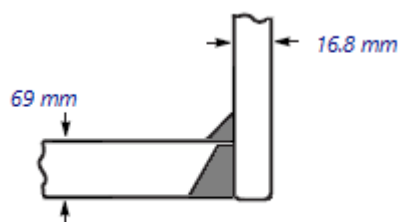
From: 50

Pressure for Reinforcement Calculations	P	4.9607	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		240.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	69.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K102"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.96 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.96)$
 $= 67.0872 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([4.96 / (115.0 \cdot 1.0)] - 1))$
 $= 1.1637 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	220.4000	mm
Parallel to Vessel Wall	Rn+tn+tt	110.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K102".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.1637	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 68.0872	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 68.0872	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(68.0872, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1637, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K102"

Intermediate Calc. for nozzle/shell Welds Tmin 15.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4880 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.4880 mm

PV Elite 28 U2 Licensee: SL User
FileName : 142-D1-D2-Rev 01
Nozzle Calcs.: K10_2"

Nozl: 69 5:12pm Aug 12, 2026

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Input, Nozzle Desc: M124"

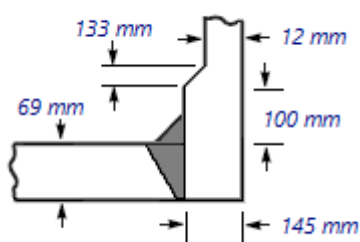
From: 50

Pressure for Reinforcement Calculations	P	4.9521	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1200.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	133.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		145.0000	mm
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		69.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M124"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.95 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.95)$
= 66.9691 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4.95 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 4.95)$
= 10.7830 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 133.0/0.5773
= 230.3629 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

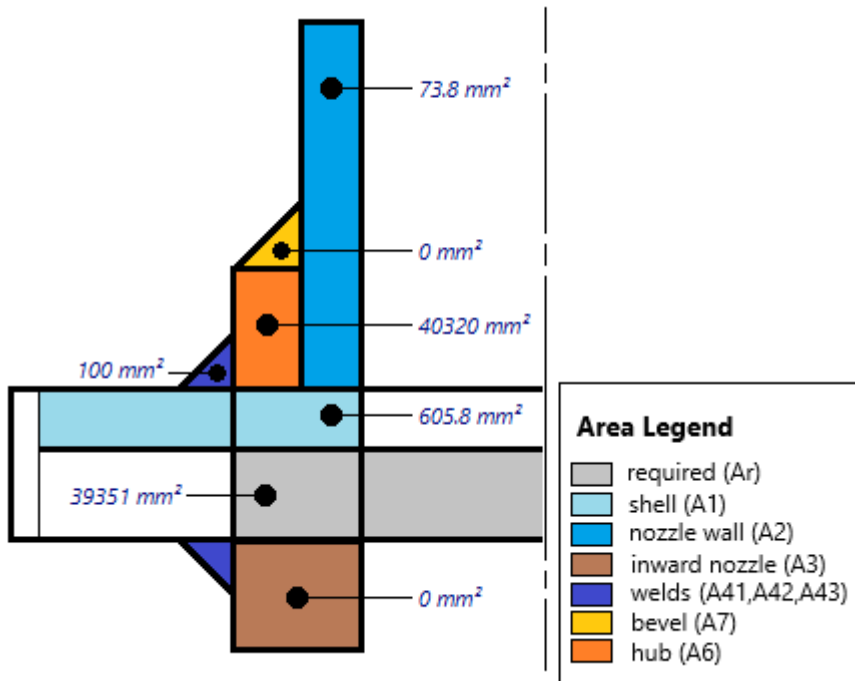
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1175.2001	mm
Parallel to Vessel Wall, opening length	d	587.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	170.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	39351.023	NA	NA
Area in Shell	A1	605.780	NA	NA
Area in Nozzle Wall	A2	73.794	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	40320.008	NA	NA
TOTAL AREA AVAILABLE	Atot	41099.582	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (587.6 \cdot 66.9691 \cdot 1.0 + 2 \cdot 144.0 \cdot 66.9691 \cdot 1.0 \cdot (1 - 1.0))$
 $= 39351.023 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 587.6(1.0 \cdot 68.0 - 1.0 \cdot 66.9691) - 2 \cdot 144.0$
 $(1.0 \cdot 68.0 - 1.0 \cdot 66.9691) \cdot (1 - 1.0)$
 $= 605.780 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 170.0)(11.0 - 10.78)1.0$
 $= 73.794 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr_2$$

$$= (2 \cdot \min(170.0, 300.0, 100.0)) \cdot (145.0 - 12.0) \cdot 1.0$$

$$= 40320.008 \text{ mm}^2 \text{ Includes Bevel Area of } 13720.008 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 11.7830 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 11.7830 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 12.0$, $t_r = 10.783$, $c = 1.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1°C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-4 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 69.0$, $t_r = 66.969$, $c = 1.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.985$, Temp. Reduction = 1°C

Min Metal Temp. w/o impact per UCS-66, Curve A	43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 42°C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 4.95/10.20 = 0.485$$

Weld Size Calculations, Description: M124"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (39351.0234 - 605.7797 + 2 \cdot 144.0 \cdot 1.0 \cdot (1.0 \cdot 68.0 - 66.9691)) \cdot 138)$$

$$= 5387262.00 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$$

$$= (40393.8008 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 5587568.50 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$



$$= (40393.8008 + 0.0 + 100.0 + (19584.0)) \cdot 138$$
$$= 8289881.50 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$
$$= (40393.8008 + 0.0 + 100.0 + 0.0 + (19584.0)) \cdot 138$$
$$= 8289881.50 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$
$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$$
$$= 647434. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$
$$= (3.1416 \cdot 299.3) \cdot (145.0 - 1.0) \cdot 0.7 \cdot 138$$
$$= 13078540. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$
$$= (3.1416/2.0) \cdot 609.6 \cdot (69.0 - 1.0) \cdot 0.74 \cdot 138$$
$$= 6648867. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (647434 + 13078540) = 13725973 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (647434 + 0 + 6648867 + 0) = 7296301 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (647434 + 6648867 + 0) = 7296301 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 13725973 N , must exceed W = 5387262 N or W1 = 5587568 N
Path 2-2 = 7296300 N , must exceed W = 5387262 N or W2 = 8289881 N
Path 3-3 = 7296300 N , must exceed W = 5387262 N or W3 = 8289881 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 25.6329 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.6330 mm

This nozzle or its diameter limit intersects weld seam # [4] on this element. The distance between the nozzle and seam is, 529.17 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 587.60 mm

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Input, Nozzle Desc: K12"

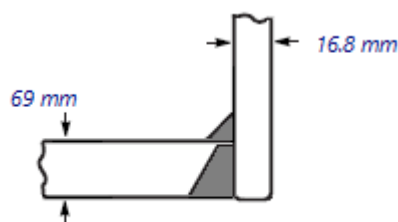
From: 60

Pressure for Reinforcement Calculations	P	4.9507	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3800.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	70.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.95)$
 $= 66.9494 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([4.95 / (115.0 \cdot 1.0)] - 1))$
 $= 1.1173 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	220.4000	mm
Parallel to Vessel Wall	Rn+tn+tt	110.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	42.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K12".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.1173	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 67.9494	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 67.9494	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(67.9494, 2.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.1173, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4.95 / 10.20 = 0.485$

Weld Size Calculations, Description: K12"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.8000	mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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PV Elite 28 U2 Licensee: SL User
FileName : 142-D1-D2-Rev 01
Nozzle Calcs.: K1_2"

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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4880 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.4880 mm

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Input, Nozzle Desc: K22"

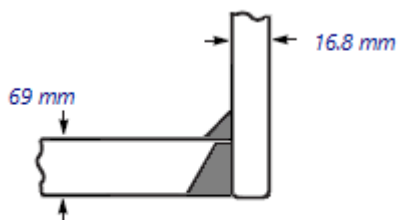
From: 60

Pressure for Reinforcement Calculations	P	4.9507	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3650.00	mm
Shell Finished (Minimum) Thickness	t	69.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3400.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	MPa
Allowable Stress At Ambient	Sna	115.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	70.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.95 \cdot 1826.0 / (138 \cdot 1.0 - 0.6 \cdot 4.95)$
= 66.9494 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $25.4 \left(\exp \left(\frac{4.95}{115.0 \cdot 1.0} \right) - 1 \right)$
= 1.1173 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	220.4000	mm
Parallel to Vessel Wall	Rn+tn+tt	110.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	42.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K22".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.1173	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 67.9494	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 67.9494	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[4.8, max(67.9494, 2.5)]
= 4.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.1173, 4.8)
= 4.8000 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.95 / 10.20 = 0.485$

Weld Size Calculations, Description: K22"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.8000	mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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PV Elite 28 U2 Licensee: SL User
FileName : 142-D1-D2-Rev 01
Nozzle Calcs.: K2_2"

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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4880 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.4880 mm

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Input, Nozzle Desc: N230"

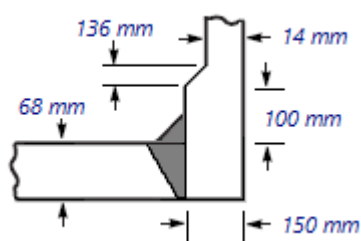
From: 70

Pressure for Reinforcement Calculations	P	4.9507	MPa
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	3650.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	68.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	136.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	68.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 30.000 in.
Actual Thickness Used in Calculation 0.551 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $4.95 \cdot 3652.0 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.95)$
= 59.1388 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4.95 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 4.95)$
= 13.4749 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 136.0/0.5773
= 235.5590 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

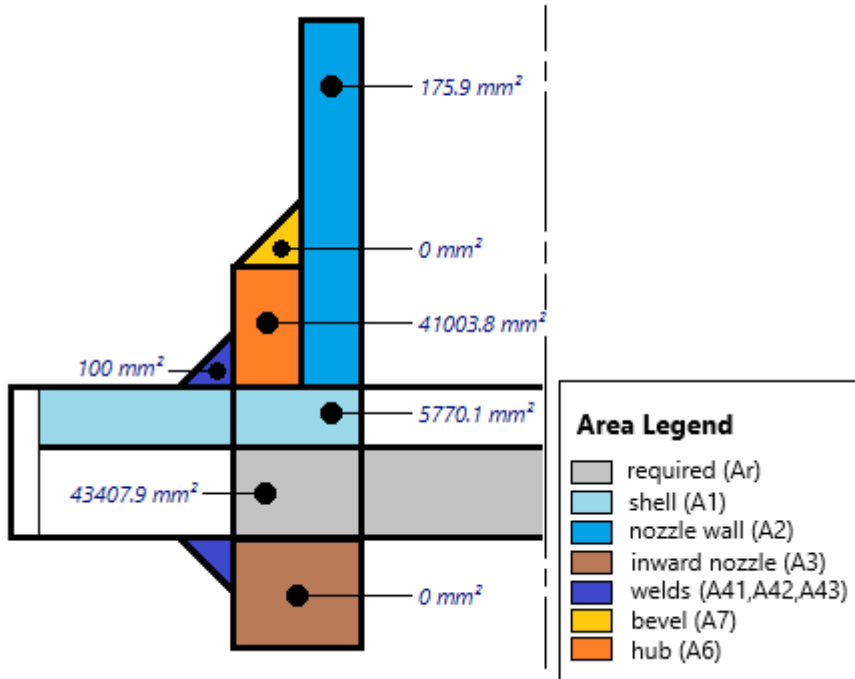
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1468.0000	mm
Parallel to Vessel Wall, opening length	d	734.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	167.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	43407.855	NA	NA
Area in Shell	A1	5770.149	NA	NA
Area in Nozzle Wall	A2	175.917	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		100.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	41003.754	NA	NA
TOTAL AREA AVAILABLE	Atot	47049.820	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (734.0 \cdot 59.1388 \cdot 1.0 + 2 \cdot 150.0 \cdot 59.1388 \cdot 1.0 \cdot (1 - 1.0))$
 $= 43407.855 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 734.0(1.0 \cdot 67.0 - 1.0 \cdot 59.139) - 2 \cdot 150.0$
 $(1.0 \cdot 67.0 - 1.0 \cdot 59.1388) \cdot (1 - 1.0)$
 $= 5770.149 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 167.5)(14.0 - 13.47)1.0$
 $= 175.917 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(167.5, 300.0, 100.0)) \cdot (150.0 - 14.0) \cdot 1.0 \\ &= 41003.754 \text{ mm}^2 \text{ Includes Bevel Area of } 13803.755 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad t_a = 13.4749 \text{ mm} \\ \text{Wall Thickness per UG16(b),} & \quad t_{r16b} = 2.5000 \text{ mm} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad t_{rb1} = 66.6962 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b1} = \max(t_{rb1}, t_{r16b}) = 66.6962 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b2} = \max(t_{rb2}, t_{r16b}) = 2.5000 \text{ mm} \\ \text{Wall Thickness per table UG-45} & \quad t_{b3} = 8.3312 \text{ mm} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ &= \min[8.331, \max(66.6962, 2.5)] \\ &= 8.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(13.4749, 8.3312) \\ &= 13.4749 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	88.0,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	257.0 MPa	Passed
Occasional	:	63.7,	Allowable	:	183.5 MPa	Passed
Shear	:	15.2,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 14.0, tr = 13.475, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.962, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	0 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 68.0, tr = 59.139, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.883, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	36 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 4.95/10.20 = 0.485

Weld Size Calculations, Description: N230"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (43407.8555 - 5770.1489 + 2 • 150.0 • 1.0 •
(1.0 • 67.0 - 59.1388))138)
= 5518889.50 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (41179.668 + 100.0 - 0.0 • 1.0) • 138
= 5696007.00 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (41179.668 + 0.0 + 100.0 + (20100.002)) • 138
= 8469522.00 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (41179.668 + 0.0 + 100.0 + 0.0 + (20100.002)) • 138
= 8469522.00 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 762.0 • 10.0 • 0.49 • 138
= 809292. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 374.0) • (150.0 - 0.0) • 0.7 • 138
= 17023660. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 762.0 • (68.0 - 1.0) • 0.74 • 138
= 8188862. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (809292 + 17023660) = 17832952 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (809292 + 0 + 8188862 + 0) = 8998154 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (809292 + 8188862 + 0) = 8998154 N

Summary of Failure Path Calculations:

Path 1-1 = 17832952N	, must exceed W = 5518889 N	or W1 = 5696007 N
Path 2-2 = 8998154 N	, must exceed W = 5518889 N	or W2 = 8469522 N
Path 3-3 = 8998154 N	, must exceed W = 5518889 N	or W3 = 8469522 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 21.8056 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 389.8058 mm

Input Echo, WRC107/537 Item 1, Description: N230" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	6570.000	mm
Vessel Thickness	Tv	68.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	1034.000	mm
Nozzle Thickness	Tn	150.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Design Internal Pressure	Dp	4.951	MPa
Include Pressure Thrust		No	

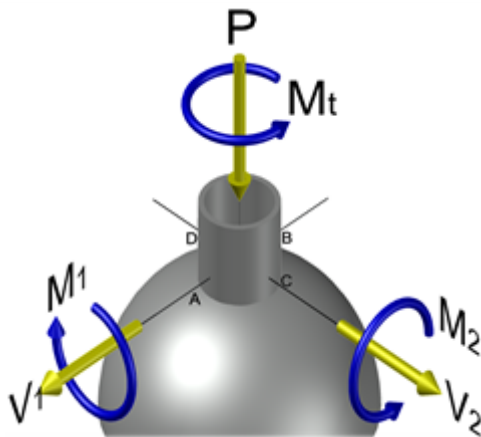
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-67473.5	N
Longitudinal Shear (SUS)	(V1) V1	48731.0	N
Circumferential Shear (SUS)	(Vc) V2	48731.0	N
Circumferential Moment (SUS)	(Mc) M1	112501976.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	112501976.0	N-mm
Torsional Moment (SUS)	Mt	129377072.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1034.0 + 2 \cdot 1.65 \cdot \sqrt{(3319.5 (68.0 - 1.0))}$$

$$= 2590.280 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-67473.5	N
Circumferential Shear	(VC)	V2	48731.0	N
Longitudinal Shear	(VL)	V1	48731.0	N
Circumferential Moment	(MC)	M1	112501976.0	N-mm
Longitudinal Moment	(ML)	M2	112501976.0	N-mm
Torsional Moment		MT	129377072.0	N-mm

Dimensionless Param: U = 1.10 TAU = 5.00 (2.95) RHO = 0.45

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03686	(A,B,C,D)
$M(x) / P$	SP 1	0.03843	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04204	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.06657	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04204	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.06657	(A,B,C,D)
$N(y) * T / P$	SP 1	0.05872	(A,B,C,D)
$M(y) / P$	SP 1	0.01100	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.05792	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.02094	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.05792	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.02094	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Rad. Bend.	P	3.5	-3.5	3.5	-3.5	3.5	-3.5	3.5	-3.5
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-21.2	21.2	21.2	-21.2
Rad. Memb.	ML	-2.2	-2.2	2.2	2.2	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-21.2	21.2	21.2	-21.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-19.4	16.1	27.5	-21.9	-19.4	16.1	27.5	-21.9



Tang. Memb. P	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tang. Bend. P	1.0	-1.0	1.0	-1.0	1.0	-1.0	1.0	-1.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-3.1	-3.1	3.1	3.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-6.7	6.7	6.7	-6.7
Tang. Memb. ML	-3.1	-3.1	3.1	3.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-6.7	6.7	6.7	-6.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-7.9	3.5	11.6	-3.7	-7.9	3.5	11.6	-3.7
Shear VC	0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Tot. Shear	1.6	1.6	0.7	0.7	0.7	0.7	1.6	1.6
Str. Int.	19.6	16.3	27.5	21.9	19.5	16.1	27.6	22.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	121.4	121.4	121.4	121.4	121.4	121.4	121.4	121.4
Rad.	Pl (SUS)	-1.7	-1.7	2.8	2.8	-1.7	-1.7	2.8	2.8
Rad.	Q (SUS)	-17.8	17.8	24.7	-24.7	-17.8	17.8	24.7	-24.7
Long.	Pm (SUS)	121.4	121.4	121.4	121.4	121.4	121.4	121.4	121.4
Long.	Pl (SUS)	-2.2	-2.2	4.0	4.0	-2.2	-2.2	4.0	4.0
Long.	Q (SUS)	-5.7	5.7	7.7	-7.7	-5.7	5.7	7.7	-7.7
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear	Q (SUS)	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
	Pm (SUS)	121.4	121.4	121.4	121.4	121.4	121.4	121.4	121.4
	Pm+Pl (SUS)	120.0	120.0	125.5	125.5	120.0	120.0	125.5	125.5
	Pm+Pl+Q (Total)	113.8	137.7	148.9	117.8	113.6	137.5	149.1	117.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.44	138.00	Passed
Pm+Pl (SUS)	125.55	207.00	Passed
Pm+Pl+Q (TOTAL)	149.07	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: D2N33"

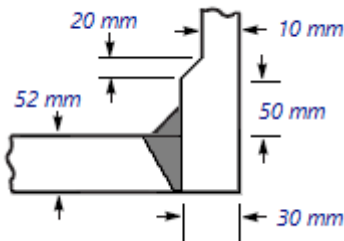
From: 110

Pressure for Reinforcement Calculations	P	4.5655	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	2900.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2000.0	mm
Centerline Distance to Tangent		1000.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2N33"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.394 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 4.57 \cdot 2906.0 \cdot 0.8982 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.57)$
 $= 43.3214 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.57 \cdot 38.1 / (138 \cdot 1.0 + 0.4 \cdot 4.57)$
 $= 1.2440 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.0 / 0.5773$
 $= 34.6410 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

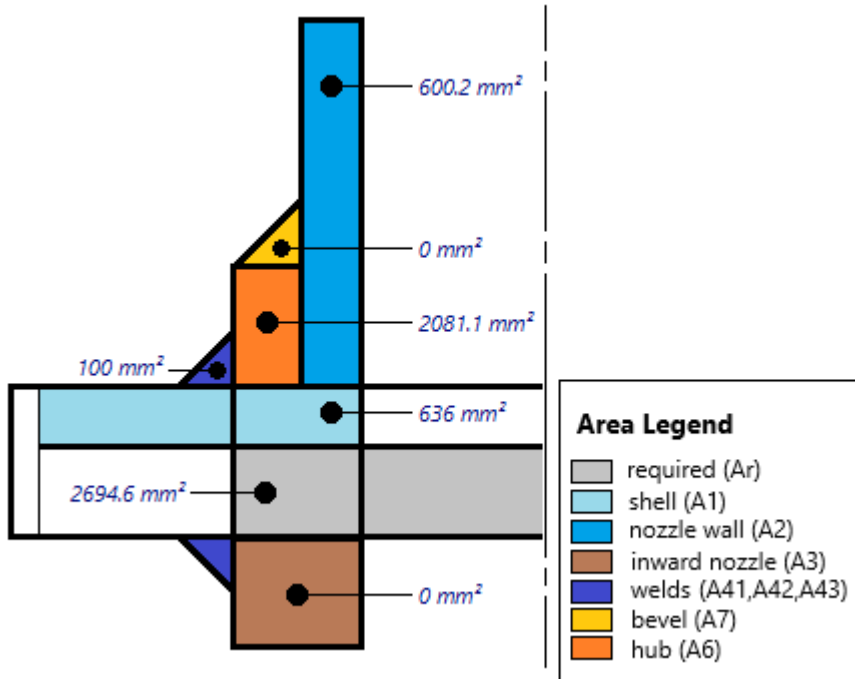
Parallel to Vessel Wall (Diameter Limit)	D1	174.2000 mm
Parallel to Vessel Wall	Rn+tn+t	87.1000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.1410 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$

= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2694.590	NA	NA
Area in Shell	A1	636.005	NA	NA
Area in Nozzle Wall	A2	600.246	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	2081.057	NA	NA
TOTAL AREA AVAILABLE	Atot	3417.308	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (62.2 \cdot 43.3214 \cdot 1.0 + 2 \cdot 27.0 \cdot 43.3214 \cdot 1.0 \cdot (1 - 1.0))$
 $= 2694.590 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 112.0(1.0 \cdot 49.0 - 1.0 \cdot 43.321) - 2 \cdot 27.0$
 $(1.0 \cdot 49.0 - 1.0 \cdot 43.3214) \cdot (1 - 1.0)$
 $= 636.005 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 52.14)(7.0 - 1.24)1.0$
 $= 600.246 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$



= 100.000 mm²

Area Available in the Hub Section [A6]:

= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(52.1,150.0,50.0)) • (30.0 - 10.0) • 1.0)
= 2081.057 mm² Includes Bevel Area of 81.057 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2440 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 51.0974 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 51.0974 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[7.8, max(51.0974, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(4.244, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 57.6,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 287.4 MPa	Passed
Occasional	: 9.1,	Allowable	: 183.5 MPa	Passed
Shear	: 38.3,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 1.244, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.178, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 30.0, tr = 1.244, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.046, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 4.57/10.20 = 0.448

Weld Size Calculations, Description: D2N33"

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (2694.5898 - 636.0048 + 2 \cdot 27.0 \cdot 1.0 \cdot (1.0 \cdot 49.0 - 43.3214)) \cdot 138) \\
 &= 326374.06 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (2681.303 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\
 &= 383787.16 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (2681.303 + 0.0 + 100.0 + (2646.0)) \cdot 138 \\
 &= 748904.06 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (2681.303 + 0.0 + 100.0 + 0.0 + (2646.0)) \cdot 138 \\
 &= 748904.06 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 76.2 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 80931. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 34.6) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 138 \\
 &= 283485. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 76.2 \cdot (52.0 - 3.0) \cdot 0.74 \cdot 138 \\
 &= 598887. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (80931 + 283485) = 364416 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (80931 + 0 + 598887 + 0) = 679818 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (80931 + 598887 + 0) = 679818 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 364415 N	, must exceed W = 326374 N	or W1 = 383787 N
Path 2-2 = 679817 N	, must exceed W = 326374 N	or W2 = 748904 N
Path 3-3 = 679817 N	, must exceed W = 326374 N	or W3 = 748904 N

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 5 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.2738 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 202.2737 mm

Input Echo, WRC107/537 Item 1, Description: D2N33" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	5220.000	mm
Vessel Thickness	Tv	52.000	mm
Design Temperature	T1	110.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	116.200	mm
Nozzle Thickness	Tn	30.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Design Internal Pressure	Dp	4.566	MPa
Include Pressure Thrust		No	

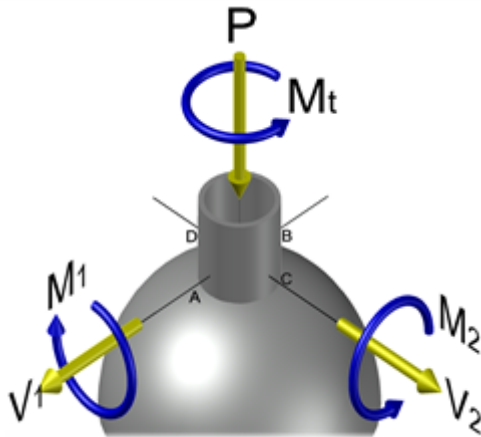
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-5997.6	N
Longitudinal Shear (SUS)	(V1) V1	3998.4	N
Circumferential Shear (SUS)	(Vc) V2	3998.4	N
Circumferential Moment (SUS)	(Mc) M1	895959.6	N-mm
Longitudinal Moment (SUS)	(Ml) M2	895959.6	N-mm
Torsional Moment (SUS)	Mt	1279929.6	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 116.2 + 2 \cdot 1.65 \cdot \sqrt{(2637.5 (52.0 - 3.0))}$$

$$= 1302.537 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-5997.6	N
Circumferential Shear	(VC)	V2	3998.4	N
Longitudinal Shear	(VL)	V1	3998.4	N
Circumferential Moment	(MC)	M1	895959.6	N-mm
Longitudinal Moment	(ML)	M2	895959.6	N-mm
Torsional Moment		MT	1279929.6	N-mm

Dimensionless Param: U = 0.16 TAU = 5.00 (1.65) RHO = 1.81

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.15264	(A,B,C,D)
$M(x) / P$	SP 2	0.14005	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.47520	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.54578	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.47520	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.54578	(A,B,C,D)
$N(y) * T / P$	SP 2	0.21328	(A,B,C,D)
$M(y) / P$	SP 2	0.11841	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.07351	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.68415	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.07351	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.68415	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Rad. Bend.	P	2.1	-2.1	2.1	-2.1	2.1	-2.1	2.1	-2.1
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-3.4	3.4	3.4	-3.4
Rad. Memb.	ML	-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-3.4	3.4	3.4	-3.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.4	1.2	6.4	-4.6	-1.4	1.2	6.4	-4.6



Tang. Memb. P	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Tang. Bend. P	1.8	-1.8	1.8	-1.8	1.8	-1.8	1.8	-1.8
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-4.3	4.3	4.3	-4.3
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-4.3	4.3	4.3	-4.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-2.0	2.9	6.6	-5.4	-2.0	2.9	6.6	-5.4
Shear VC	0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Tot. Shear	1.7	1.7	0.8	0.8	0.8	0.8	1.7	1.7
Str. Int.	3.4	4.0	7.3	5.9	2.6	3.2	8.2	6.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		121.8	121.8	121.8	121.8	121.8	121.8	121.8	121.8
Rad. Pl (SUS)		-0.1	-0.1	0.9	0.9	-0.1	-0.1	0.9	0.9
Rad. Q (SUS)		-1.3	1.3	5.5	-5.5	-1.3	1.3	5.5	-5.5
Long. Pm (SUS)		121.8	121.8	121.8	121.8	121.8	121.8	121.8	121.8
Long. Pl (SUS)		0.5	0.5	0.6	0.6	0.5	0.5	0.6	0.6
Long. Q (SUS)		-2.5	2.5	6.0	-6.0	-2.5	2.5	6.0	-6.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear Q (SUS)		1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Pm (SUS)		121.8	121.8	121.8	121.8	121.8	121.8	121.8	121.8
Pm+Pl (SUS)		122.5	122.5	123.0	123.0	122.5	122.5	123.0	123.0
Pm+Pl+Q (Total)		121.7	125.7	129.1	117.6	120.9	125.0	130.0	118.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.76	138.00	Passed
Pm+Pl (SUS)	122.97	207.00	Passed
Pm+Pl+Q (TOTAL)	129.95	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: D2N1 24"

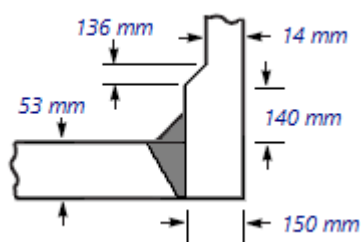
From: 120

Pressure for Reinforcement Calculations	P	4.5507	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9525.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	140.0000	mm
Height of Beveled Transition	L'	136.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	2700.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2N1 24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.551 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 48.8814 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4.55 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 4.55)$
= 9.9203 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 136.0/0.5773
= 235.5590 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

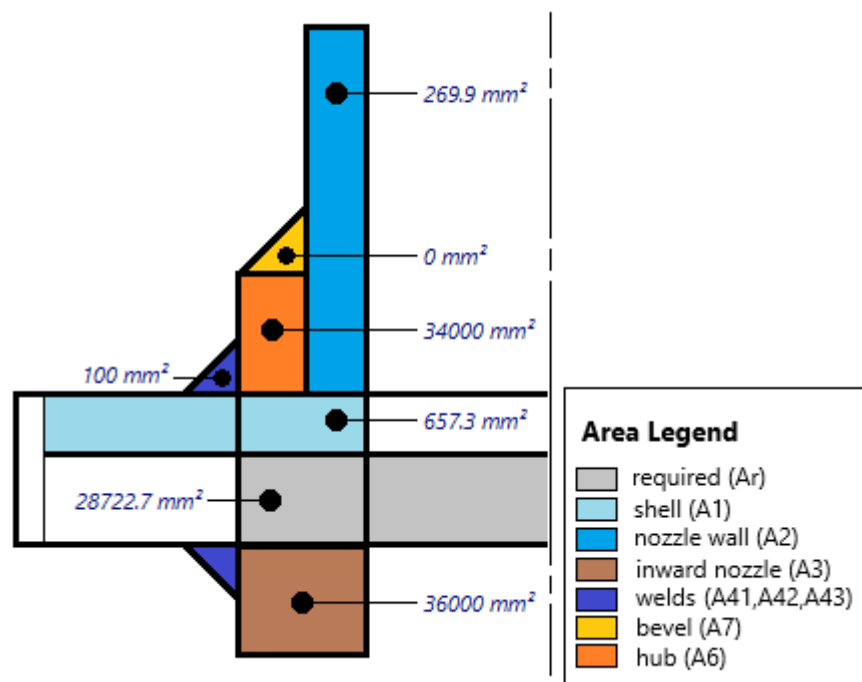
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1175.2001	mm
Parallel to Vessel Wall, opening length	d	587.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	125.0000	mm
Normal to Vessel Wall, Inward		20.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	28722.719	NA	NA
Area in Shell	A1	657.282	NA	NA
Area in Nozzle Wall	A2	269.935	NA	NA
Area in Inward Nozzle	A3	36000.000	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	33999.996	NA	NA
TOTAL AREA AVAILABLE	Atot	71027.211	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (587.6 \cdot 48.8814 \cdot 1.0 + 2 \cdot 147.0 \cdot 48.8814 \cdot 1.0 \cdot (1 - 1.0))$
 $= 28722.719 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 587.6(1.0 \cdot 50.0 - 1.0 \cdot 48.8814) - 2 \cdot 147.0$
 $(1.0 \cdot 50.0 - 1.0 \cdot 48.8814) \cdot (1 - 1.0)$
 $= 657.282 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 125.0)(11.0 - 9.92)1.0$
 $= 269.935 \text{ mm}^2$

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 144.0 \cdot (125.0) \cdot 1.0$
 $= 36000.000 \text{ mm}^2$



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 100.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(125.0, 300.0, 140.0)) \cdot (150.0 - 14.0) \cdot 1.0$$

$$= 33999.996 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 12.9203 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 51.8814 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 51.8814 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(51.8814, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(12.9203, 11.3312)$$

$$= 12.9203 \text{ mm}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 80.9,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 264.1 MPa	Passed
Occasional	: 59.7,	Allowable	: 183.5 MPa	Passed
Shear	: 16.2,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 14.0, tr = 9.92, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.902, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-4 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 53.0, tr = 48.881, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.978, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	38 °C
Min Metal Temp. at Required thickness (UCS 66.1)	36 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 4.55/5.10 = 0.892

Weld Size Calculations, Description: D2N1_24"



Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (28722.7188 - 657.2822 + 2 \cdot 147.0 \cdot 1.0 \cdot (1.0 \cdot 50.0 - 48.8814)) \cdot 138)$
 $= 3918080.25 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (34269.9336 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 4742647.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (34269.9336 + 36000.0 + 100.0 + (14700.0)) \cdot 138$
 $= 11738651.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (34269.9336 + 36000.0 + 100.0 + 0.0 + (14700.0)) \cdot 138$
 $= 11738651.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 647445. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 299.3) \cdot (150.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 13351009. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (53.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 4888872. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (647445 + 13351009) = 13998454 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (647445 + 0 + 4888873 + 0) = 5536318 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (647445 + 4888873 + 0) = 5536318 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 13998454N	, must exceed W = 3918080 N	or W1 = 4742647 N
Path 2-2 = 5536317 N	, must exceed W = 3918080 N	or W2 = 11738651 N
Path 3-3 = 5536317 N	, must exceed W = 3918080 N	or W3 = 11738651 N



Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 32.3974 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 3053.0000 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 587.60 mm

Input Echo, WRC297 Item 1, Description: D2N1 24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2900.000	mm
Shell Thickness	53.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	881.600	mm
Nozzle Thickness	150.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc 138.00	MPa
Nozzle Hot S.I. Allowable	SNmh 138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	4.55	MPa
Radial Load	P	-44982.18	N
Circumferential Shear	Vc	29988.19	N
Longitudinal Shear	Vl	29988.19	N
Circumferential Moment	Mc	0.504E+08	N-mm
Longitudinal Moment	Ml	0.504E+08	N-mm
Torsional Moment	Mt	0.720E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.877) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned} &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\ &= 881.6 + 2 \cdot 1.65 \cdot \sqrt{(1478.0(53.0 - 3.0))} \\ &= 1778.690 \text{ mm} \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2956.000 mm
Nozzle Outside Diameter	(d) =	881.600 mm



Vessel Thickness used (T) = 50.000 mm
 Nozzle Thickness used (t) = 147.000 mm
 T / t = 0.340
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 2.293$$

Nr/P = 0.065
 Mr/P = 0.130
 M0/P = 0.039
 N0/P = 0.082
 MrD/Mc = 0.280
 NrDL/Mc = 0.096
 M0d/Mc = 0.086
 N0DL/Mc = 0.079
 MrD/Ml = 0.103
 NrDL/Ml = 0.070
 M0D/Ml = 0.031
 N0DL/Ml = 0.104

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-4	4	4	-4
Normal Pressure Stress	130	134	130	134
Outplane Stress Summary	129	133	142	129

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-14	14	14	-14
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	64	64	95	39

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	129	133	142	129

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----



(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-11	11	11	-11
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64
Outplane Stress Summary	57	72	84	52

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-38	38	38	-38
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130	134	130	134
Inplane Stress Summary	104	157	185	85

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	1	1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	104	157	185	85

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	14	8	14
Hoop Stress Summary	7	13	12	18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	3	3	7	3

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	7	13	12	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	14	8	14
Hoop Stress Summary	8	13	12	17

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	1	6	10	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	8	13	12	17

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130	134	130	134	130	134	130	134
Circ. Pl (SUS)	-1	-1	3	3	-1	-1	3	3
Circ. Q (SUS)	0	0	8	-8	-25	25	51	-51
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	0	0	2	2	0	0	2	2
Long. Q (SUS)	-1	1	27	-27	-7	7	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	130.0	134.0	130.0	134.0	130.0	134.0	130.0	134.0
Pm+Pl (SUS)	129.0	133.0	133.0	137.0	129.0	133.0	133.0	137.0
Pm+Pl+Q (Total)	129.0	133.0	141.0	129.0	104.0	158.0	184.0	86.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	134.00	138.00	Passed
Pm+Pl (SUS)	137.00	207.00	Passed
Pm+Pl+Q (TOTAL)	184.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	14	8	14	8	14	8	14
Circ. Pl (SUS)	-1	-1	3	3	0	0	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	2	-2	-2	2	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	8.0	14.0	8.0	14.0	8.0	14.0	8.0	14.0
Pm+Pl (SUS)	7.0	13.0	11.0	17.0	8.0	14.0	10.0	16.0
Pm+Pl+Q (Total)	7.0	13.0	11.0	17.0	8.0	14.0	10.0	16.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	138.00	Passed
Pm+Pl (SUS)	17.00	207.00	Passed
Pm+Pl+Q (TOTAL)	17.00	414.00	Passed



*Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.*

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Input, Nozzle Desc: D2K12"

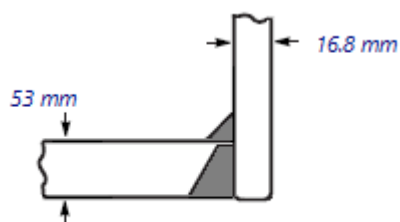
From: 120

Pressure for Reinforcement Calculations	P	4.5507	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9500.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		250.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2K12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 48.8814 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 0.9554 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D2K12".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9554	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.8814	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.8814	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.8814, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9554, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.55 / 5.10 = 0.892$

Weld Size Calculations, Description: D2K12"



Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6131 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6131 mm

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Input, Nozzle Desc: D2K22"

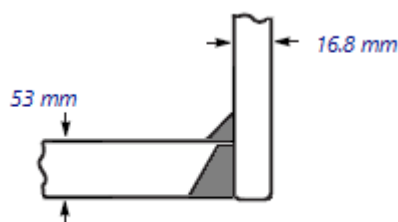
From: 120

Pressure for Reinforcement Calculations	P	4.5547	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8250.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		250.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2K22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 48.9254 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 0.9563 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D2K22".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9563	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.9254	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.9254	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.9254, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9563, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.55 / 5.10 = 0.893$

Weld Size Calculations, Description: D2K22"



Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6131 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6131 mm

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Input, Nozzle Desc: D2K32"

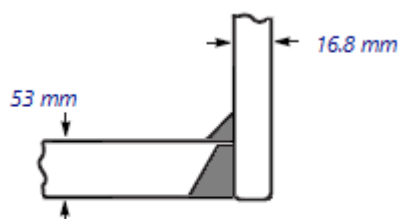
From: 120

Pressure for Reinforcement Calculations	P	4.5507	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9500.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		260.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2K32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
 $= 48.8814 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.55 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
 $= 0.9554 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D2K32".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9554	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.8814	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.8814	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(51.8814, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.9554, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4.55 / 5.10 = 0.892$

Weld Size Calculations, Description: D2K32"



Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6131 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6131 mm

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Input, Nozzle Desc: D2K42"

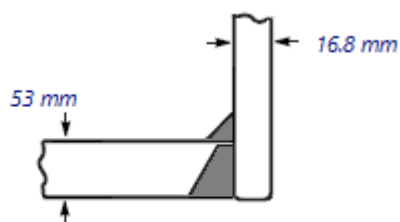
From: 120

Pressure for Reinforcement Calculations	P	4.5547	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8250.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		260.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2K42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 48.9254 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 0.9563 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D2K42".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9563	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.9254	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.9254	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.9254, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9563, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.55 / 5.10 = 0.893$

Weld Size Calculations, Description: D2K42"



Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6131 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6131 mm

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Input, Nozzle Desc: D2K52"

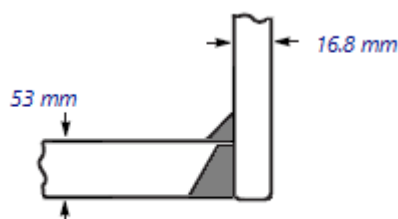
From: 120

Pressure for Reinforcement Calculations	P	4.5507	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9500.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2K52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 48.8814 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 0.9554 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D2K52".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9554	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.8814	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.8814	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.8814, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9554, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.55 / 5.10 = 0.892$

Weld Size Calculations, Description: D2K52"



Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6131 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6131 mm

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Input, Nozzle Desc: D2K62"

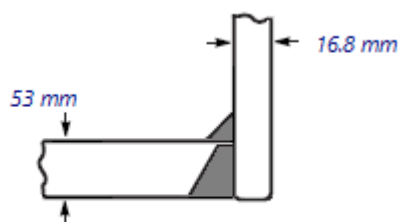
From: 120

Pressure for Reinforcement Calculations	P	4.5547	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8250.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2K62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 48.9254 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 0.9563 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D2K62".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9563	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.9254	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.9254	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.9254, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9563, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.55 / 5.10 = 0.893$

Weld Size Calculations, Description: D2K62"



Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6131 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6131 mm

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Input, Nozzle Desc: D2K72"

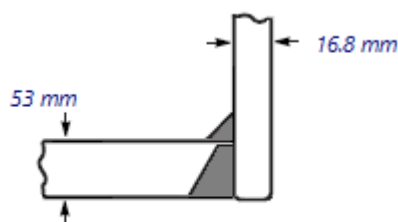
From: 120

Pressure for Reinforcement Calculations	P	4.5507	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9500.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		280.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2K72"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 48.8814 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 0.9554 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D2K72".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9554	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.8814	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.8814	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.8814, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9554, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.55 / 5.10 = 0.892$

Weld Size Calculations, Description: D2K72"



Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6131 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6131 mm

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Input, Nozzle Desc: D2K82"

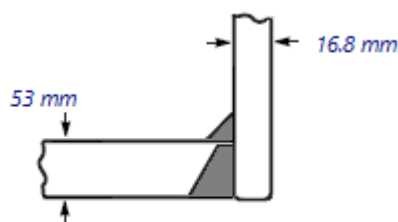
From: 120

Pressure for Reinforcement Calculations	P	4.5547	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8250.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		280.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2K82"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
 $= 48.9254 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.55 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
 $= 0.9563 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D2K62".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9563	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.9254	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.9254	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(51.9254, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.9563, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4.55 / 5.10 = 0.893$

Weld Size Calculations, Description: D2K82"



Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6131 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6131 mm

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Input, Nozzle Desc: D2K92"

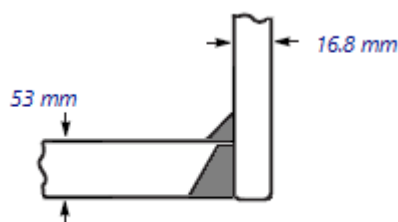
From: 120

Pressure for Reinforcement Calculations	P	4.5507	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		10500.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2K92"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 48.8814 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 0.9554 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D2K9_2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9554	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.8814	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.8814	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.8814, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9554, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.55 / 5.10 = 0.892$

Weld Size Calculations, Description: D2K9_2"



Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6131 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6131 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 92.16 mm

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Input, Nozzle Desc: D2K102"

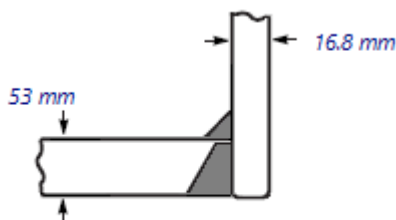
From: 120

Pressure for Reinforcement Calculations	P	4.5507	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		10300.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2K102"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 48.8814 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 0.9554 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	184.3280	mm
Parallel to Vessel Wall	Rn+tn+tt	92.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D2K10_2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.9554	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 51.8814	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 51.8814	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(51.8814, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.9554, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.55 / 5.10 = 0.892$

Weld Size Calculations, Description: D2K10_2"



Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6131 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.6131 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 92.16 mm

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Input, Nozzle Desc: D2M124"

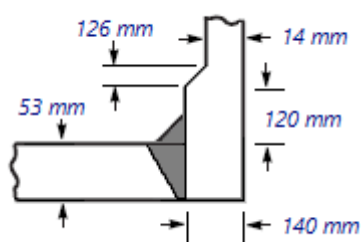
From: 120

Pressure for Reinforcement Calculations	P	4.5507	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	53.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8750.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	120.0000	mm
Height of Beveled Transition	L'	126.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		140.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		53.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D2M124"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.551 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.55 \cdot 1453.0 / (138 \cdot 1.0 - 0.6 \cdot 4.55)$
= 48.8814 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4.55 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 4.55)$
= 9.9203 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 126.0/0.5773
= 218.2384 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

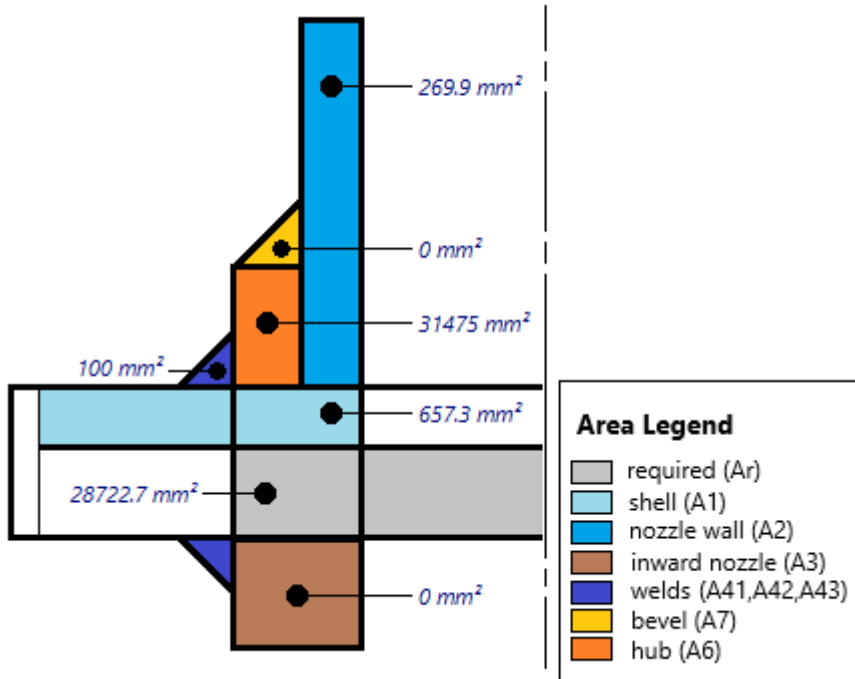
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1175.2001	mm
Parallel to Vessel Wall, opening length	d	587.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	125.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	28722.719	NA	NA
Area in Shell	A1	657.282	NA	NA
Area in Nozzle Wall	A2	269.935	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	31475.016	NA	NA
TOTAL AREA AVAILABLE	Atot	32502.232	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (587.6 \cdot 48.8814 \cdot 1.0 + 2 \cdot 137.0 \cdot 48.8814 \cdot 1.0 \cdot (1 - 1.0))$
 $= 28722.719 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 587.6(1.0 \cdot 50.0 - 1.0 \cdot 48.8814) - 2 \cdot 137.0$
 $(1.0 \cdot 50.0 - 1.0 \cdot 48.8814) \cdot (1 - 1.0)$
 $= 657.282 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 125.0)(11.0 - 9.92)1.0$
 $= 269.935 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(125.0, 300.0, 120.0)) \cdot (140.0 - 14.0) \cdot 1.0 \\ &= 31475.016 \text{ mm}^2 \text{ Includes Bevel Area of } 1235.014 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 12.9203 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 12.9203 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 14.0$, $t_r = 9.92$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.902$, Temp. Reduction = $5 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	2 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-4 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: A

Govrn. thk, $t_g = 53.0$, $t_r = 48.881$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.978$, Temp. Reduction = $1 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	38 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	36 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : 36 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4.55/5.10 = 0.892$

Weld Size Calculations, Description: D2M124"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (28722.7188 - 657.2822 + 2 \cdot 137.0 \cdot 1.0 \cdot (1.0 \cdot 50.0 - 48.8814)) \cdot 138) \\ &= 3914993.00 \text{ N} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_4 - (W_i - C_{an} / .707)^2 \cdot fr_2) \cdot S_v \\ &= (31744.9492 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\ &= 4394229.50 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (31744.9492 + 0.0 + 100.0 + (13700.001)) \cdot 138 \end{aligned}$$



= 6284668.50 N

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\ &= (31744.9492 + 0.0 + 100.0 + 0.0 + (13700.001)) \cdot 138 \\ &= 6284668.50 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 647445. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 299.3) \cdot (140.0 - 3.0) \cdot 0.7 \cdot 138 \\ &= 12442777. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 609.6 \cdot (53.0 - 3.0) \cdot 0.74 \cdot 138 \\ &= 4888872. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (647445 + 12442777) = 13090223 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (647445 + 0 + 4888873 + 0) = 5536318 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (647445 + 4888873 + 0) = 5536318 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 13090223N	, must exceed W = 3914993 N	or W1 = 4394229 N
Path 2-2 = 5536317 N	, must exceed W = 3914993 N	or W2 = 6284668 N
Path 3-3 = 5536317 N	, must exceed W = 3914993 N	or W3 = 6284668 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 32.3974 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 385.3974 mm

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Input, Nozzle Desc: N224"

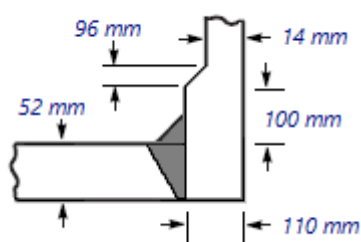
From: 130

Pressure for Reinforcement Calculations	P	4.5507	MPa
Temperature for Internal Pressure	Temp	110	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	2900.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	52.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	96.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N224"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.551 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $4.55 \cdot 2906.0 \cdot 0.8982 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.55)$
= 43.1803 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4.55 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 4.55)$
= 9.9203 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 96.0/0.5773
= 166.2769 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

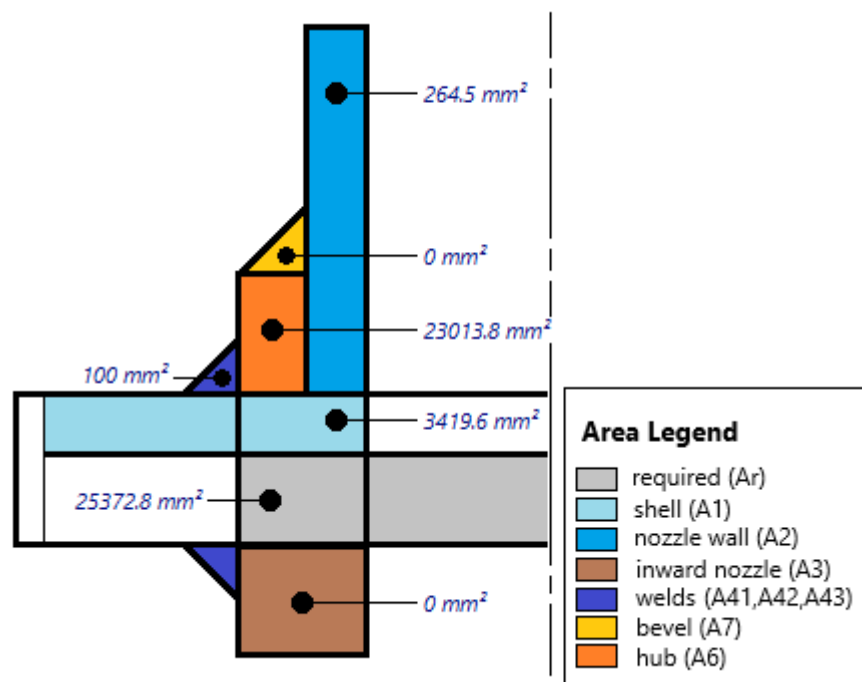
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1175.2001	mm
Parallel to Vessel Wall, opening length	d	587.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	122.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	25372.775	NA	NA
Area in Shell	A1	3419.627	NA	NA
Area in Nozzle Wall	A2	264.537	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	23013.756	NA	NA
TOTAL AREA AVAILABLE	Atot	26797.920	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (587.6 \cdot 43.1803 \cdot 1.0 + 2 \cdot 107.0 \cdot 43.1803 \cdot 1.0 \cdot (1 - 1.0))$
 $= 25372.775 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 587.6(1.0 \cdot 49.0 - 1.0 \cdot 43.18) - 2 \cdot 107.0$
 $(1.0 \cdot 49.0 - 1.0 \cdot 43.1803) \cdot (1 - 1.0)$
 $= 3419.627 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 122.5)(11.0 - 9.92)1.0$
 $= 264.537 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(122.5, 300.0, 100.0)) \cdot (110.0 - 14.0) \cdot 1.0 \\ &= 23013.756 \text{ mm}^2 \text{ Includes Bevel Area of } 3813.756 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad t_a = 12.9203 \text{ mm} \\ \text{Wall Thickness per UG16(b),} & \quad t_{r16b} = 4.5000 \text{ mm} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad t_{rb1} = 50.9409 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b1} = \max(t_{rb1}, t_{r16b}) = 50.9409 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b2} = \max(t_{rb2}, t_{r16b}) = 4.5000 \text{ mm} \\ \text{Wall Thickness per table UG-45} & \quad t_{b3} = 11.3312 \text{ mm} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ &= \min[11.331, \max(50.9409, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(12.9203, 11.3312) \\ &= 12.9203 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	80.9,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	264.1 MPa	Passed
Occasional	:	59.7,	Allowable	:	183.5 MPa	Passed
Shear	:	4.2,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 14.0, tr = 9.92, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.902, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-4 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 52.0, tr = 43.18, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.881, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	31 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 31 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 4.55/5.10 = 0.892

Weld Size Calculations, Description: N224"

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

$$\begin{aligned} \text{Weld Load [W]:} \\ &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (25372.7754 - 3419.6272 + 2 \cdot 107.0 \cdot 1.0 \cdot \\ &\quad (1.0 \cdot 49.0 - 43.1803)) \cdot 138) \\ &= 3201127.75 \text{ N} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

$$\begin{aligned} \text{Weld Load [W1]:} \\ &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (23278.293 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\ &= 3225930.00 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W2]:} \\ &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (23278.293 + 0.0 + 100.0 + (10486.0)) \cdot 138 \\ &= 4672874.50 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W3]:} \\ &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (23278.293 + 0.0 + 100.0 + 0.0 + (10486.0)) \cdot 138 \\ &= 4672874.50 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

$$\begin{aligned} \text{Shear, Outward Nozzle Weld [Sonw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 647445. \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Shear, Nozzle Wall [Snw]:} \\ &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 299.3) \cdot (110.0 - 3.0) \cdot 0.7 \cdot 138 \\ &= 9718080. \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Tension, Shell Groove Weld [Tngw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 609.6 \cdot (52.0 - 3.0) \cdot 0.74 \cdot 138 \\ &= 4791095. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

$$\begin{aligned} \text{Failure Path [Path 1-1]:} \\ &= (\text{SONW} + \text{SNW}) = (647445 + 9718080) = 10365526 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 2-2]:} \\ &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (647445 + 0 + 4791095 + 0) = 5438540 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 3-3]:} \\ &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (647445 + 4791095 + 0) = 5438540 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 10365526N	, must exceed W = 3201127 N	or W1 = 3225930 N
Path 2-2 = 5438540 N	, must exceed W = 3201127 N	or W2 = 4672874 N
Path 3-3 = 5438540 N	, must exceed W = 3201127 N	or W3 = 4672874 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 17.5766 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 369.5766 mm

Input Echo, WRC107/537 Item 1, Description: N224" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	5220.000	mm
Vessel Thickness	Tv	52.000	mm
Design Temperature	T1	110.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	801.600	mm
Nozzle Thickness	Tn	110.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Design Internal Pressure	Dp	4.551	MPa
Include Pressure Thrust		No	

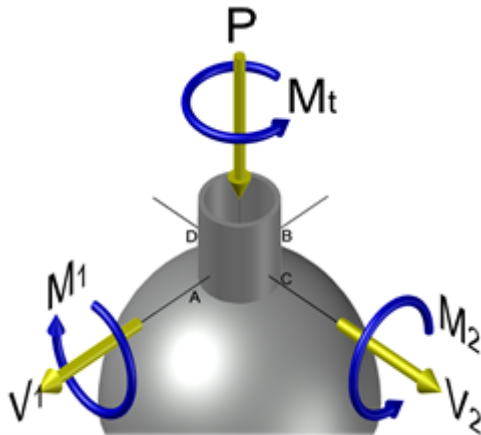
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-44982.2	N
Longitudinal Shear (SUS)	(V1) V1	29988.2	N
Circumferential Shear (SUS)	(Vc) V2	29988.2	N
Circumferential Moment (SUS)	(Mc) M1	50400792.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	50400792.0	N-mm
Torsional Moment (SUS)	Mt	-72000960.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 801.6 + 2 \cdot 1.65 \cdot \sqrt{(2637.5 (52.0 - 3.0))}$$

$$= 1987.937 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-44982.2	N
Circumferential Shear	(VC)	V2	29988.2	N
Longitudinal Shear	(VL)	V1	29988.2	N
Circumferential Moment	(MC)	M1	50400792.0	N-mm
Longitudinal Moment	(ML)	M2	50400792.0	N-mm
Torsional Moment		MT	-72000960.0	N-mm

Dimensionless Param: U = 1.11 TAU = 5.00 (3.25) RHO = 0.46

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03626	(A,B,C,D)
$M(x) / P$	SP 1	0.03750	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04110	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.06412	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04110	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.06412	(A,B,C,D)
$N(y) * T / P$	SP 1	0.05768	(A,B,C,D)
$M(y) / P$	SP 1	0.01074	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.05685	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.02023	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.05685	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.02023	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Rad. Bend.	P	4.2	-4.2	4.2	-4.2	4.2	-4.2	4.2	-4.2
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-2.4	-2.4	2.4	2.4
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-22.5	22.5	22.5	-22.5
Rad. Memb.	ML	-2.4	-2.4	2.4	2.4	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-22.5	22.5	22.5	-22.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-20.0	16.5	29.8	-23.6	-20.0	16.5	29.8	-23.6

Tang. Memb. P	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Tang. Bend. P	1.2	-1.2	1.2	-1.2	1.2	-1.2	1.2	-1.2
Tang. Memb. MC	0.0	0.0	0.0	0.0	-3.3	-3.3	3.3	3.3
Tang. Bend. MC	0.0	0.0	0.0	0.0	-7.1	7.1	7.1	-7.1
Tang. Memb. ML	-3.3	-3.3	3.3	3.3	0.0	0.0	0.0	0.0
Tang. Bend. ML	-7.1	7.1	7.1	-7.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-8.1	3.6	12.7	-3.9	-8.1	3.6	12.7	-3.9
Shear VC	0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5
Tot. Shear	-1.0	-1.0	-1.9	-1.9	-1.9	-1.9	-1.0	-1.0
Str. Int.	20.0	16.6	30.0	23.8	20.3	16.8	29.8	23.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		121.4	121.4	121.4	121.4	121.4	121.4	121.4	121.4
Rad. Pl (SUS)		-1.7	-1.7	3.1	3.1	-1.7	-1.7	3.1	3.1
Rad. Q (SUS)		-18.2	18.2	26.7	-26.7	-18.2	18.2	26.7	-26.7
Long. Pm (SUS)		121.4	121.4	121.4	121.4	121.4	121.4	121.4	121.4
Long. Pl (SUS)		-2.2	-2.2	4.4	4.4	-2.2	-2.2	4.4	4.4
Long. Q (SUS)		-5.9	5.9	8.3	-8.3	-5.9	5.9	8.3	-8.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5
Pm (SUS)		121.4	121.4	121.4	121.4	121.4	121.4	121.4	121.4
Pm+Pl (SUS)		119.9	119.9	125.9	125.9	119.9	119.9	125.9	125.9
Pm+Pl+Q (Total)		113.3	138.0	151.3	117.7	113.6	138.2	151.2	117.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.37	138.00	Passed
Pm+Pl (SUS)	125.92	207.00	Passed
Pm+Pl+Q (TOTAL)	151.33	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The computed vessel MAWP seems to be less than the design pressure!**

Please review all reports carefully!

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